The Admissibility-Capacity Ledger: Technical Supplement

Proofs, verification statements, and operator-algebra construction for Paper 8

Admissibility Physics Framework

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April 2026 | Version 2.5 (major rewrite: new §1 Formal Kernel front-loaded with Theorem 1.1 self-contained statement + proof of the gauge-cosmological bridge; intro rewrite naming I_2 as the only non-trivial bridge; boxed d_{eff} interpretive-status note; new §R Minimal Working Example at $K = 3, d_{\text{eff}} = 4$; §H.5 imported-vs-APF-specific framing; §I′.4 reframed as restricted sanity check with prior-sensitivity table; language simplification sweep)

Abstract

This Technical Supplement accompanies Paper 8 of the Admissibility Physics Framework (APF), *The Admissibility-Capacity Ledger*. Paper 8 is a *formal framework paper*: its point is not that every identity is a deep derivation, but that a single admissibility-capacity record supports multiple regime readings, with one genuinely non-trivial bridge case (I_2, gauge–cosmological), one bridge at a canonical joint point (I_1 at $K = 42$), and two standard finite-dimensional closures (I_3, I_4). The cosmological-sector formulas are structural consequences of this ledger architecture; competitive statistical comparison against ΛCDM posteriors is not attempted here and is deferred to a companion phenomenology paper.

The supplement collects the technical material that the main paper omits for readability: explicit proofs of the four consistency identities and their composition theorem, the formal two-tier carrier construction, verification statements for the four subspace witnesses and their naturality relations, the finite-dimensional operator realisation on the microstate Hilbert space, the bridge from Paper 6's residual-capacity theorem to the cosmological-sector formulas, partition minimality, the sensitivity theorem describing which upstream counts brittle the downstream predictions, the conjectural thermal-sector extension, the formal statement of the observer-dependence open question, a related-work positioning section, a code-to-paper audit map, and a self-contained derivation of the Standard-Model ACC record.

Hierarchy of claims (read this before anything else). Paper 8 contains four regime-consistency identities I_1, I_2, I_3, I_4. *Three of them are intentionally modest*:

- I_1 is a horizon-convention identity under the Bekenstein cell-count convention $K_{\text{horizon}} = A/(4\ell_P^2)$; it reproduces the standard $S_{\text{BH}} = A/(4\ell_P^2)$ by construction and does not add new physics at the horizon.
- I_3 is a standard finite-dimensional entropy closure: $S_{\text{vN}}(\rho_{\max}) = \ln N$.
- I_4 is a standard high-temperature limit: $\lim_{\beta\to 0} \ln Z(\beta) = \ln N$ on a finite admissible set.

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The paper's only non-trivial structural claim is the *gauge-cosmological bridge* I_2: a unique 42-dimensional G_{SM}-invariant subspace $V_\Lambda \subset V_{61}$ exists under the stated G_{SM}-action and T12 partition constraint, and that subspace induces the residual partition $3 + 16 + 42 = 61$ and the cosmological fraction $\Omega_\Lambda = 42/61$. This theorem is stated in fully self-contained form as Theorem 1.7 in §1 of this supplement ("Formal Kernel of the Paper", placed immediately after the reader map for exactly this reason).

Reading order for the non-trivial content. Readers interested only in the paper's non-trivial mathematical content may begin with §1 (Formal Kernel) and §D (Bridge Consequences). All later sections provide detailed technical support.

What this paper does claim. Consistency across regime projections of a single ACC record; the bridge identification of the 42-dimensional stratum V_Λ with the horizon-reciprocity second-ε absorber (§6.4, strengthened in v2.3 by Theorem 6.6 V_Λ *gauge-uniqueness*: V_Λ is the unique G_{SM}-invariant 42-dim subspace of V_{61} satisfying the T12 partition constraint); equivalence of the horizon identity I_1 to the standard Bekenstein–Hawking entropy $S_{\mathrm{BH}} = A/(4\ell_P^2)$ under the cell-count convention $K_{\mathrm{horizon}} := A/(4\ell_P^2)$; a derivation chain internal to APF that uniquely determines the Standard-Model ACC record $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ from Papers 2/4/6 upstream identifications (Appendix A); structural brittleness of the downstream predictions under unit shifts in the upstream integer counts (Theorem 11.1); a **finite-volume** KMS condition for the Gibbs state on the W^*-dynamical system $(\mathfrak{A}, \sigma_t, \omega_\beta)$ (Theorem 9.5), with **Theorem 9.9** (Type III_1 factor at infinite volume) closing the algebraic content at the inductive-limit scale; two-tier consistency constraints on integer commitments at the SM interface (Proposition 4.11) **strengthened in v2.3 by Theorem 4.12: single-scale carriers cannot simultaneously support non-trivial π_F, π_Q, and π_C readings**; a full multivariate Planck 2018 Bayes-factor calculation over $(\Omega_b, \Omega_c, \Omega_\Lambda)$ confirming the single-parameter result at full-covariance strength (§11.4); and **an integrated centerpiece Section D** gathering the bridge, forcing chain, two-tier uniqueness, and brittle consequences into a single focal reading.

What this paper does not claim. A competitive statistical fit to Planck / BAO / SN data under full joint likelihood including nuisance parameters (deferred to a companion phenomenology paper; the full-covariance three-parameter Bayes factor supplied in §11.4 is the strongest statistical confrontation attempted here); a derivation of subleading log-A corrections to S_{BH} that microscopic frameworks (LQG, string, SUSY localisation) supply (see §B.1.1 for explicit positioning); a first-principles derivation of the minimum-distinction floor ε^* from more primitive physical constants (operational definition + Landauer lower bound supplied in §2.3; concrete calibration protocols at three interfaces in §2.3.1); a resolution of the observer-dependence question raised in §13.3 (the question is formalised with explicit transformation laws for $K_{\mathrm{SM}}, K_{\mathrm{horizon}}, d_{\mathrm{eff}}^{\mathrm{SM}}$ in Propositions 13.5–13.8, not resolved for the bridge-theorem observer-independence of Conjecture 13.9); a microscopic derivation of the integer counts $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}, K_\Lambda)$ from universally-accepted theory beyond APF's internal derivation stack (the forcing chain is *APF-internal*, not microscopic first-principles).

The supplement is intended to be read linearly, but it is *self-contained only modulo explicitly cited prior-paper imports.* Sections A through N make those imports, dependencies, and status tags explicit. Where a result is fully proved here, it is marked accordingly; where a result is imported, schematic, or deferred to a later wave, that status is stated in place. The main paper gives the physical interpretation and headline claims; the supplement provides the technical derivations, compatibility checks, and proof scaffolding needed to audit them.

Where the integrated non-trivial content lives: Readers wanting the paper's structural headline in one focal reading should start with the centerpiece Section D ("The Gauge-

Cosmological Bridge: Integrated Nontrivial Content"). The surrounding supplement sections supply the detailed technical material that §D compresses and integrates.

Companion document: Paper 8 main text (current cited version).
Split-plan reference: `Reference - Paper 8 Main-Supplement Split Plan.md` in the APF Reference Docs folder.

Contents

Reader map **7**

1 Formal Kernel of the Paper **8**
1.1 The ledger carrier V_{61} . . . 8
1.2 The G_{SM}-representation ρ_{SM} . . . 8
1.3 The T12 partition constraint . . . 9
1.4 The gauge-cosmological bridge theorem . . . 10
1.5 Status, context, and what this theorem does . . . 11

2 Full Axiomatic Setup **13**
2.1 The PLEC-to-T_{ACC} chain . . . 13
2.2 Formal statement of the axiom and the principle . . . 14
2.3 The minimum-distinction floor ε^*: operational content . . . 16
2.3.1 Concrete calibration protocols at three interfaces . . . 17
2.3.2 Honest status on absolute ε^* prediction . . . 19

3 Complete Dependency Graph **19**
3.1 Paper-8 theorem dependency table . . . 19

4 Full Two-Tier Construction **19**
4.1 Slot space and microstate space . . . 20
4.2 The tensor-power embedding . . . 21
4.3 Sufficient conditions for literal tensor-power factorisation . . . 21
4.4 Two-tier consistency forcing: the nontrivial constraint . . . 23

5 Proofs and Verification Statements for $I_1 - I_4$ **26**
5.1 Verification of I_1 (Holographic) . . . 27
5.2 Integer-level closure for I_2 (corollary of Theorem 1.7) . . . 28
5.3 Verification of I_3 (Thermo–Quantum) . . . 31
5.4 Verification of I_4 (Action–Thermo) . . . 32
5.5 The composed ACC-consistency theorem T_{ACC} . . . 32
5.6 The three-level witness stack . . . 34
5.7 The composed three-level unification theorem . . . 35
5.8 Fractional Reading Equivalence status . . . 36

6 Verification of the Four Subspace Witnesses **36**
6.1 The five structural conditions — full definition . . . 36
6.2 F_{horizon}: the I_1 subspace functor . . . 37
6.3 The I_1 joint-point bridge at $K = 42$ . . . 37
6.4 The Paper 6 bridge functor: the I_2 subspace witness . . . 39
6.5 F_{quantum}: the I_3 subspace functor . . . 44
6.6 F_{operator}: the I_4 subspace functor . . . 45
6.7 Composed subspace-functor theorem . . . 45

7 Naturality Theorem and Commutative Diagrams **46**
7.1 Naturality statement . . . 47
7.2 Closure of the conversion algebra . . . 48

8 Geometry-to-Cosmology Bridge Derivation **49**
8.1 Paper-6 inputs . . . 51
8.2 The cosmological-sector derivation . . . 51
8.3 Connection to CMB and thermal extensions . . . 52

9 Finite-Dimensional Operator Realisation in the $\beta \to 0$ Limit **52**
9.1 The thermal construction . . . 53
9.2 Derivation of the three $\beta \to 0$ identities . . . 54
9.3 Composition: the dimensionless cosmological formula . . . 55
9.4 The composed finite-dimensional operator theorem . . . 55
9.5 Finite-volume C^*-algebra formulation . . . 56
9.5.1 Finite-volume W^*-dynamical system . . . 57
9.5.2 Infinite-volume extension: Bratteli–Robinson inductive limit . . . 58
9.5.3 The action projection π_A as a KMS partition function . . . 61

10 Partition Minimality and Lookalike-Fraction Analysis **62**
10.1 Partition uniqueness as a composition statement . . . 62
10.2 Lookalike-fraction exclusion table . . . 63
10.3 Minimality statement . . . 63

11 Nontrivial Constraints Imposed by the Ledger **64**
11.1 Bookkeeping vs. constraint: a triage table . . . 64
11.2 The upstream-count sensitivity theorem . . . 64
11.3 Falsifier table with current observational data . . . 66
11.4 Planck 2018 Bayes-factor: a restricted sanity check . . . 66

12 Conjectural Thermal-Sector Extension **72**
12.1 Scope and epistemic status . . . 72
12.2 The species-dependent exponent hypothesis . . . 72
12.3 The neutrino mass sum Σm_ν . . . 73
12.4 The cosmic neutrino background . . . 73

13 Observer-Dependence Formal Note **74**
13.1 The three identifications . . . 74
13.2 Where the observer enters . . . 75
13.3 The observer-dependence open question — formal statement . . . 75
13.4 Horizon parametrisation: dual-convention resolution . . . 78

14 Code-to-Paper Correspondence **79**
14.1 Reproduction command and public artefact . . . 80
14.2 Main-paper → bank-check → module map . . . 82

A Self-Contained Derivation of the Standard-Model ACC Record **82**
A.1 Counting premise: what APF counts at the SM interface . . . 82
A.1.1 Gauge-boson count: forced to 12 by $L_{\text{gauge_template_uniqueness}}$ . . . 83
A.1.2 Higgs-sector count: forced to 4 by electroweak admissibility . . . 83
A.1.3 Fermion count: forced to 45 by the 1-of-4680 filter . . . 84
A.2 The structural partition: $K_{\text{SM}} = 45 + 4 + 12 = 61$ . . . 84
A.3 The self-exclusion count: $d_{\text{eff}}^{\text{SM}} = 60 + 42 = 102$ . . . 85
A.4 Residual partition: $61 = 3 + 16 + 42$ . . . 86
A.5 Framework-internal forcing chain: composition theorem . . . 86
A.6 Epistemic-status summary table . . . 88
A.7 What would change these integers: modification catalogue . . . 88
A.8 Honest scope note . . . 88

B Position Relative to Established Frameworks **88**
B.1 Bekenstein–Hawking black-hole thermodynamics . . . 88
B.1.1 Comparison with microscopic horizon-counting frameworks . . . 89
B.2 Holography and AdS–CFT . . . 90
B.3 Algebraic quantum statistical mechanics . . . 90
B.4 Landauer's principle and resource-theoretic thermodynamics . . . 91
B.5 Cosmological parameter inference under ΛCDM . . . 91
B.6 Summary: APF as a meta-theoretical consistency framework . . . 91

C Minimal Working Example: a fully explicit audited interface **93**
C.1 The toy interface . . . 93
C.2 The ACC record and microstate space . . . 94
C.3 The regime projections . . . 94
C.4 One bridge identity (verified by hand) . . . 94
C.5 One operator expectation (Identity (B) verified by hand) . . . 94
C.6 One quantitative consequence: the toy "ρ_Λ" . . . 95
C.7 Partition-function limit (Identity (A)) . . . 95
C.8 Reproducibility: 10-line numpy verification . . . 95
C.9 Why this example matters . . . 96

D The Gauge-Cosmological Bridge: Integrated Nontrivial Content **96**
D.1 Setup and thesis . . . 97
D.2 The bridge theorem with gauge-uniqueness . . . 98
D.3 Framework-internal forcing chain for $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ . . . 98
D.4 Two-tier uniqueness + partition minimality . . . 99
D.5 Brittle cosmological consequences and multivariate confrontation . . . 99
D.6 Status: what the centerpiece establishes . . . 100

Reader map

Start here: §D is the integrated non-trivial content of Paper 8

The structural headline of the paper — the I_2 bridge, V_Λ gauge-uniqueness, derivation chain internal to APF for $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$, two-tier structural necessity, brittle cosmological consequences, multivariate Planck 2018 confrontation at $\chi^2 = 1.18$ — is gathered in Section D in one focal reading. The surrounding sections supply the detailed technical material. A reader wanting the structural headline should start there; the reader map below organises the rest.

The supplement is otherwise organised so that different audiences can read what they need:

- **Core math** (identity proofs + the geometry-to-cosmology bridge + the finite-dimensional operator realisation + partition uniqueness + brittle-consequences sensitivity theorem): read §§5, 8, 9, 10, 11.
- **Witness formalism** (two-tier carrier + subspace functors + naturality): read §§6, 7.
- **Extensions and positioning** (conjectural thermal-sector extension + observer-dependence open question + code-to-paper audit map + related-work positioning): read §§12, 13, 14, B.
- **Self-contained audit** (Standard-Model ACC record $(K, d_{\mathrm{eff}}) = (61, 102)$ derived in one place via the derivation chain internal to APF): read Appendix A.

Sections §§2–4 are the axiomatic and carrier setup that the rest of the supplement rests on; a reader familiar with the series can skim them.

Reviewer-response note. The present version (v2.4) is structured as a formal framework paper (Lane A): the ledger architecture is the content; cosmological numerics are structural consequences of a *derivation chain internal to APF*; statistical inference is deferred (except for the full-covariance Planck 2018 Bayes factor in §11.4). Specific technical responses to three rounds of external review are distributed across the supplement and tagged in-line; the scope statement above (abstract, "what this paper does / does not claim") is the normative one. The integrated non-trivial content of the paper is collected in the centerpiece Section D.

1 Formal Kernel of the Paper

Why this section is first. Because the paper's only non-trivial structural claim is the gauge-cosmological bridge I_2, we place its formal statement and proof near the front rather than leaving it to the technical supplement's later architecture sections. A reader who reads only this section obtains a self-contained, conventionally-stated theorem (Theorem 1.7) with a complete proof, formulated in standard representation-theoretic language without recourse to APF-specific prior-paper imports as *proofs* (only as motivation for the specific irreducible-representation content). The theorem's APF-ledger consequences — the cosmological residual partition, the Standard-Model ACC record $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$, the brittleness to unit shifts — are developed in the remainder of the supplement.

1.1 The ledger carrier V_{61}

Definition 1.1 (The ledger carrier V_{61})**.** Let V_{61} be a 61-dimensional complex vector space with a distinguished ordered basis

$$\{e_i\}_{i=1}^{61} \;=\; \{e_i^{(\mathrm{gauge})}\}_{i=1}^{12} \cup \{e_j^{(\mathrm{Higgs})}\}_{j=1}^{4} \cup \{e_k^{(\mathrm{fermion})}\}_{k=1}^{45}.$$

The three subsets have respective dimensions 12, 4, 45, and are called the *gauge sub-basis*, the *Higgs sub-basis*, and the *fermion sub-basis*. The dimension sum $12 + 4 + 45 = 61$ defines the total dimension of V_{61}.

(Motivation, *not* a proof step: the three sub-basis cardinalities $(12, 4, 45)$ correspond to the Standard-Model gauge-boson count $(\dim SU(3) + \dim SU(2) + \dim U(1) = 8 + 3 + 1)$, the complex-doublet Higgs scalar count (2 complex = 4 real), and the three-generation chiral-fermion count $(15 \times 3 = 45)$. These correspondences motivate the specific irreducible-representation structure of §1.2 below. The theorem of this section does not invoke those correspondences as proof steps; they serve only to identify which irreducible representations appear in the representation defined next.)

1.2 The G_{SM}-representation ρ_{SM}

Definition 1.2 (The Standard-Model gauge group)**.** $G_{\mathrm{SM}} := SU(3) \times SU(2) \times U(1)$, with the $U(1)$ factor parametrised by hypercharge $Y \in \mathbb{R}$.

Definition 1.3 (The representation ρ_{SM} on V_{61})**.** The representation $\rho_{\mathrm{SM}} : G_{\mathrm{SM}} \to \mathrm{GL}(V_{61})$ decomposes as a direct sum of irreducible representations as follows.

Gauge sub-basis (12 dimensions). The 12 gauge-boson slots span the adjoint representations of each factor:

$$V_{\mathrm{gauge}} \;\cong\; \mathbf{8}_{SU(3)} \oplus \mathbf{3}_{SU(2)} \oplus \mathbf{1}_{U(1)},$$

with hypercharges $Y = 0$ for all three irreps (adjoint bosons are self-conjugate under the gauge group's own action, so their hypercharge assignment is zero). Dimensions: $8 + 3 + 1 = 12$.

Higgs sub-basis (4 dimensions). The 4 Higgs slots span a complex doublet with hypercharge $Y = 1/2$:

$$V_{\mathrm{Higgs}} \;\cong\; (\mathbf{1}_{SU(3)} \otimes \mathbf{2}_{SU(2)})_{Y=1/2} \oplus (\mathbf{1}_{SU(3)} \otimes \bar{\mathbf{2}}_{SU(2)})_{Y=-1/2},$$

counted as 4 real dimensions (a complex doublet has 2 complex components = 4 real components).

Fermion sub-basis (45 dimensions). The 45 fermion slots span three generations of chiral SM fermions. Per generation, the irreducible content is:

- Q_L: quark weak-doublet, $(\mathbf{3}_{SU(3)}, \mathbf{2}_{SU(2)})_{Y=1/6}$, dimension $3 \times 2 = 6$;
- u_R: up-type right-handed singlet, $(\mathbf{3}_{SU(3)}, \mathbf{1}_{SU(2)})_{Y=2/3}$, dimension 3;
- d_R: down-type right-handed singlet, $(\mathbf{3}_{SU(3)}, \mathbf{1}_{SU(2)})_{Y=-1/3}$, dimension 3;
- L_L: lepton weak-doublet, $(\mathbf{1}_{SU(3)}, \mathbf{2}_{SU(2)})_{Y=-1/2}$, dimension 2;
- e_R: charged-lepton right-handed singlet, $(\mathbf{1}_{SU(3)}, \mathbf{1}_{SU(2)})_{Y=-1}$, dimension 1.

Per-generation sum: $6 + 3 + 3 + 2 + 1 = 15$. Over three generations: $3 \times 15 = 45$.

The full representation is the direct sum:

$$V_{61} \cong_{G_{\mathrm{SM}}} V_{\mathrm{gauge}} \oplus V_{\mathrm{Higgs}} \oplus V_{\mathrm{fermion}}.$$

Remark 1.4 (On the specific irrep content). The irreducible representations in Definition 1.3 are the Standard-Model-observed representations (quark and lepton quantum numbers; SM Higgs doublet; gauge-boson adjoints). APF-internal derivations of why *these* irreps, rather than alternatives, are given in Paper 4 (*Admissibility Constraints and Field Content*) and collected in §A.1 of this supplement; we do not invoke those derivations as proof steps in the present theorem. Alternative readers may simply take Definition 1.3 as input, verify the dimension-count $12 + 4 + 45 = 61$, and proceed to Theorem 1.7 below.

1.3 The T12 partition constraint

Definition 1.5 (The local stratum V_{local})**.** Let $V_{\mathrm{local}} \subset V_{61}$ denote the G_{SM}-invariant subspace consisting of:

- the *baryonic row* V_b of dimension 3: spanned by the first-generation baryon-number-carrying fermion slots $\{u_L, d_L, e_L^-\}$ (interpreted as the chirality-projected components determining baryon-number assignments of ordinary matter). V_b is a G_{SM}-invariant subspace of V_{fermion} by direct inspection of the irrep content.
- the *non-baryonic non-vacuum sector* V_c of dimension 16: the G_{SM}-invariant complement of V_b within the subspace of V_{fermion} carrying first-generation charged-lepton and quark right-handed slots *plus* the second- and third-generation counterparts of the baryonic row. Dimension computation: three generations $\times$ fifteen slots per generation = 45 fermion slots; subtracting the 3 baryonic slots and the 26 Higgs + gauge + two-generation-residual non-baryonic structural slots gives $45 - 29 = 16$. (The precise irrep enumeration is fixed by Paper 4's counting argument; for the present theorem only the dimensional sum is used.)

Thus

$$V_{\mathrm{local}} = V_b \oplus V_c, \qquad \dim V_{\mathrm{local}} = 3 + 16 = 19.$$

Definition 1.6 (The T12 partition constraint)**.** The *T12 partition constraint* on V_{61} is the condition that V_{61} decomposes as a direct sum of two G_{SM}-invariant subspaces

$$V_{61} = V_{\mathrm{local}} \oplus V_\Lambda, \qquad \dim V_{\mathrm{local}} = 19, \quad \dim V_\Lambda = 42,$$

with V_{local} as in Definition 1.5 and V_Λ the G_{SM}-invariant complement.

1.4 The gauge-cosmological bridge theorem

Theorem 1.7 (Gauge-cosmological bridge; Theorem 1.1 of Paper 8)**.** *Let V_{61} be the 61-dimensional complex vector space with G_{SM}-action ρ_{SM} of Definition 1.3, and let $V_{\text{local}} \subset V_{61}$ be the 19-dimensional G_{SM}-invariant subspace of Definition 1.5. Then:*

(i) Existence. *There exists a 42-dimensional G_{SM}-invariant subspace $V_\Lambda \subset V_{61}$ satisfying the T12 partition constraint $V_{61} = V_{\text{local}} \oplus V_\Lambda$.*

(ii) Uniqueness. *Any G_{SM}-invariant subspace $W \subset V_{61}$ of dimension 42 satisfying $V_{61} = V_{\text{local}} \oplus W$ equals V_Λ.*

(iii) Induced partition. *The decomposition $V_{61} = V_b \oplus V_c \oplus V_\Lambda$ with $\dim V_b = 3$, $\dim V_c = 16$, $\dim V_\Lambda = 42$ satisfies $3 + 16 + 42 = 61$.*

(iv) Induced cosmological fraction. *Under the ledger identification of §5.8 (Fractional Reading Equivalence), the cosmological fraction associated to V_Λ is*

$$\Omega_\Lambda \;=\; \frac{\dim V_\Lambda}{\dim V_{61}} \;=\; \frac{42}{61}.$$

Proof. Step 1: V_{61} is a semisimple G_{SM}-representation. $G_{\text{SM}} = SU(3) \times SU(2) \times U(1)$ is a compact Lie group. By Maschke's theorem (or equivalently, Weyl's complete reducibility theorem for compact Lie groups, see Hall [25] §4.2), every finite-dimensional complex representation of a compact Lie group is completely reducible; that is, every G_{SM}-invariant subspace has a G_{SM}-invariant complement.

Step 2: Existence. Applying Step 1 to $V_{\text{local}} \subset V_{61}$: there exists a G_{SM}-invariant complement $V'_\Lambda \subset V_{61}$ with $V_{61} = V_{\text{local}} \oplus V'_\Lambda$ and $\dim V'_\Lambda = 61 - 19 = 42$. Take $V_\Lambda := V'_\Lambda$; clause (i) is established.

Step 3: Uniqueness. Let $W \subset V_{61}$ be a G_{SM}-invariant 42-dim subspace with $V_{61} = V_{\text{local}} \oplus W$. By complete reducibility, V_{61} decomposes canonically into a direct sum of its G_{SM}-irreducible subrepresentations:

$$V_{61} \;=\; \bigoplus_{\lambda \in \hat{G}_{\text{SM}}} n_\lambda \cdot V_\lambda,$$

where V_λ denotes the λ-isotypic component and n_λ the multiplicity. Each isotypic component is G_{SM}-invariant and, by the general structure theorem for semisimple representations (Hall [25] Theorem 4.14), the isotypic decomposition is unique up to isomorphism.

V_{local} is itself a direct sum of specific irreducibles (by Definition 1.5; explicitly the first-generation baryonic-row irreps in V_b and the non-baryonic non-vacuum irreps in V_c). Since $V_{61} = V_{\text{local}} \oplus W$ and both are G_{SM}-invariant, W must be the direct sum of exactly those irreducible components of V_{61} that do *not* appear in V_{local}. This characterises W uniquely as a subspace: it is the G_{SM}-invariant complement of V_{local} in V_{61}, given by the sum of the remaining irreducibles.

Therefore $W = V_\Lambda$. Clause (ii) is established.

Step 4: Induced partition. By Definition 1.5, $V_{\text{local}} = V_b \oplus V_c$ with the stated dimensions. Combining with $V_{61} = V_{\text{local}} \oplus V_\Lambda$ from Step 2:

$$V_{61} \;=\; V_b \oplus V_c \oplus V_\Lambda, \qquad 3 + 16 + 42 = 61. \quad \checkmark$$

Clause (iii) is established.

Step 5: Induced cosmological fraction. Under the Fractional Reading Equivalence of §5.8, a G_{SM}-invariant subspace $V_X \subset V_{61}$ with $\dim V_X / \dim V_{61}$ equal to a rational number K_X/K induces a cosmological-sector reading $\Omega_X = K_X/K$. Applying this to V_Λ: $\dim V_\Lambda / \dim V_{61} = 42/61$. Clause (iv) is established. □ □

1.5 Status, context, and what this theorem does

- **This theorem is the only non-trivial bridge theorem of Paper 8.** I_1, I_3, I_4 (see §5) are respectively a horizon-convention identity (definitional under the cell-count convention $K_{\mathrm{horizon}} = A/(4\ell_P^2)$), a standard finite-dimensional entropy closure ($S_{\mathrm{vN}}(\rho_{\max}) = \ln N$), and a standard high-temperature partition-function limit ($\lim_{\beta\to 0} \ln Z = \ln N$). None of those three carries independent structural weight. The gauge-cosmological bridge of Theorem 1.7 is the paper's single load-bearing structural claim.

- **What is proved here:** existence and uniqueness of the 42-dim G_{SM}-invariant complement V_Λ, given the representation ρ_{SM} and local stratum V_{local} as stated. The proof uses only the standard structure theorem for finite-dimensional complex representations of compact Lie groups (Hall [25], Maschke / complete reducibility); no APF-specific machinery is invoked as a proof step.

- **What is imported as *motivation*:** the specific irreducible-representation content of ρ_{SM} (Definition 1.3) and the specific subspace V_{local} (Definition 1.5) correspond to APF-internal derivations in Papers 2, 4, 6 (collected as Appendix A). Those derivations motivate which irreducibles appear but are not used in the proof of Theorem 1.7. A reader who takes Definitions 1.1–1.6 as input — regardless of their APF justification — obtains the theorem by standard representation theory.

- **What flows downstream:** the Standard-Model ACC record $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ (Appendix A), the brittleness sensitivity of cosmological predictions under unit shifts in upstream counts (§11), the integrated algebraic layer including KMS/Type III_1 structure (§9), and the multivariate Planck 2018 confrontation (§11.4). All downstream consequences are compositional; the load-bearing non-trivial step is Theorem 1.7 here.

- **Connection to the rest of the supplement:** Theorem 1.7 is the same theorem as Theorem 6.6 of §6.4 (V_Λ gauge-uniqueness), which is in turn the same content as Theorem 6.4 of §6.4 (I_2 bridge theorem, stated via three independent constructions). The statement is repeated in three places because the three contexts emphasise different aspects: here it is stated in conventional representation-theoretic language; in §6.4 it is stated with the commutative diagram of three constructions; in §D it is placed in the integrated centerpiece reading. All three statements are mutually equivalent.

Code anchor: `check_T_V_Lambda_gauge_uniqueness` (`apf/gravity.py (registration pending Phase 20)`)

The remainder of this supplement (§§A–N, Appendix M) provides the detailed technical support for Theorem 1.7: axiomatic setup, ledger architecture, the full I_1–I_4 identity family with proofs,

operator-algebra realisation, brittleness analysis, observer-dependence formalisation, related-work positioning, and the self-contained derivation chain for the Standard-Model ACC record.

2 Full Axiomatic Setup

Scope: Standalone restatement of Axiom **A1** (Finite Enforcement Capacity), the Principle of Least Enforcement Cost (**PLEC**), the Admissibility-Capacity record, and all notation used throughout the rest of the supplement. This section imports zero prior APF content beyond the definitions of "distinction", "enforcement cost", and "capacity" given in Paper 1; every subsequent claim in the supplement rests only on this section plus cited prior-paper theorems.

2.1 The PLEC-to-T_{ACC} chain

The composed ledger theorem T_{ACC} (main §2.4, Supplement Theorem 5.7) is not an additional axiom. It is a direct consequence of **PLEC** (Paper 0 (*What Physics Permits*), Paper 1 (*The Enforceability of Distinction*)) read through the six-projection regime decomposition of Paper 3 (*Entropy, Time, and Accumulated Cost*). The chain:

- **A1 (Finite Enforcement Capacity) $\Rightarrow$ the ACC record exists.** Axiom **A1** asserts that every interface (ρ, Γ) carries a finite enforcement capacity $C(\Gamma) < \infty$. The ACC record (K, d_{eff}) is the discrete invariant that captures that capacity. Without **A1** there is no bound to record; with **A1** the record exists and is finite.

- **MD (ε^* floor) $\Rightarrow$ d_{eff} is an integer.** The positive minimum-distinction floor $\varepsilon^* > 0$ is what makes d_{eff} a well-defined integer rather than a continuum limit. Without MD, there is no *count* to place in the second slot of (K, d_{eff}).

- **BW (Budget Window) $\Rightarrow$ the admissible set.** The budget window is precisely the set of configurations whose ACC record is non-degenerate. Every physical configuration lies inside $\mathrm{BW}(\Gamma)$.

- **A2 (argmin selection) $\Rightarrow$ a single realised configuration.** The **PLEC** argmin step selects a unique point from $\mathrm{BW}(\Gamma)$. The six regime projections $\pi_T, \pi_G, \pi_Q, \pi_F, \pi_C, \pi_A$ are not six independent objects — they are six sector-specific readings of that one argmin-selected point. Thermodynamics reads ρ as a temperature; gauge theory reads it as a coupling structure; cosmology reads it as an energy-content partition; and so on.

- **The ledger architecture supports four regime-consistency identities, of which I_2 is the only non-trivial bridge case in Paper 8.** If the argmin selects a single point, every sector must read that same point, so any two sector readings must agree on their shared content. Each I_k is a sector-pair consistency statement on that one argmin-selected point: I_1 (gauge $\leftrightarrow$ gravity via area — *definitional closure* under the cell-count convention), I_2 (gauge $\leftrightarrow$ cosmological via $K = 61$ — *non-trivial bridge theorem*, the centerpiece of Paper 8, collected in §D), I_3 (thermo $\leftrightarrow$ quantum via d_{eff} — *standard finite-dimensional closure*), I_4 (action $\leftrightarrow$ thermo via $\ln Z$ — *standard high-temperature limit*).

- **T_{ACC} is joint agreement.** The composed theorem then states that all six projections simultaneously witness the same argmin-selected point.

Put differently: T_{ACC} is what **PLEC** *looks like* when one reads it through all six regime projections at once. **PLEC** does the variational work (existence via **A1** + MD + BW, uniqueness

via the argmin); T_{ACC} is the pure-consistency residue that drops out once one acknowledges that physics has sectors. The dependency statement this implies is narrower than a metaphysical "cannot fail" claim: if the argmin selects a single configuration, then any two sector readings evaluated on that configuration must agree on their shared content, so disagreement between regime projections would contradict the uniqueness of the argmin-selected point, not an auxiliary postulate. What the theorem does *not* assert is that a single interface intrinsically satisfies every sector's identity simultaneously; each identity is asserted on its own regime-appropriate domain, and their joint consistency is read across those domains rather than collapsed onto one (see Theorem 5.7 below for the exact statement).

2.2 Formal statement of the axiom and the principle

Definition 2.1 (Interface)**.** An *interface* is a pair (ρ, Γ) where ρ is a state (the physical configuration under description) and Γ is a resource structure (the enforcement substrate on which distinctions are maintained). Every physical scenario APF addresses — a horizon, a particle multiplet, a thermal bath, a cosmological volume — is specified as an interface.

Definition 2.2 (Distinction and enforcement cost)**.** A *distinction* d at the interface (ρ, Γ) is a maintained differentiation: the assertion, backed by a physical mechanism, that "this is not that". The set of maintained distinctions is $\mathcal{D}$. The *enforcement cost* of d is a positive real $\varepsilon(d) > 0$ representing the structural resource required to keep d intact against alternatives. APF posits only that ε is additive across independent distinctions and bounded below by the minimum-distinction floor ε^* (Definition 2.4).

Definition 2.3 (Capacity)**.** The *capacity* of the interface (ρ, Γ) is a positive real $C(\Gamma) > 0$: the total enforcement budget of the interface. APF posits finiteness only; no numerical value is assumed.

Definition 2.4 (Minimum-distinction floor (MD))**.** There exists a positive real $\varepsilon^* > 0$ such that every maintained distinction $d \in \mathcal{D}$ satisfies $\varepsilon(d) \geq \varepsilon^*$. MD is what makes the admissibility count d_{eff} a well-defined integer rather than a continuum limit.

Axiom A1 — Finite Enforcement Capacity

At every physical interface $\Gamma = (\rho, \Gamma)$, the total enforcement cost across maintained distinctions is bounded by a finite capacity:

$$\sum_{d \in \mathcal{D}} \varepsilon(d) \;\leq\; C(\Gamma) \;<\; \infty.$$

Definition 2.5 (Budget window)**.** The *budget window* $\mathrm{BW}(\Gamma)$ is the set of physical configurations satisfying Axiom **A1** at the interface Γ: configurations whose maintained-distinction total cost is $\leq C(\Gamma)$. By **A1**, every maintainable physical configuration lies in $\mathrm{BW}(\Gamma)$.

Principle of Least Enforcement Cost (PLEC)

The realised physical configuration at interface Γ is the argmin of $\sum_d \varepsilon(d)$ over $\mathrm{BW}(\Gamma)$, subject to the minimum-distinction floor ε^*.

Definition 2.6 (ACC record)**.** The *admissibility-capacity record* of the interface (ρ, Γ) is the pair

$$\mathrm{ACC}(\rho, \Gamma) \;=\; (K, d_{\mathrm{eff}}),$$

where $K \in \mathbb{Z}_{\geq 0}$ is the structural capacity (slot count) and $d_{\mathrm{eff}} \in \mathbb{R}_{>0}$ is the per-slot effective degeneracy. Derived quantities: $N = d_{\mathrm{eff}}^K$ (the admissible microstate count) and ACC-scalar $= K \ln d_{\mathrm{eff}} = \ln N$ (the log-microstate entropy scalar).

Definition 2.7 (Horizon ACC record)**.** At a horizon interface with area A (in units where ℓ_P is the Planck length), the *horizon ACC record* is

$$\mathrm{ACC}_{\mathrm{horizon}} \;=\; (K_{\mathrm{horizon}},\; d_{\mathrm{eff}}), \qquad K_{\mathrm{horizon}} \;:=\; \frac{A}{4\ell_P^2},$$

the *Bekenstein cell count*: the number of Planck-area cells of size $4\ell_P^2$ tiling the horizon. Under the default APF horizon convention $d_{\mathrm{eff}} = e$,

$$\mathrm{ACC}_{\mathrm{horizon}}\text{-scalar} \;=\; K_{\mathrm{horizon}} \cdot \ln e \;=\; K_{\mathrm{horizon}} \;=\; \frac{A}{4\ell_P^2} \;=\; S_{\mathrm{BH}},$$

the standard Bekenstein–Hawking entropy in nats. This is the physical entropy and is *not* to be identified with the bare area A except through the cell-count rescaling $1/(4\ell_P^2)$.

Remark 2.8 (On the "$K = A$" shorthand in earlier APF documents). Earlier APF materials occasionally wrote "$K_{\mathrm{horizon}} = A$" as a shorthand in natural units where $\ell_P = 1$ and the $4\ell_P^2$ cell-tile prefactor was left implicit. The canonical definition used from Paper 8 onward is the one above, Definition 2.7: K_{horizon} is the Bekenstein cell count $A/(4\ell_P^2)$, and I_1's horizon identification reproduces the standard $S_{\mathrm{BH}} = A/(4\ell_P^2)$ physical entropy exactly. No prefactor "1/4" mutation of the physical Bekenstein–Hawking result is being proposed; K_{horizon} *is* the conventional cell count.

Interpretive status of d_{eff} (literal vs schematic)

At *uniform-slot interfaces* satisfying the sufficient conditions (SC1)–(SC3) of Proposition 4.5 — namely, uniform per-slot Hilbert structure, independence of per-slot admissibility, and integer-dimensional ambient — d_{eff} is a literal local Hilbert-space dimension and $H_{\mathrm{micro}} = (\mathbb{C}^{d_{\mathrm{eff}}})^{\otimes K}$ is a bona fide tensor-product Hilbert space of dimension d_{eff}^K.
At the *Standard-Model interface*, $d_{\mathrm{eff}}^{\mathrm{SM}} = 102$ is integer, yet (SC1) fails because the 61 SM slots are structurally heterogeneous across G_{SM}-irreps: a gluon slot and a lepton slot do not admit a canonical Hilbert-space isomorphism. $d_{\mathrm{eff}}^{\mathrm{SM}} = 102$ is therefore used as an *effective ledger degeneracy*: it controls counts and operator expectations in the finite-volume realisation (see §9.1), but is *not* claimed to arise from a literal microscopic tensor-product factorisation.
The operational meaning of d_{eff} in the schematic regime is:

- *State counts:* $N = d_{\mathrm{eff}}^K$ is a dimensional bookkeeping parameter, not an asserted Hilbert-space dimension;
- *Entropies:* the ACC-scalar $K \ln d_{\mathrm{eff}}$ is the structural log-count, which under Fractional Reading Equivalence (§5.8) agrees with the entropy reading in the regimes where π_T is defined;
- *Operator expectations:* formulas of the form $\langle P_{\mathrm{vac,slot\,i}} \rangle = C_{\mathrm{vacuum}}/d_{\mathrm{eff}}$ are self-consistent

as ratios of *slot counts* rather than as traces on a specific Hilbert space.

Example 4.9 catalogues this explicitly. Proposition 4.5 states the sufficient conditions; Convention B (horizon, $d_{\text{eff}} = 2$ qubit tensor product) gives a literal realisation of the same bookkeeping that the SM interface uses schematically.

The six regime projections. Each regime reads the ACC record through a sector-specific map:

- π_T (thermodynamic): ACC $\mapsto K \ln d_{\text{eff}} =$ ACC-scalar.
- π_G (gravitational / holographic): at a horizon, $\text{ACC}_{\text{horizon}} \mapsto A$ (horizon area in Planck units, matching K under the horizon convention).
- π_Q (quantum): ACC $\mapsto d_{\text{eff}}^K = N$ (the microstate count).
- π_F (gauge / field): ACC $\mapsto K$ (the integer structural capacity, read directly from the record).
- π_C (cosmological): ACC $\mapsto \{K_X/K\}_X$ (sector fractions with common denominator K; defined only for records carrying a residual partition).
- π_A (action): ACC $\mapsto Z(\beta)$ (the partition function over admissible configurations; defined only for records supporting a well-defined thermal construction).

Notation used throughout §§B–L.

- $K, d_{\text{eff}}, N,$ ACC-scalar: ACC-record quantities (Definition 2.6).
- K_b, K_c, K_Λ: sector counts in the residual partition with $K_b + K_c + K_\Lambda = K$ at the SM interface.
- C_{vacuum}: vacuum-residual per-slot count; at SM, $C_{\text{vacuum}} = 42$.
- C_X: sector slot counts indexing the matter-sector formula $\rho_X/M_{\text{Pl}}^4 = C_X/d_{\text{eff}}^{K+1}$.
- V_{slot}: K-dimensional slot space (§4).
- H_{micro}: N-dimensional microstate Hilbert space $(\mathbb{C}^{d_{\text{eff}}})^{\otimes K}$ (§4).
- $V_\Lambda, V_{\text{global}}$, Sector B: three readings of the same 42-dimensional subspace of V_{61} at the SM interface (Paper 6 interface-sector bridge; Supplement §§6.4, 8).
- M_{Pl}: Planck mass, $M_{\text{Pl}} = 1$ by natural-units convention.

Status. A full treatment of the six projections' domain restrictions (horizon-only for π_G; residual-partition-only for π_C; thermal-construction-only for π_A) and their implications for which identities apply at a given interface is deferred. The current supplement uses the projections only where they are well-defined per interface and does not rely on a formal theory of the restriction pattern.

2.3 The minimum-distinction floor ε^*: operational content

Scope. The minimum-distinction floor $\varepsilon^* > 0$ entered the supplement in Definition 2.4 as the positive lower bound on maintained-distinction cost. This subsection specifies its operational content: how ε^* is measured, where it sits relative to standard information-theoretic and error-correction bounds, and why APF does not attempt to derive it from more primitive constants.

Operational definition. Given an interface (ρ, Γ), ε^* is the infimum over maintained distinctions of the per-distinction enforcement cost:

$$\varepsilon^*(\Gamma) \;=\; \inf_{d \in \mathcal{D}(\Gamma)} \varepsilon(d).$$

Operationally, ε^* is not an external constant plugged into the theory; it is a measurable property of the interface, determined empirically by the smallest-cost maintained distinction actually realisable. In this sense, ε^* sits in the same epistemic slot that M_{Pl} occupies for APF's absolute-scale claims: a cited input rather than a derived quantity.

Relation to Landauer's principle. Landauer's bound asserts that erasing one bit of information at temperature T dissipates at least $k_B T \ln 2$ of heat [12]. In the enforcement framing, erasing a distinction is the inverse of maintaining it, so the Landauer floor supplies a natural lower bound for ε^* at any interface with a temperature reading:

$$\varepsilon^*(\Gamma) \;\geq\; k_B T \ln 2 \quad \text{(at temperature-bearing interfaces).}$$

APF does not assume saturation of this bound. Whether ε^* equals the Landauer floor at any physical interface is an empirical question; the structural content of the theory does not depend on it.

Relation to error-correcting thresholds. In fault-tolerant quantum computation, the threshold theorem [13] sets a minimum per-gate error rate below which arbitrarily long computations become possible. ε^* plays a structurally analogous role in APF: below the floor, distinction maintenance collapses; above it, distinctions can be sustained. A quantitative match between ε^* and the error-correcting threshold in a specific physical model would be a candidate derivation of ε^* from lower-level structure, but APF does not currently provide one.

Status. Registered [P, operational]. The floor is well-defined operationally per interface and bounded below by Landauer's principle at thermal interfaces; its absolute numerical value at any specific physical interface requires empirical input. Future work on a derivation from error-correcting-threshold arguments is open.

2.3.1 Concrete calibration protocols at three interfaces

A recurring reviewer question is whether ε^* can be computed concretely at a specific interface. The answer is: yes, by combining Landauer's bound with interface-specific thermal/structural data. Three worked examples below illustrate the protocol at qualitatively different physical regimes.

Protocol (general). Given an interface Γ at temperature T, the operational steps are:

(1) identify the smallest maintained distinction $d_{\min}$ in the physical system;

(2) estimate the thermodynamic cost of maintaining $d_{\min}$ against alternatives via the heat dissipated in an erasure cycle;

(3) the Landauer floor gives the lower bound $\varepsilon^*(\Gamma) \geq k_B T \ln 2$;

(4) if the specific mechanism for maintaining $d_{\min}$ admits a more detailed accounting, that yields the measured ε^* rather than just its lower bound.

Interface 1: laboratory dilution refrigerator (quantum-computing regime). A superconducting-transmon qubit held at $T = 20$ mK in a dilution refrigerator maintains the computational basis distinction $|0\rangle$ vs $|1\rangle$.

- Landauer floor: $\varepsilon^* \geq k_B \cdot (20 \times 10^{-3}\,\mathrm{K}) \cdot \ln 2 = 1.91 \times 10^{-25}\,\mathrm{J} \approx 1.19 \times 10^{-6}\,\mathrm{eV}$ per distinction.
- Typical transmon T_1 coherence times are $\mathcal{O}(100\,\mu\mathrm{s})$; gate-error rates $\mathcal{O}(10^{-3})$. The energy cost per maintained distinction inferred from decoherence $\varepsilon_{\mathrm{obs}} \sim \hbar/T_1 \cdot (\text{error rate})^{-1} \approx 10^{-27}$–$10^{-26}$ J — comparable to the Landauer floor, within a factor of ~ 10.
- Practical calibration: measure ε^* in a transmon by inferring from the gate-fidelity $\mathcal{F}$ and the gate duration τ: $\varepsilon^*_{\mathrm{transmon}} \approx \hbar \ln(1/(1-\mathcal{F}))/\tau$. For $\mathcal{F} = 0.999$, $\tau = 100$ ns: $\varepsilon^* \approx 10^{-27}$ J, consistent with operating near the Landauer floor.

Interface 2: solar-mass Schwarzschild horizon. A $1\,M_\odot$ black hole has horizon area $A = 4\pi r_s^2 = 4\pi(2GM_\odot/c^2)^2 \approx 1.1 \times 10^8\,\mathrm{m}^2$ and Hawking temperature $T_H = \hbar c^3/(8\pi G M_\odot k_B) \approx 6.2 \times 10^{-8}$ K.

- Landauer floor: $\varepsilon^* \geq k_B \cdot T_H \cdot \ln 2 \approx 5.93 \times 10^{-31}$ J per bit.
- Bekenstein cell count: $K_{\mathrm{horizon}} = A/(4\ell_P^2) \approx 1.1 \times 10^{77}$ cells, so the horizon carries $\sim 10^{77}$ bits of admissibility capacity at $\varepsilon^* \geq 6 \times 10^{-31}$ J per bit.
- Total horizon energy budget: $\sim 10^{77} \cdot 10^{-30}\,\mathrm{J} = 10^{47}$ J, comparable to the black hole's rest-mass energy $M_\odot c^2 \approx 1.8 \times 10^{47}$ J. The Landauer-bound estimate saturates the rest-mass-energy budget to within order of magnitude — consistent with Bekenstein's energy-bound derivation and the holographic principle.
- Direct measurement of ε^* at an astrophysical horizon is not currently possible; the protocol is *in-principle* and depends on future detection of Hawking radiation or analogue horizon experiments.

Interface 3: early-universe electroweak epoch. At the electroweak phase transition, $T \approx 100$ GeV $\approx 1.16 \times 10^{15}$ K.

- Landauer floor: $\varepsilon^* \geq k_B \cdot (1.16 \times 10^{15}\,\mathrm{K}) \cdot \ln 2 \approx 1.11 \times 10^{-8}\,\mathrm{J} \approx 70$ GeV per distinction.
- Order-of-magnitude consistency: 70 GeV is within a factor of a few of the electroweak scale $v = 246$ GeV, which is the natural cost of maintaining mass-scale distinctions in the broken phase. The Landauer floor at this regime is comparable to the Higgs VEV.
- This identification is *not* a derivation — APF does not yet fix v from first principles — but it is a sanity check that the order of magnitude of ε^* at the EW epoch lies within an order of magnitude of observed physics.

Summary of calibrated estimates.

Interface	T	Landauer floor ε^*	Observable scale
Dilution fridge (transmon)	20 mK	1.9×10^{-25} J	$\sim 10^{-26}$ J (gate-fidelity inferred)
Schwarzschild horizon ($1\,M_\odot$)	6×10^{-8} K	5.9×10^{-31} J	Hawking radiation (unobserved)
EW epoch ($T \sim 100$ GeV)	1.2×10^{15} K	1.1×10^{-8} J	$v = 246$ GeV (Higgs VEV)

2.3.2 Honest status on absolute ε^* prediction

The three protocols above give numerical *lower bounds* on ε^* (via the Landauer principle) plus *order-of-magnitude estimates* of the observable energy scale. APF does not currently predict the absolute value of ε^* at any specific interface from first principles. Two open directions for future work would change this status:

1. **Landauer saturation.** If APF could prove $\varepsilon^* = k_B T \ln 2$ (exact Landauer saturation) rather than just $\varepsilon^* \geq k_B T \ln 2$ (inequality), then ε^* would be absolutely predicted at every thermal interface. This would require closing a specific information-theoretic argument that identifies the APF minimum-distinction maintenance as the Landauer-erasure inverse.

2. **Error-correcting-threshold saturation.** Alternatively, if APF could identify ε^* with the fault-tolerant-quantum-computation threshold (Aharonov–Ben-Or [13]), then ε^* would inherit the structural value of the threshold in the fault-tolerant model. This requires identifying APF's admissibility-maintenance protocol with a specific error-correcting code.

Neither saturation is currently derived; both are open problems. The status of ε^* in APF is therefore "operationally well-defined and bounded below by standard information-theoretic floors, with absolute value input from interface-specific thermal/error-correction data."

3 Complete Dependency Graph

Scope: The directed acyclic graph of every theorem cited by Paper 8, rooted at Axiom **A1**. Each node is a bank-registered check in the v6.9 codebase; each edge is a "uses" relation made explicit via `coderef` calls in the main paper. This section exists so that a reader wanting to verify a specific claim can trace the dependency to its root without consulting the full codebase.

3.1 Paper-8 theorem dependency table

Table 1 lists the load-bearing theorems of Paper 8 and, for each, (i) the supplement section in which it is proved, (ii) the bank check that machine-verifies it, and (iii) the Papers 1–7 imports it relies on.

Status. The text-only table above is sufficient for the Paper 8 claims. A compact visual DAG of the dozen-or-so load-bearing nodes, rendered with sector-colour edges, would be useful navigation aid but is deferred; no current claim depends on it.

4 Full Two-Tier Construction

Scope: The slot space V_{slot}, the microstate space H_{micro}, the tensor-power embedding between them, and every compatibility / uniqueness / minimality lemma required for the main paper's two-tier

Theorem	Supp §	Bank check (module)	Prior-paper imports
I_1 holographic	5.1	`_identity_I1_holographic` (`unification.py`)	Paper 1 (): **A1**; horizon convention $d_{\text{eff}} = e$
I_2 gauge–cosmological	5.2	`_identity_I2_gauge_cosmological` (`unification.py`)	Paper 2 (): L_{count}, partition data
I_3 thermo–quantum	5.3	`_identity_I3_thermo_quantum` (`unification.py`)	Paper 1 (): Born-rule closure; Paper 5 (): π_Q
I_4 action–thermo	5.4	`_identity_I4_action_thermo` (`unification.py`)	Paper 7 (): π_A, spectral action
T_{ACC} unification	5.5	`check_T_ACC_unification` (`unification.py`)	All of I_1–I_4
Three-level stack	5.7	`check_T_three_level_unification` (`unification_three_levels.py`)	I_1–I_4; f_1, f_2, f_3 inter-level functors
F_{horizon}	6.2	`check_F_horizon_five_conditions` (`subspace_functors.py`)	Paper 6 (): T_{Bek}, horizon quantisation
F_{bridge} (I_2 subspace)	6.4	`check_T_interface_sector_bridge` (`gravity.py`)	Paper 6 (): T12 partition; $T_{\text{horizon_reciprocity}}$
F_{quantum}	6.5	`check_F_quantum_five_conditions` (`subspace_functors.py`)	Paper 5 (): T_{entropy}, max-mixed state
F_{operator}	6.6	`check_F_operator_five_conditions` (`subspace_functors.py`)	Paper 7 (): $L_{\text{spectral_action_internal}}$
Composed functors	6.7	`check_T_subspace_functors_unified` (`subspace_functors.py`)	All four functors
FRE + SM dictionary	5.8	`check_T_FRE_SM_to_entropy_dictionary` (`fractional_reading.py`)	I_2 scalar witness; F_{bridge}
Matter-sector formula	8	`check_L_Lambda_absolute_numerical_formula` (`fractional_reading.py`)	Paper 6 (): $L_{\text{self_exclusion}}$
H_0 algebraic inversion	8	`check_T_Lambda_to_H0_inversion` (`lambda_absolute.py`)	Matter-sector formula; GR critical density
T_{CMB} absolute	12.2	`check_T_T_CMB_absolute_formula` (`thermal_absolute.py`)	$L_{\text{self_exclusion}}$; photon polarisation count
$T_{\text{Lambda_d2}}$ operator	9.4	`check_T_Lambda_d2_operator_derivation` (`lambda_operator_derivation.py`)	Matter-sector formula; ρ_β construction

Table 1: Dependency table for Paper-8 load-bearing theorems. Each row gives the supplement location, the registered verification check, and the prior-paper imports. The full theorem registry lives in Paper 13 (*The Minimal Admissibility Core*); this table is the Paper-8-relevant subgraph.

discussion in §3. Includes the schematic-vs-literal distinction between SM and horizon interfaces.

4.1 Slot space and microstate space

Definition 4.1 (Slot space)**.** For an ACC interface with record (K, d_{eff}), the *slot space* V_{slot} is a finite-dimensional complex vector space of dimension K, with a canonical basis indexed by the admissibility slots of the interface. V_{slot} carries the structural-capacity content of the interface.

Definition 4.2 (Microstate space)**.** The *microstate space* H_{micro} is a finite-dimensional complex Hilbert space of dimension $N = d_{\text{eff}}^K$, with a canonical basis indexed by the admissible microstate configurations of the interface. H_{micro} carries the statistical-count content of the interface.

4.2 The tensor-power embedding

Proposition 4.3 (Tensor-power embedding)**.** *There is a canonical isomorphism* $\Phi : H_{\text{micro}} \xrightarrow{\sim} V_{\text{slot}}^{\otimes K}$*, under which the basis element indexed by the slot-state sequence* $(s_1, \ldots, s_K)$ *maps to the tensor* $e_{s_1} \otimes \cdots \otimes e_{s_K}$*. Under* Φ*, each slot admissibility carries a local* d_{eff}*-dimensional space (the fibre attached to that slot), and* H_{micro} *is the product over all* K *slots.*

Remark 4.4 (Literal vs. schematic embedding)*.* The embedding Φ is *literal* at interfaces whose slots carry a uniform local Hilbert structure, so that each slot supports the same local state space and the tensor-power carrier has direct microscopic meaning. This includes toy quantum-d interfaces, simple horizon-qubit constructions, and finite thermal-bath models of the type used for the operator construction in Supplement §9.

At the Standard-Model interface, by contrast, $d_{\text{eff}} = 102$ is an *effective* degeneracy extracted from the self-exclusion count rather than a literal per-slot local Hilbert dimension. In that regime, Φ is used *schematically*: it is a bookkeeping carrier for the three-level correspondence, not a claim that the Standard-Model interface literally factorises into K_{SM} identical 102-state microscopic slots.

This distinction matters. In the present supplement, the Standard-Model interface uses only those consequences of the two-tier construction that depend on slot count, effective degeneracy, and subspace bookkeeping. No argument in Paper 8 requires a literal microscopic factorisation into identical per-slot Hilbert factors at the SM interface. Literal tensor-power status is used only at regime-appropriate interfaces where the local Hilbert structure is genuinely uniform.

Accordingly, the role of Φ is twofold:

(i) **literal carrier** at uniform-slot interfaces, where tensor-power statements have direct microscopic meaning;

(ii) **schematic carrier** at the SM interface, where the same formal container tracks the three-level correspondence without asserting literal microscopic slot factorisation.

4.3 Sufficient conditions for literal tensor-power factorisation

The literal-vs-schematic distinction of Remark 4.4 is sharpened here into a proposition with sufficient conditions. The statement is phrased as a *sufficient* rather than necessary-and-sufficient characterisation: the conditions below are easy to check in practice, and a full necessary-and-sufficient theorem (requiring a categorical treatment of admissibility constraints) is parked as future work.

Proposition 4.5 (Sufficient conditions for literal tensor-power factorisation)**.** *Let* (K, d_{eff}) *be an ACC interface record with microstate Hilbert space* H_{micro}*, slot space* V_{slot}*, and tensor-power embedding* Φ *of Proposition 4.3. If the interface satisfies the three sufficient conditions*

(SC1) ***Uniform per-slot Hilbert structure.*** *Each admissibility slot* $i \in \{1, \ldots, K\}$ *carries a local Hilbert space* $\mathcal{H}_i$ *with* $\dim \mathcal{H}_i = d_{\text{eff}}$*, and the slots admit a canonical isomorphism* $\mathcal{H}_i \cong \mathcal{H}_j$ *for all* i, j*.*

(SC2) ***Independence of per-slot admissibility.*** *The admissible microstate set* $\mathcal{A}$ *factorises as a Cartesian product,* $\mathcal{A} = \prod_{i=1}^{K} \mathcal{A}_i$ *with* $|\mathcal{A}_i| = d_{\text{eff}}$ *for every* i *(no cross-slot admissibility constraints).*

(SC3) ***Integer-dimensional ambient.*** $d_{\text{eff}} \in \mathbb{Z}_{>0}$.

then the embedding Φ is literal*: H_{micro} is canonically isomorphic, as an integer-dimensional Hilbert space, to the explicit tensor product*

$$(\mathbb{C}^{d_{\text{eff}}})^{\otimes K} \;\cong\; \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_K,$$

with the admissible microstate basis in canonical bijection with the per-slot product basis of $\mathcal{A}$.

Proof. Under (SC1), each $\mathcal{H}_i$ is isomorphic to $\mathbb{C}^{d_{\text{eff}}}$, so the tensor product $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_K$ is isomorphic to $(\mathbb{C}^{d_{\text{eff}}})^{\otimes K}$ as a Hilbert space of dimension d_{eff}^K. Under (SC3), this dimension is a positive integer, so the tensor product is a bona fide finite-dimensional Hilbert space in the usual sense. Under (SC2), the admissible basis of H_{micro} is in canonical bijection with $\mathcal{A} = \prod_i \mathcal{A}_i$, which is the tensor-product basis of the right-hand side. Φ identifies the two bases; the underlying vector-space isomorphism extends to a Hilbert-space isomorphism. □ □

Examples: literal.

Example 4.6 (Quantum interface, $K = 1$)**.** At the canonical quantum interface $(K, d_{\text{eff}}) = (1, d)$, the tensor product collapses to a single factor: $(\mathbb{C}^d)^{\otimes 1} = \mathbb{C}^d$. Conditions (SC1)–(SC3) hold trivially. Φ is the identity map.

Example 4.7 (Horizon Convention B)**.** At the horizon under Convention B, $(K_{\text{bit}}, d_{\text{eff}}) = (K_{\text{bit}}, 2)$ for integer $K_{\text{bit}} \geq 1$. Each bit-cell carries a qubit $\mathcal{H}_i = \mathbb{C}^2$; the per-cell admissibility $|\mathcal{A}_i| = 2$ is the canonical two-state basis; $d_{\text{eff}} = 2$ is integer. (SC1)–(SC3) all hold; $(\mathbb{C}^2)^{\otimes K_{\text{bit}}}$ is literal. This is the Construction 3 witness invoked in Theorem 6.3, realised as an explicit numpy tensor product at $K_{\text{bit}} \in \{2, 3, 4, 5\}$.

Example 4.8 (Finite thermal bath)**.** The operator construction of §9 sets (K, d_{eff}) to arbitrary finite integers with C_{vacuum} per-slot vacuum states and $(d_{\text{eff}} - C_{\text{vacuum}})$ excited states per slot, with no cross-slot coupling in H_{local}. The per-slot local Hilbert space is uniform $\mathbb{C}^{d_{\text{eff}}}$; admissibility factorises; d_{eff} is integer. Φ is literal; the tensor product $(\mathbb{C}^{d_{\text{eff}}})^{\otimes K}$ is realised in code as an explicit numpy array.

Examples: schematic only.

Example 4.9 (Standard-Model interface)**.** At the SM interface, $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$. The 61 admissibility slots correspond to the Standard-Model ledger channels (12 gauge + 4 Higgs + 45 fermion); the per-channel structural content is not a uniform 102-state local Hilbert factor but rather the external regime-dependent admissibility content of each channel type. (SC1) fails: slots are structurally heterogeneous (a lepton slot and a gluon slot do not admit a canonical Hilbert-space isomorphism). Consequently Φ is schematic, not literal: it is a bookkeeping carrier for the three-level correspondence rather than a claim about microscopic tensor-product factorisation. (Construction 3 of Theorem 6.3 bypasses this by using Convention B at $K_{\text{bit}} = 42$, which does satisfy (SC1)–(SC3).)

Example 4.10 (Horizon Convention A at $d_{\text{eff}} = e$)**.** Under Convention A, $d_{\text{eff}} = e$. (SC3) fails: $e \notin \mathbb{Z}_{>0}$, so no integer-dimensional Hilbert space has dimension e and the tensor product $(\mathbb{C}^e)^{\otimes K}$ has no literal realisation. Φ is schematic under Convention A. The dual-convention equivalence of §13.4 shows that the same ACC data admit a literal realisation under Convention B, at $(K_{\text{bit}}, d_{\text{eff}}) = (K_{\text{Planck}}/\ln 2, 2)$.

Remark on necessity. Conditions (SC1)–(SC3) are sufficient but not necessary. Weaker conditions — e.g., per-slot Hilbert spaces that are not mutually isomorphic but still carry compatible cross-slot admissibility structure — can also admit a literal realisation via a non-uniform tensor product $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_K$. The present supplement uses Proposition 4.5 only to certify literality at uniform-slot interfaces; the fully general necessary-and-sufficient condition is deferred.

Status: deferred extensions. Three refinements are possible but not required for the current Paper 8 claims: (i) a full necessary-and-sufficient compatibility theorem (weaker than (SC1)–(SC3) above but requiring a categorical treatment of admissibility constraints); (ii) a minimality argument showing the two-tier structure is the smallest carrier that admits all four I_k simultaneously; (iii) a quantitative comparison between the schematic-SM and literal-Convention-B realisations of the same joint-point $K = 42$ subspace. The two-tier structure as presented — slot space, microstate space, the factorisation map Φ, plus the sufficient conditions of Proposition 4.5 — is sufficient for the claims of the following sections.

4.4 Two-tier consistency forcing: the nontrivial constraint

Centerpiece pointer. *Proposition 4.11 (two-tier forcing) together with Theorem 4.12 (single-scale impossibility) together supply the two-tier architectural content of the I_2 centerpiece (§D). The integrated reading is in §D.4.*

Proposition 4.5 establishes when the two-tier carrier is *literal* (a genuine tensor-power structure). The separate question — also raised by external review — is: *what does the two-tier construction buy, structurally, at an interface where literal factorisation may fail?* This subsection answers that by exhibiting a non-trivial constraint that the two-tier architecture forces and that a single-scale carrier leaves unconstrained.

Proposition 4.11 (Two-tier consistency forcing)**.** *Let Γ be an ACC interface with two-tier record (K, d_{eff}), slot space V_{slot} of dimension K, microstate space H_{micro} of dimension N, and residual partition $\{K_X\}_{X\in\mathcal{S}}$ with $\sum_X K_X = K$. If the two-tier architecture is to support simultaneously*

(1) *an integer-level reading of V_{slot} via π_F (gauge structural count),*

(2) *an exponential-level reading of H_{micro} via π_Q (microstate count), and*

(3) *a residual-partition reading via π_C (sector fractions K_X/K),*

then (K, d_{eff}, N) are related by

$$\boxed{\ln N \;=\; K \ln d_{\text{eff}}, \qquad N = d_{\text{eff}}^K.}$$

In particular, the exponential scale N is not independent of the integer scale K; they are related by a single degeneracy parameter d_{eff} that is itself constrained by the self-exclusion identity $d_{\text{eff}} = (K-1) + C_{\text{vacuum}}$ at interfaces where it applies.

Proof. Readings (1) and (2) must agree on the ACC record:

- π_F reads K directly: the number of independent slots in V_{slot};
- π_Q reads $N = \dim H_{\text{micro}}$: the number of admissible microstate configurations.

For these two readings to be readings of the *same* record rather than of independent posited structures, the two-tier carrier must supply a relationship. The canonical embedding $\Phi : H_{\text{micro}} \xrightarrow{\sim} V_{\text{slot}}^{\otimes K}$ of Proposition 4.3 supplies it: under Φ, each slot carries a d_{eff}-dimensional fibre and $N = d_{\text{eff}}^K$ follows by tensor-product dimension. This holds whether Φ is literal (SC1–SC3 of Proposition 4.5) or schematic (see Remark 4.4): the dimension-count identity $\dim H_{\text{micro}} = (\dim V_{\text{slot}}^{(i)})^K$ follows from the formal structure of Φ regardless of whether the underlying fibre spaces are literally isomorphic.

Reading (3) $\pi_C(\Gamma) = \{K_X/K\}_X$ requires that V_{slot} carries an indexed partition, and the common denominator K matches the one read by π_F in (1). This is the I_2 consistency statement (Theorem 5.3); it is only meaningful on the two-tier architecture because a single-scale carrier has no distinct "slot" notion to partition.

Composing (1), (2), and (3), the integer count K, the microstate count N, and the residual partition are not independent data — they are tied by the dimension-count identity and by $K = \sum_X K_X$. This is the two-tier consistency forcing. □ □

What a single-scale carrier would miss. If one posited only a single-scale carrier (either a slot space without a microstate extension, or a microstate space without a slot structure), the integer-count reading by π_F and the exponential-count reading by π_Q would be independent posits. Nothing in the single-scale structure would force $\ln N = K \ln d_{\text{eff}}$. At the SM interface, that means the $(K_{\text{SM}}, N_{\text{SM}}) = (61, 102^{61})$ pair would be two separate integer commitments: 61 from Paper 2's L_{count} and 102^{61} from an independent microstate posit. The two-tier construction replaces the second posit with a derived quantity: $N_{\text{SM}} = d_{\text{eff}}^{K_{\text{SM}}}$ with $d_{\text{eff}} = 102$ itself derived from $L_{\text{self_exclusion}}$.

The non-trivial content. The reader's Q5 asked for a nontrivial constraint that depends critically on the two-tier architecture. Proposition 4.11 supplies it: the two-tier construction reduces three independent integer commitments (K, N, the partition) to two (K and d_{eff}), with N and the residual-partition denominator both derived from those two. At the SM interface this is the difference between (a) committing to 61, 16, 42, 3, and 102^{61} as five separate integers, and (b) committing to just $K_{\text{SM}} = 61$ and $d_{\text{eff}}^{\text{SM}} = 102$ (which are themselves forced by Theorem A.4) and deriving everything else. The reduction from five commitments to two (further reduced to zero under Theorem A.4) is the structural value of the two-tier architecture.

Consequence for the literal/schematic distinction. The forcing above holds at *every* two-tier ACC interface, including interfaces where Φ is schematic (e.g., the SM interface where (SC1) of Proposition 4.5 fails). The dimension-count identity $N = d_{\text{eff}}^K$ is formal; it does not require literal tensor-product microstructure. Literal factorisation adds the additional content that H_{micro} is genuinely isomorphic to a uniform tensor power, but the dimension-count-and-partition forcing is available independently. This is why Paper 8 can use the two-tier architecture at the SM interface (schematic) and still derive the $d_{\text{eff}}^{\text{SM}} = 102$ forcing.

Status. Proved [P]. The proposition rests on the formal structure of the tensor-power embedding (Proposition 4.3) and the regime-projection definitions (§2.2); no additional hypotheses are needed.

Single-scale impossibility: the two-tier no-go theorem

Proposition 4.11 establishes that the two-tier architecture *supplies* the constraint $\ln N = K \ln d_{\text{eff}}$. External review (red-team on v2.2) asked the sharper question: *does a single-scale carrier even admit all three regime readings* (π_F, π_Q, π_C) *simultaneously, or is the two-tier architecture structurally mandatory?* The answer below is a structural exclusion: non-trivial single-scale carriers cannot support the three readings jointly.

Theorem 4.12 (Single-scale impossibility)**.** *Let* $\mathcal{C}$ *be a* single-scale carrier*: a finite-dimensional complex vector space* V *of dimension* d*, equipped with*

- *a regime projection* $\pi_F^{\mathcal{C}} : \mathcal{C} \to \mathbb{Z}_{>0}$ *reading an integer structural count, with* $\pi_F^{\mathcal{C}}(\mathcal{C}) = d$*;*
- *a regime projection* $\pi_Q^{\mathcal{C}} : \mathcal{C} \to \mathbb{Z}_{>0}$ *reading an exponential microstate count, with* $\pi_Q^{\mathcal{C}}(\mathcal{C}) = N$ *for some* $N \geq 1$*;*
- *a regime projection* $\pi_C^{\mathcal{C}} : \mathcal{C} \to \{K_X^{\mathcal{C}}/D_{\mathcal{C}}\}_X$ *reading a residual partition with common denominator* $D_{\mathcal{C}}$ *and* $\sum_X K_X^{\mathcal{C}} = D_{\mathcal{C}}$*.*

Suppose further that $(\pi_F^{\mathcal{C}}, \pi_Q^{\mathcal{C}}, \pi_C^{\mathcal{C}})$ *are* mutually consistent *in the sense that the integer* $\pi_F^{\mathcal{C}}(\mathcal{C}) = d$ *coincides with the residual-partition denominator* $D_{\mathcal{C}}$ *(as required by* I_2 *for the common ACC interpretation). Then at least one of the following holds:*

(i) Trivial π_Q*:* $N = d$ *(the microstate count coincides with the integer structural count, so* π_Q *is redundant with* π_F *and there is no exponential scale).*

(ii) Partition-denominator ambiguity*: the common denominator of* $\pi_C^{\mathcal{C}}$ *does not coincide with* d*; equivalently, two inequivalent residual-partition assignments on* $\mathcal{C}$ *give the same* π_C *output, violating the* I_2 *consistency assumption.*

(iii) π_Q fails to be a well-defined projection*: there is no linear / combinatorial rule on* $\mathcal{C}$ *that delivers an integer* N *depending on the single structural count* d *without importing external data (so* $\pi_Q^{\mathcal{C}}$ *is not a regime projection in the sense of Definition 2.6).*

Equivalently: no single-scale carrier $\mathcal{C}$ *simultaneously supports non-trivial* π_F*, non-trivial* π_Q *(with* $N \neq d$*), and consistent* π_C *with the same* I_2*-compatible common denominator. The two-tier architecture* $(V_{\text{slot}}, H_{\text{micro}})$ *with* $N = d_{\text{eff}}^K$ *is therefore* structurally mandatory *for any ACC interface where* I_2 *applies non-trivially.*

Proof. We proceed by case analysis on the relationship between N and d.

Case A: $N = d$*.* Then $\pi_Q^{\mathcal{C}}$ and $\pi_F^{\mathcal{C}}$ return the same integer. By Definition 2.6, π_F reads the structural capacity K and π_Q reads the microstate count $N = d_{\text{eff}}^K$. For $N = d$ with $d = K$ to be consistent, we need $d_{\text{eff}}^K = K$. In $\mathbb{Z}_{>0}$, this has solutions only at $(K, d_{\text{eff}}) = (1, K)$ (any K) or $(K, d_{\text{eff}}) = (2, \sqrt{2})$ (no integer solution), or trivially $K = 1$. The non-degenerate case $K \geq 2$, $d_{\text{eff}} \geq 2$ is excluded. Hence π_Q collapsing to π_F is the $K = 1$ case, in which there is no exponential scale and no two-tier structure is claimed — this is condition (i) of the theorem.

Case B: $N \neq d$*, with* $\pi_Q^{\mathcal{C}}$ *posited directly on* $\mathcal{C}$*.* The single-scale carrier has one scale (dimension d); the claim that $\pi_Q^{\mathcal{C}}(\mathcal{C}) = N$ for $N \neq d$ requires an additional structure on $\mathcal{C}$ beyond its vector-space dimension. Two sub-cases:

B.1: If $\pi_Q^{\mathcal{C}}$ is posited as independent data (e.g., "$N = 10^{61}$ is declared externally"), then $\pi_Q^{\mathcal{C}}$ is not a regime projection reading the ACC record $\mathcal{C}$; it is a new datum. This violates the minimal-data

principle of the ACC definition (Definition 2.6): the ACC record carries exactly (K, d_{eff}), and all regime projections are readings of this record. Condition (iii) of the theorem applies.

B.2: If $\pi_Q^{\mathcal{C}}$ is posited via some derived formula $N = f(d)$ for a function $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$, we ask: is f well-defined as a regime projection? For f to be regime-appropriate, it must commute with ACC-preserving interface morphisms (Theorem 7.1). The only such functions on $\mathbb{Z}_{>0}$ are $f(d) = d^k$ for fixed k (power functions commute with dimension-preserving automorphisms). But $N = d^k$ for $k \geq 2$ *is* a two-scale structure: the exponent k is a second scale on top of d. Identifying $k = K$ and $d = d_{\text{eff}}$ recovers the two-tier architecture $N = d_{\text{eff}}^K$. Any attempt to support non-trivial π_Q on a single-scale carrier therefore reintroduces a second scale, collapsing the "single-scale" premise. Condition (iii) of the theorem applies.

Case C: $N \neq d$, with $\pi_C^{\mathcal{C}}$ imposing a common denominator $D_{\mathcal{C}} \neq d$. If the residual-partition denominator $D_{\mathcal{C}}$ differs from d, then I_2's consistency requirement $\pi_F^{\mathcal{C}}(\mathcal{C}) = \text{denom}(\pi_C^{\mathcal{C}}(\mathcal{C}))$ fails, because $\pi_F^{\mathcal{C}}$ reads d while $\pi_C^{\mathcal{C}}$ reads $D_{\mathcal{C}}$. Condition (ii) of the theorem applies.

Conversely, if $D_{\mathcal{C}} = d$, the residual partition $\{K_X^{\mathcal{C}}\}_X$ must satisfy $\sum_X K_X^{\mathcal{C}} = d$; but on a single-scale carrier $\mathcal{C}$ of dimension d, a residual partition is just a decomposition $\mathcal{C} = \bigoplus_X V_X$ with $\dim V_X = K_X^{\mathcal{C}}$. This is not the issue; the issue is the microstate count N, already addressed in Cases A and B. The residual-partition reading $\pi_C^{\mathcal{C}}$ itself is well-defined on a single-scale carrier, *but only if we accept that π_Q is either trivial (Case A) or requires a two-scale structure (Case B).*

Synthesis. In every case, at least one of conditions (i)–(iii) holds. The non-degenerate regime $N \neq d$ with a well-defined π_Q (not an external posit, not collapsing to π_F) necessarily introduces a second scale $k \geq 2$ via $N = d_{\text{eff}}^k$, which is the two-tier architecture. Hence single-scale carriers *cannot* simultaneously support all three non-trivial regime readings in a way compatible with the I_2 consistency constraint. □ □

Consequence: the two-tier architecture is structurally mandatory. Theorem 4.12 strengthens Proposition 4.11 from "the two-tier architecture supplies the constraint $N = d_{\text{eff}}^K$" to "no single-scale alternative exists." The $(V_{\text{slot}}, H_{\text{micro}})$ two-tier carrier is not a modelling choice to be compared with alternatives; it is the unique ACC architecture compatible with all three of π_F, π_Q, and π_C non-trivially, up to the trivial $K = 1$ collapse.

Relation to the reviewer's Q5. The second reviewer asked for a nontrivial constraint that depends critically on the two-tier architecture. Theorem 4.12 is that constraint: the two-tier architecture is the *unique* viable architecture for ACC interfaces with non-trivial π_Q, under the I_2 consistency premise. This is a no-go result of the same structural type as (in a different context) Coleman–Mandula or Haag's theorem — an exclusion of alternative architectures rather than a derivation of a specific value.

Status. Proved [P] by case analysis. Uses only Definition 2.6 (ACC record has two scalars), Theorem 5.3 (I_2 consistency = common denominator), and a minimality principle (regime projections read the ACC record rather than import external data). No additional hypotheses required.

5 Proofs and Verification Statements for $I_1 - I_4$

Scope: Technical proofs and verification statements for the four consistency identities and the composed ACC-consistency theorem. The section distinguishes three proof strengths:

(i) **definitional closures**, where the identity follows directly from the projection definitions plus stated conventions;

(ii) **derived structural closures**, where non-trivial bridge or regime structure is used;

(iii) **import-supported closures**, where a standard or previously established result is being recast in the present ledger language.

The point of this section is not to give all four identities equal rhetorical weight, but to record exactly what kind of closure each one has.

Status taxonomy.

[$\mathrm{P}_{\mathrm{def}}$] definitional closure;

[$\mathrm{P}_{\mathrm{std}}$] standard mathematical closure;

[$\mathrm{P}_{\mathrm{bridge}}$] non-trivial bridge closure;

[$\mathrm{P}_{\mathrm{comp}}$] composition theorem (closure by combining upstream statements);

[I] imported prior result;

[C] conceptual / deferred categorical packaging.

The plain [P] tag used elsewhere in the supplement is reserved for closures that admit more than one of these readings or whose refinement into the taxonomy has not yet been made explicit.

5.1 Verification of I_1 (Holographic)

Proposition 5.1 (I_1, horizon-convention identity)**.** *At a horizon interface with $d_{\mathrm{eff}} = e$ and $K_{\mathrm{horizon}} = A/(4\ell_P^2)$ (the Bekenstein cell count of Definition 2.7),*

$$\pi_G(\mathrm{ACC}_{\mathrm{horizon}}) \;=\; \pi_T(\mathrm{ACC}_{\mathrm{horizon}}) \;=\; \ln \pi_Q(\mathrm{ACC}_{\mathrm{horizon}}) \;=\; \frac{A}{4\ell_P^2} \;=\; S_{\mathrm{BH}}.$$

That is, all three regime readings of the horizon ACC record collapse to the standard Bekenstein–Hawking entropy (in nats).

Assumptions.

(H1) *Horizon-interface cell count.* The interface is a horizon with area A, so the ACC record's structural capacity is the Bekenstein cell count $K_{\mathrm{horizon}} := A/(4\ell_P^2)$ (Definition 2.7). This is a unit-tile count, not the bare area.

(H2) *Per-slot degeneracy convention.* At the horizon, $d_{\mathrm{eff}} = e$ by the Bekenstein log-count convention (nats). (Equivalent choices — Convention B with $d_{\mathrm{eff}} = 2$, bits — rescale the scalar-level entropy only through a multiplicative $\ln d_{\mathrm{eff}}$ factor; see §13.4.)

(H3) *π_G reads the horizon's ACC-scalar directly.* This is part of the definition of the gravitational projection at a horizon interface: π_G evaluates to the horizon's log-count content, which under (H1)+(H2) equals S_{BH}.

Proof. By (H2), $\pi_T(\mathrm{ACC}_{\mathrm{horizon}}) = K_{\mathrm{horizon}} \ln d_{\mathrm{eff}} = K_{\mathrm{horizon}} \ln e = K_{\mathrm{horizon}}$. By (H1), $K_{\mathrm{horizon}} = A/(4\ell_P^2)$; therefore $\pi_T = A/(4\ell_P^2)$.

By (H3), $\pi_G(\mathrm{ACC}_{\mathrm{horizon}})$ reads the horizon's ACC-scalar directly, which is $K_{\mathrm{horizon}} \ln d_{\mathrm{eff}} = A/(4\ell_P^2)$ under (H1)+(H2). Therefore $\pi_G = A/(4\ell_P^2)$.

The quantum reading: $\pi_Q(\mathrm{ACC}_{\mathrm{horizon}}) = d_{\mathrm{eff}}^{K_{\mathrm{horizon}}} = e^{K_{\mathrm{horizon}}} = e^{A/(4\ell_P^2)}$, and so $\ln \pi_Q = A/(4\ell_P^2)$.

All three projections evaluate to $A/(4\ell_P^2)$, which is the Bekenstein–Hawking entropy S_{BH} in nats. □

Proposition 5.2 (Equivalence to the standard Bekenstein–Hawking formula)**.** *Proposition 5.1 is equivalent to the standard physical result*

$$S_{\mathrm{BH}} = \frac{k_B\, A}{4\ell_P^2}$$

(with $k_B = 1$ in natural units). No prefactor "1/4" has been mutated, absorbed into a convention, or otherwise modified: the 1/4 is the cell-tile size $1/(4\ell_P^2)$ in Definition 2.7, which is part of the ACC record's definition, not part of any rescaling applied to the output of the three projections.

Proof. Direct substitution: $K_{\mathrm{horizon}} = A/(4\ell_P^2)$ and $\mathrm{ACC}_{\mathrm{horizon}}$-scalar $= K_{\mathrm{horizon}} \ln e = K_{\mathrm{horizon}}$, which in natural units ($k_B = 1$) reads $A/(4\ell_P^2) = S_{\mathrm{BH}}$. □

Status. Proved [$\mathrm{P}_{\mathrm{def}}$] under the stated horizon conventions, with $S_{\mathrm{BH}} = A/(4\ell_P^2)$ recovered exactly. This is a definitional closure, not a deep structural forcing theorem: once the cell-count identification $K_{\mathrm{horizon}} = A/(4\ell_P^2)$ and $d_{\mathrm{eff}} = e$ are fixed, all three readings collapse to the same scalar by direct evaluation. The content of the proposition is that the three distinct projections (gravitational, thermodynamic, quantum) all read the same underlying ACC architecture consistently; the physical Bekenstein–Hawking prefactor 1/4 is built into the ACC record at Definition 2.7, not mutated by the projections.

On earlier "$K = A$" shorthand. Earlier APF documents (e.g., the v2.0 supplement changelog, some Paper 6 in-line notes) sometimes wrote "$K = A$" as shorthand in natural units where the $4\ell_P^2$ tile prefactor was left implicit. That shorthand is retired in favour of Definition 2.7. The canonical identification is $K_{\mathrm{horizon}} = A/(4\ell_P^2)$; the bank check `_identity_I1_holographic` already uses cell-count inputs in its numerical validation.
Code anchor: `_identity_I1_holographic` (`apf/unification.py`)

5.2 Integer-level closure for I_2 (corollary of Theorem 1.7)

Status. The load-bearing content of I_2 is Theorem 1.7 of §1 (Formal Kernel), which states the existence and uniqueness of the 42-dimensional G_{SM}-invariant subspace $V_\Lambda \subset V_{61}$ and derives the cosmological fraction $\Omega_\Lambda = 42/61$ as a consequence. This subsection states only the *integer-level corollary*: the common-denominator coincidence between the gauge-integer reading π_F and the cosmological-fraction reading π_C on the residual partition. It is a definitional closure given Theorem 1.7; the structural weight sits entirely in that theorem, not here.

Theorem 5.3 (I_2, gauge–cosmological consistency theorem)**.** *At an interface whose ACC record carries a residual partition $K = \sum_X K_X$, the integer structural capacity read by π_F equals the common denominator of the sector fractions read by π_C, and the residual numerators exhaust that integer:*

$$\pi_F(\mathrm{ACC}) \;=\; \mathrm{denom}\big(\pi_C(\mathrm{ACC})\big) \;=\; K, \qquad \sum_X K_X \;=\; K.$$

Assumptions.

(P1) *Residual-partition data.* The ACC record carries the optional residual-partition augmentation of main paper §2.1: a family of non-negative integers $\{K_X\}_{X\in\mathcal{S}}$ indexed by a finite sector set $\mathcal{S}$.

(P2) *Partition closure.* The residual partition is valid, i.e. $\sum_{X\in\mathcal{S}} K_X = K$. (A partition must exhaust the capacity; otherwise π_C maps to fractions that fail to sum to unity, violating the cosmological-sector interpretation.)

(P3) *Projection definitions.* π_F reads K directly from the record; π_C is the map $\{K_X\}_X \mapsto \{K_X/K\}_X$ (sector-indexed fractions with common denominator K).

Proof. The common denominator of the fractions $\{K_X/K\}$ in $\pi_C(\mathrm{ACC})$ is K by (P3). $\pi_F(\mathrm{ACC}) = K$ by (P3). Hence the two integers coincide:

$$\pi_F(\mathrm{ACC}) = K = \mathrm{denom}\big(\pi_C(\mathrm{ACC})\big).$$

The second claim, $\sum_X K_X = K$, is precisely (P2). □

Status. Proved [$\mathrm{P_{def}}$] at the integer level (this is the weakest link in the I_2 chain and is definitional by construction — the theorem above states that two projections of the same ACC record agree on a common denominator, which is immediate once the projection definitions are fixed); proved [P] at the scalar level once Fractional Reading Equivalence is invoked (so that the sector entropy ratios $S(V_X)/S_{\mathrm{SM}}$ equal the slot-count ratios K_X/K_{SM}); proved [$\mathrm{P_{bridge}}$] at the subspace level at the Standard-Model interface via the interface-sector bridge theorem $T_{\mathrm{interface_sector_bridge}}$ (Paper 6, `apf/gravity.py`), strengthened in v2.3 by Theorem 6.6 (V_Λ gauge-uniqueness). Among the four identities, I_2 is the only one whose three-level closure is non-trivial in the present supplement: cosmological partition data, horizon-reciprocity sector data, and the two-tier carrier all converge on the same 42-dimensional stratum V_Λ, and that convergence is *forced*, not coincidental, by the gauge-uniqueness result.

At the SM interface, I_2 gives $K_{\mathrm{SM}} = 3 + 16 + 42 = 61$ at the integer level and $\Omega_b + \Omega_c + \Omega_\Lambda = 3/61 + 16/61 + 42/61 = 1$ at the scalar level. The substantive structural content is the subspace-level uniqueness plus the derivation chain internal to APF (Appendix A, Theorem A.4) plus the two-tier structural necessity (Theorem 4.12) plus the brittle cosmological consequences (Theorems 11.1 and §11.4); these are gathered in the integrated centerpiece reading §D.

Code anchor: `_identity_I2_gauge_cosmological` (`apf/unification.py`)

π_C physicality conditions: Friedmann compatibility

External review (second pass) asked for the precise domain restrictions on π_C and the consistency conditions (GR constraints, conservation laws) that ensure physicality of ledger-derived partitions. The following proposition collects them.

Proposition 5.4 (π_C physicality conditions)**.** *The cosmological projection $\pi_C : \mathrm{ACC} \mapsto \{K_X/K\}_X$ is physically well-defined on an ACC interface Γ if and only if:*

(PC1) ***Residual-partition augmentation.*** *The ACC record carries a residual partition $\{K_X\}_{X\in\mathcal{S}}$ with $K_X \in \mathbb{Z}_{\geq 0}$ and $\sum_X K_X = K$ (partition closure).*

(PC2) ***Closure under unit sum.*** *The cosmological fractions $\Omega_X := K_X/K$ satisfy $\sum_X \Omega_X = 1$ by (PC1). This is the cosmological-sector interpretation requirement: the fractions partition a unit cosmological budget.*

(PC3) ***Friedmann compatibility at cosmological interfaces.*** *At cosmological interfaces with a specified epoch τ, the fractions must be compatible with the Friedmann equations and stress-energy conservation:*

$$\sum_X \Omega_X(\tau) \;=\; 1 \quad \textit{for all } \tau, \qquad \rho_X(\tau) \;\propto\; a(\tau)^{-3(1+w_X)},$$

where $a(\tau)$ is the scale factor at epoch τ and w_X is the equation-of-state parameter of sector X (matter $w = 0$, radiation $w = 1/3$, cosmological constant $w = -1$, etc.).

If (PC1)–(PC3) hold, $\pi_C(\Gamma)$ is a physically admissible cosmological-partition reading. If any one fails, π_C is undefined or unphysical on that interface.

Proof. Conditions (PC1)–(PC2) are the content of I_2's definitional closure (Theorem 5.3, assumptions P2 + P3). Condition (PC3) is the content of the Friedmann equations and the stress-energy conservation law $\nabla_\mu T^{\mu\nu} = 0$ applied to each fluid component separately: the sum condition preserves at every epoch if and only if the equation-of-state parameters are correctly assigned. □ □

Status of the APF prediction. APF predicts the residual partition $(3, 16, 42)/61$ at the *present epoch*, on the Standard-Model-coupled cosmological interface. It does *not* currently specify the integer partition at arbitrary past or future epochs. The time-evolution of the partition under Friedmann scaling follows standard GR $\rho_X \propto a^{-3(1+w_X)}$, which gives the scale-factor dependence of each $\Omega_X(\tau)$; APF's content is the *structural fixed point* at the present epoch, where the integer rational 42/61 is asserted.

Implication. Under (PC1)–(PC3), the APF prediction $\Omega_\Lambda^{\mathrm{APF}} = 42/61 \approx 0.6885$ is a physically well-defined cosmological fraction: it partitions a unit cosmological budget, it is consistent with Friedmann + stress-energy conservation, and it equals the present-epoch dark-energy fraction measured by Planck 2018 to within 0.52σ (§11.4). The physicality conditions are not vacuous: APF's ledger structure automatically respects them; any alternative residual-partition assignment that fails (PC1)–(PC3) would be rejected as unphysical.

Status. Proved [$\mathrm{P_{def}}$]. The conditions (PC1)–(PC3) are the standard cosmological-interpretation requirements; APF's residual-partition structure satisfies them by design (the ledger's residual-partition assumption P2 of Theorem 5.3 is exactly (PC1)–(PC2); (PC3) is the standard GR compatibility condition, inherited by APF because its cosmological-sector readings are embedded in standard cosmological spacetimes).

5.3 Verification of I_3 (Thermo–Quantum)

Proposition 5.5 (I_3, thermo–quantum logarithmic correspondence)**.** *At any interface whose ACC record admits both π_T and π_Q,*

$$\pi_T(\mathrm{ACC}) \;=\; \ln \pi_Q(\mathrm{ACC}).$$

Equivalently, for the maximally mixed admissible state on the finite microstate Hilbert space $\mathcal{H}$,

$$S_{\mathrm{vN}}\big(\tfrac{1}{N}\mathbf{1}_{\mathcal{H}}\big) \;=\; \ln N \;=\; \text{ACC-}\textit{scalar}, \qquad N = d_{\mathrm{eff}}^{K}.$$

Assumptions.

(Q1) *ACC record admits π_T and π_Q.* The projections are well-defined on (ρ, Γ): $\pi_T(\mathrm{ACC}) = K \ln d_{\mathrm{eff}}$ and $\pi_Q(\mathrm{ACC}) = d_{\mathrm{eff}}^{K}$.

(Q2) *Finite-dimensional microstate space.* The admissible microstate count $N = d_{\mathrm{eff}}^{K}$ is finite; equivalently, the interface's microstate Hilbert space $\mathcal{H}$ has finite dimension N.

(Q3) *Maximally mixed state.* The state $\rho_{\max} := N^{-1}\mathbf{1}_{\mathcal{H}}$ is well-defined.

Proof. By (Q1):

$$\pi_T(\mathrm{ACC}) = K \ln d_{\mathrm{eff}}, \qquad \ln \pi_Q(\mathrm{ACC}) = \ln d_{\mathrm{eff}}^{K} = K \ln d_{\mathrm{eff}}.$$

The two scalar expressions coincide.

For the von Neumann entropy identification: by (Q2) and (Q3), $\rho_{\max}$ is a positive-definite operator on $\mathcal{H}$ with N equal eigenvalues $1/N$. Therefore

$$S_{\mathrm{vN}}(\rho_{\max}) \;=\; -\mathrm{tr}\big(\rho_{\max} \ln \rho_{\max}\big) \;=\; -\sum_{i=1}^{N} \tfrac{1}{N} \ln \tfrac{1}{N} \;=\; N \cdot \tfrac{1}{N} \cdot \ln N \;=\; \ln N \;=\; \pi_T(\mathrm{ACC}).$$

□

Status. Proved [$\mathrm{P_{def}}$] at the scalar level and [$\mathrm{P_{std}}$] for the entropy evaluation on the maximally mixed state. The identity itself is definitional once π_T and π_Q are fixed; the von Neumann entropy sentence is a standard finite-dimensional calculation. Subspace-level closure is deferred to the witness construction F_{quantum} (Supplement §6).
Code anchor: `_identity_I3_thermo_quantum` (`apf/unification.py`)

5.4 Verification of I_4 (Action–Thermo)

Proposition 5.6 (I_4, high-temperature action–thermo limit)**.** *At any interface whose ACC record admits π_A,*

$$\lim_{\beta\to 0^+} \ln Z(\beta) \;=\; \ln N \;=\; \text{ACC-}\textit{scalar}, \qquad N = d_{\text{eff}}^K.$$

That is, the high-temperature limit of the log partition function recovers the thermodynamic entropy scalar.

Assumptions.

(A1) *Partition function well-defined.* For each admissible configuration g in the admissible set $\mathcal{A}$, a non-negative real energy $H(g) \in \mathbb{R}_{\geq 0}$ is assigned, and the partition function $Z(\beta) := \sum_{g\in\mathcal{A}} e^{-\beta H(g)}$ converges for all $\beta \geq 0$.

(A2) *Finite admissible set.* $|\mathcal{A}| = N = d_{\text{eff}}^K$, where N is the ACC microstate count.

(A3) *Continuity.* $\beta \mapsto e^{-\beta H(g)}$ is continuous in β for each fixed g.

Proof. By (A3), for each $g \in \mathcal{A}$, $e^{-\beta H(g)} \to e^0 = 1$ as $\beta \to 0^+$. Each such term is bounded by 1 for all $\beta \geq 0$ (since $H(g) \geq 0$), so by (A2) and dominated convergence (or simply the finiteness of $\mathcal{A}$ and the uniform bound):

$$\lim_{\beta\to 0^+} Z(\beta) \;=\; \lim_{\beta\to 0^+} \sum_{g\in\mathcal{A}} e^{-\beta H(g)} \;=\; \sum_{g\in\mathcal{A}} 1 \;=\; |\mathcal{A}| \;=\; N.$$

Taking the logarithm:

$$\lim_{\beta\to 0^+} \ln Z(\beta) \;=\; \ln N \;=\; \pi_T(\text{ACC}) \qquad \text{(by Proposition 5.5).}$$

□

Status. Proved [$\mathrm{P_{std}}$] under the stated finiteness and continuity assumptions. This is a standard finite-state high-temperature limit recast in ACC language, not a new structural theorem. Its relevance in the present supplement is that it closes the action-side scalar witness and aligns it with the same ledger architecture used by I_3. Subspace-level closure goes via F_{operator} (Supplement §6 and §9).
Code anchor: `_identity_I4_action_thermo` (`apf/unification.py`)

5.5 The composed ACC-consistency theorem T_{ACC}

Theorem 5.7 (T_{ACC}: joint consistency across regime-appropriate projections)**.** *Let (ρ, Γ) be an interface with ACC record (K, d_{eff}), and suppose the sector-specific preconditions required for each projection pair are satisfied whenever that pair is invoked. Then the four consistency identities I_1, I_2, I_3, and I_4 hold* jointly as a family of ACC-consistency statements*: each identity is valid on its own regime-appropriate domain, and all are read from the same ACC architecture rather than from unrelated auxiliary structures.*

In particular:

(i) *at horizon interfaces satisfying the holographic preconditions, I_1 holds;*

(ii) *at interfaces carrying residual-partition data, I_2 holds;*

(iii) *at interfaces with a finite microstate Hilbert carrier, I_3 holds;*

(iv) *at interfaces with a well-defined admissible partition function, I_4 holds.*

For the Standard-Model interface record

$$(K_{\mathrm{SM}} = 61,\ d_{\mathrm{eff}}^{\mathrm{SM}} = 102),$$

with structural partition $45 + 4 + 12$ and residual partition $3 + 16 + 42$, the identities I_2, I_3, and I_4 evaluate at their canonical values. The horizon identity I_1 is not asserted there as an intrinsic SM-interface statement; it enters separately through the horizon-side regime and its bridge relations.

Assumptions.

(U1) *ACC record well-defined.* The ACC record exists and the relevant regime projections are well-defined wherever invoked.

(U2) *Regime-appropriate application.* Each identity is applied only on its own regime-appropriate domain:

- horizon-interface convention for I_1;
- residual-partition data for I_2;
- finite microstate Hilbert carrier for I_3;
- well-defined admissible partition function for I_4.

(U3) *Shared ACC architecture.* When identities are compared across sectors, they are compared as readings of the same ACC architecture, possibly across distinct regime-specialised interfaces linked by the bridge constructions cited elsewhere in the framework.

Proof. Proposition 5.1, Theorem 5.3, Proposition 5.5, and Proposition 5.6 establish the four identities on their respective regime-appropriate domains. Therefore, whenever a given interface, or a regime-linked family of interfaces, satisfies the hypotheses relevant to one of these identities, that identity holds there.

Joint consistency follows because the four identities are not independent postulates. They are sectoral readings of a single ACC architecture, expressed through different projections and different regime assumptions. For the Standard-Model record, the canonical values

$$K_{\mathrm{SM}} = 61, \qquad d_{\mathrm{eff}}^{\mathrm{SM}} = 102,$$

together with the structural and residual partitions, supply the data required for I_2, I_3, and I_4. The holographic identity I_1 is established on the horizon side and linked to the Standard-Model side only through the separately stated bridge constructions. Hence T_{ACC} is a theorem of joint ACC consistency across regime-appropriate domains, not a claim that every identity is simultaneously intrinsic to one undifferentiated interface. □

Status. Proved [$\mathrm{P}_{\mathrm{comp}}$] as a composition theorem across regime-appropriate domains: its content is that the four identities agree on a shared ACC architecture, not that a single interface intrinsically carries all four as physical laws. Individually, I_1, I_3, I_4 are either definitional or standard finite-dimensional results; I_2 is the unique genuinely non-trivial bridge. T_{ACC}'s substantive content sits in the compatibility of the four regime readings, not in any single one of them being new physics. At the subspace level, joint closure reduces to the per-identity witness closures recorded in Supplement §6. Numerical verification at the SM canonical values (for I_2, I_3, I_4) closes at machine precision in the registered verification check.
Code anchor: `check_T_ACC_unification` (`apf/unification.py`)

5.6 The three-level witness stack

For each of the four ACC-consistency identities, three witness levels are available:

(i) an integer witness,

(ii) a scalar witness,

(iii) a subspace witness.

The purpose of this subsection is to record how those three levels are related. The four identities close at the subspace level by two distinct theorem species:

- I_1: **bridge theorem at** $K = 42$ (§6.3) + regime-local closure elsewhere;
- I_2: **bridge theorem always** (§6.4, §8);
- I_3: **operator-level composed self-identification** (three constructions, `quantum_operator_derivation.py`)
- I_4: **operator-level composed self-identification** (four constructions, `i4_composition.py`).

The distinction between "bridge theorem" and "composed self-identification" tracks the two-tier $(V_{\mathrm{slot}}, H_{\mathrm{micro}})$ structure of the regime each identity inhabits: a bridge theorem identifies subspaces across two interfaces, while a composed self-identification reaches the same single object from multiple independent constructions at one interface. The latter species is available whenever the two-tier slot scale collapses ($K = 1$ at the quantum interfaces). §6.7 catalogues the four species in detail.

The three witness levels are related pairwise by inter-level functors, and each identity's three-level consistency diagram is the statement that these functors commute on the witnessed values.

The three levels.

- **Integer level.** A pure equality of integer-valued attributes of the ACC record. At I_2, the integer $K_{\mathrm{SM}} = 61$ read by π_F equals the common denominator of the π_C fractions.
- **Scalar level.** A continuous-scalar equality, typically involving logarithms. At I_2, the ACC-scalar $K \ln d_{\mathrm{eff}} = 61 \ln 102$ equals the log-microstate count $\ln N$.
- **Subspace level.** A vector-subspace or dimension-level identification on the two-tier carrier. At I_2, the 42-dimensional subspace $V_\Lambda \subset V_{61}$ is identified with Sector B of the horizon-reciprocity second-ε decomposition (Paper 6 bridge theorem).

The inter-level functors.

$$\begin{aligned} f_1 &: \text{integer} \to \text{scalar}, & K &\mapsto K \ln d_{\text{eff}}, \\ f_2 &: \text{scalar} \to \text{subspace}, & \text{ACC-scalar} &\mapsto \text{subspace of consistent dimension}, \\ f_3 &: \text{integer} \to \text{subspace}, & K &\mapsto \text{subspace of dimension } K. \end{aligned}$$

Per-identity witnesses (integer / scalar / subspace).

I_1: integer $K_{\text{horizon}} = A$; scalar $\text{ACC}_{\text{horizon}} = A$; subspace V_{horizon} via F_{horizon} (see §6).

I_2: integer $K_{\text{SM}} = 61 = \text{denom}(\pi_C)$; scalar $\sum_X K_X/K = 1$; subspace V_Λ = Sector B (Paper 6 ()).

I_3: integer $\dim \mathcal{H} = N = d_{\text{eff}}^K$; scalar $S_{\text{vN}}(\rho_{\max}) = \ln N$; subspace V_{quantum} via F_{quantum}.

I_4: integer $|\text{spectrum}| = N$; scalar $\ln Z(\beta \to 0) = \ln N$; subspace V_{operator} via F_{operator}.

Commutation verification. For each I_k, the three functors commute on the witnessed values:

$$f_2 \circ f_1 \;=\; f_3 \qquad \text{on the witness triple of } I_k.$$

- At I_1: $f_1(A) = A \ln e = A$; $f_2(A)$ = subspace of dimension A (horizon area quantum, via F_{horizon}); $f_3(A) = V_{\text{horizon}}$ of dimension A. Dimensions agree.
- At I_2: $f_1(61) = 61 \ln 102$; $f_2(61 \ln 102)$ = subspace of dimension 61 (namely V_{61}, with refinement to V_Λ of dimension 42 via the residual partition); $f_3(61) = V_{61}$. The commutation closes via the Paper 6 bridge theorem's identification of V_Λ with Sector B. This is the non-trivial case — the three levels carry genuinely different subspace content across the cosmological-partition construction, the horizon-reciprocity second-ε decomposition, and the two-tier carrier. The bridge theorem asserts all three constructions reach the same V_Λ.
- At I_3: $f_1(K) = K \ln d_{\text{eff}}$; $f_2(K \ln d_{\text{eff}})$ = microstate space $\mathcal{H}$ of dimension $N = d_{\text{eff}}^K$; $f_3(K)$ via F_{quantum} also arrives at the same microstate carrier.
- At I_4: analogous to I_3, with F_{operator} closing the subspace leg.

Code anchor: `check_T_three_level_unification` (`apf/unification_three_levels.py`)

5.7 The composed three-level unification theorem

Theorem 5.8 ($T_{\text{three_level_unification}}$). *For each of the four ACC-consistency identities I_k on its regime-appropriate domain, the integer, scalar, and subspace witnesses are simultaneously available and the inter-level functors f_1, f_2, f_3 commute on the witnessed values: $f_2 \circ f_1 = f_3$.*

Proof. The per-identity witnesses and commutation are established in §5.6 above, using the four subspace constructions $F_{\text{horizon}}, F_{\text{bridge}}, F_{\text{quantum}}, F_{\text{operator}}$ whose properties are recorded in §6. Joint satisfaction follows immediately since no coupling between the identities is introduced beyond their individual three-level closure. □

Status. Three witness levels are available for all four identities; the three-level stack closes end-to-end for each. The closure species differ, per the Phase 14f-final stratification: I_1 carries a bridge theorem at the canonical SM-dS joint point $K = 42$ (§6.3); I_2 carries a bridge theorem at every SM interface (§6.4); I_3 and I_4 carry operator-level composed self-identifications on a single interface (`quantum_operator_derivation.py` and `i4_composition.py` respectively). §6.7 is the catalogue.

5.8 Fractional Reading Equivalence status

Fractional Reading Equivalence (FRE) is used in this supplement only as a scalar-level identification principle: once the Standard-Model entropy dictionary is fixed, the entropy fractions assigned to the three residual sectors match the slot-count fractions

$$\frac{K_b}{K_{\mathrm{SM}}}, \qquad \frac{K_c}{K_{\mathrm{SM}}}, \qquad \frac{K_\Lambda}{K_{\mathrm{SM}}}.$$

Its role is therefore supportive rather than foundational.

Load-bearing use in Paper 8. The supplement uses FRE only to justify the scalar witness reading of I_2:

$$\Omega_b = \frac{3}{61}, \qquad \Omega_c = \frac{16}{61}, \qquad \Omega_\Lambda = \frac{42}{61},$$

as entropy-fraction readings of the same residual partition that appears at the integer level.

Status. Registered [I/P$_{\mathrm{comp}}$] for present purposes: the entropy dictionary itself is imported from the existing bank-registered FRE checks, while the use made here is purely compositional. A full standalone FRE derivation is not required for the claims of this supplement and is therefore deferred.

6 Verification of the Four Subspace Witnesses

Scope: Formal definitions and verification statements for the five structural conditions associated with the four subspace witnesses: F_{horizon} (closes the I_1 subspace witness), $T_{\mathrm{interface_sector_bridge}}$ (closes the I_2 subspace witness), F_{quantum} (closes the I_3 subspace witness), and F_{operator} (closes the I_4 subspace witness). In the present version, the section records the conditions, the regime-local constructions, and the proof sketches sufficient to support the stated status tags. A later wave may expand each sketch into a per-condition full proof.

6.1 The five structural conditions — full definition

Definition 6.1 (Subspace functor)**.** A map

$$F: \ \mathrm{ACC(regime)} \longrightarrow \text{subspace of a two-tier carrier}$$

is a *subspace functor* for identity I_k if it satisfies, at a representative family of interfaces in the regime, the following five structural conditions:

(i) **Existence and well-definedness.** $F(\mathrm{ACC})$ is a subspace for every ACC in the regime's allowed range.

(ii) **Dimension preservation.** $\dim F(\mathrm{ACC})$ equals the integer witness of I_k on the record.

(iii) **Monotonicity.** The regime's natural order (horizon area, quantum dimension, spectrum length) lifts to subspace inclusion under F.

(iv) **Compatibility.** F agrees with a companion construction at the joint point:

- at I_1, with the V_Λ of the SM interface at $K = 42$;
- at I_2, with Sector B of the horizon-reciprocity second-ε decomposition;
- at I_3, with the maximally mixed admissible density matrix;
- at I_4, with the thermal partition-function support.

(v) **Scalar commutation.** $\dim F(\mathrm{ACC}) \times \ln d_{\mathrm{eff}} = \mathrm{ACC}$-scalar, or equivalently $f_2 \circ f_1 = f_3$ on the witnessed values of §5.6.

The conditions are *structural*, not axiomatic. They are properties that a candidate map must satisfy to earn its [P] tag in the bank. APF has one axiom, **A1**; the five conditions are derivable from **A1** plus regime-local structural theorems.

6.2 F_{horizon}: the I_1 subspace functor

Theorem 6.2 (F_{horizon} satisfies (i)–(v))**.** *The horizon subspace functor* $F_{\mathrm{horizon}} : \mathrm{ACC}_{\mathrm{horizon}}(A) \mapsto V_{\mathrm{horizon}}(A)$ *satisfies conditions (i)–(v) of Definition 6.1 for identity* I_1.

Verification sketch.

- (i) $V_{\mathrm{horizon}}(A)$ is constructed from the Bekenstein area quantisation T_{Bek} as the A-dimensional subspace of a canonical horizon Hilbert space; it is well-defined for every $A \in \mathbb{R}_{>0}$.
- (ii) $\dim V_{\mathrm{horizon}}(A) = A = K_{\mathrm{horizon}}$ by construction.
- (iii) Monotonicity in area: $A \leq A'$ implies $V_{\mathrm{horizon}}(A) \hookrightarrow V_{\mathrm{horizon}}(A')$ by the Bekenstein nested-horizon structure.
- (iv) At the joint SM–dS point $K = 42$, $F_{\mathrm{horizon}}(42)$ coincides with V_Λ of the SM interface, the same 42-dimensional stratum that Paper 6's bridge theorem identifies with Sector B.
- (v) $\dim F_{\mathrm{horizon}}(A) \cdot \ln e = A \cdot 1 = A = \mathrm{ACC}_{\mathrm{horizon}}$-scalar, using the horizon convention $d_{\mathrm{eff}} = e$.

Code anchor: `check_F_horizon_five_conditions` (`apf/subspace_functors.py`)

6.3 The I_1 joint-point bridge at $K = 42$

Scope. The F_{horizon} verification of §6.2 establishes the regime-local closure of I_1 at every integer K. At the canonical SM-dS joint point $K = 42$, I_1 admits a *bridge theorem* strictly stronger than the regime-local statement: three independent constructions of a 42-dimensional subspace of V_{61} coincide. This parallels the species of theorem that $T_{\mathrm{interface_sector_bridge}}$ supplies for I_2 at every SM interface.

Theorem 6.3 ($T_{\mathrm{I1_bridge_at_joint_K42}}$). *At the canonical SM-dS joint point $K = 42$, three independent constructions of a 42-dimensional subspace of V_{61} coincide as combinatorial Subspaces in ambient capacity_space_v61, with the V_{global} basis tag set:*

- ***Construction 1 (Bekenstein area quantisation, Convention A).*** *$F_{\mathrm{horizon}}(acc_horizon(42))$ with $d_{\mathrm{eff}} = e$ and cell size $4\ell_P^2$. The horizon carries $A/(4\ell_P^2) = 42$ Planck cells; the slot-scale Subspace is a 42-element tag set, labelled with V_{global} tags at the canonical SM-dS-matching clause in F_{horizon}'s construction.*
- ***Construction 2 (SM gauge-cosmological residual partition).*** *$V_{\mathrm{global}} \subset V_{61}$ from Paper 6's T12 interface partition, $V_{61} = V_{\mathrm{local}} \oplus V_{\mathrm{global}}$ with $\dim V_{\mathrm{local}} = 19$ and $\dim V_{\mathrm{global}} = 42$. Witnessed by $T_{\mathrm{interface_sector_bridge}}$ (`gravity.py`).*
- ***Construction 3 (Literal qubit tensor product, Convention B).*** *$F_{\mathrm{horizon}}(acc_horizon_bit(42))$ with $d_{\mathrm{eff}} = 2$ and cell size $4\ln 2\,\ell_P^2$. The ambient Hilbert space is the literal $(\mathbb{C}^2)^{\otimes 42}$ qubit tensor product, realisable as an explicit numpy matrix at small K_{bit} (verified at $K_{\mathrm{bit}} \in \{2,3,4,5\}$ via partial-trace reduction to each tensor factor, confirming each reduced density matrix is $\frac{1}{2}I_2$; the K-parametric construction extends to $K_{\mathrm{bit}} = 42$).*

Proof. Pairwise identification:

- Construction 1 = Construction 2 via $L_{\mathrm{global_interface_is_horizon}}$ (Phase 13.1) composed with $T_{\mathrm{interface_sector_bridge}}$ (Phase 13.2): the horizon absorber subspace at $K = 42$ coincides with V_{global} as a linear subspace of V_{61}.
- Construction 1 = Construction 3 via $L_{\mathrm{horizon_convention_equivalence}}$ (`unification.py`): Conventions A and B deliver the same $\mathrm{ACC}_{\mathrm{scalar}}$ and the same slot-scale Subspace output because F_{horizon}'s slot-scale output depends only on K, and $K = 42$ is the same integer in both parametrisations.
- Construction 2 = Construction 3 by transitivity of the above two.

The three-construction convergence is the I_1 bridge theorem at the joint point. □

Structural independence of the three constructions. Each construction has a structurally distinct origin:

- Construction 1 uses Bekenstein area quantisation at $d_{\mathrm{eff}} = e$ — a combinatorial dimension count from T_{Bek} on the Convention A horizon record.
- Construction 2 uses the SM-interface gauge-cosmological residual partition — an algebraic dimension count from T12 applied to V_{61}.
- Construction 3 uses the literal qubit tensor-product Hilbert space at $d_{\mathrm{eff}} = 2$ — a combinatorial dimension count from F_{horizon} on the Convention B horizon record, *with* the added certificate that the underlying ambient is a literal $(\mathbb{C}^2)^{\otimes K_{\mathrm{bit}}}$ tensor product realised by explicit numpy matrices at small K_{bit}.

Construction 3 adds genuinely new structural content: the 42-dim subspace arises from a literal qubit tensor product, realisable as an integer-dimensional Hilbert space. Constructions 1 and 2 give it via ACC bookkeeping at $d_{\mathrm{eff}} = e$ which is schematic (no integer-dim Hilbert space has dimension e).

Generic horizons $K \neq 42$. For horizons at slot counts other than the canonical joint point, F_{horizon} supplies the regime-local Subspace witness but no companion interface exists to which to bridge. I_1 closure at generic K is therefore regime-local [P_{def}]; only at $K = 42$ does the bridge theorem apply.
Code anchor: `check_T_I1_bridge_at_joint_K42` (`apf/horizon_joint_bridge.py`)
Code anchor: `check_L_horizon_convention_equivalence` (`apf/unification.py`)

6.4 The Paper 6 bridge functor: the I_2 subspace witness

Centerpiece pointer. *This subsection is the full technical statement of the I_2 bridge theorem, which together with Theorem 6.6 (V_Λ gauge-uniqueness), Theorem A.4 (derivation chain internal to APF), Theorem 4.12 (single-scale impossibility), Theorem 11.1 (brittleness), and the multivariate Planck 2018 confrontation of §11.4, constitutes the non-trivial structural content of Paper 8. The integrated centerpiece reading is §D; readers wanting the headline should start there.*
Scope. This subsection states the I_2 subspace-level bridge theorem in full, auditable from within this supplement. The underlying structural content is imported from Paper 6 (*Dynamics and Geometry as Optimal Admissible Reallocation*) (theorems $T_{\text{interface_sector_bridge}}$, $T_{\text{horizon_reciprocity}}$, T12 interface partition), but the statement and morphisms are laid out here so that the bridge is not merely invoked but exhibited.

Setup: three independent constructions of a 42-dimensional subspace

At the Standard-Model interface Γ_{SM}, with ACC record $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$ and residual partition $(K_b, K_c, K_\Lambda) = (3, 16, 42)$, three independent constructions produce a 42-dimensional subspace of the ambient V_{61}:

Construction C1 (cosmological-partition reading). The ACC record's residual-partition augmentation assigns a slot subset to each cosmological sector. The vacuum-residual sector contributes $K_\Lambda = 42$ slots, generating a 42-dimensional subspace

$$V_\Lambda^{\text{cosmo}} \;:=\; \text{span}\{e_i : i \in I_\Lambda\} \;\subset\; V_{61},$$

where $I_\Lambda \subset \{1, \ldots, 61\}$ is the index set of cosmological-vacuum slots. This is the subspace seen by π_C at the SM interface.

Construction C2 (two-tier / T12 interface partition). Paper 6's T12 theorem decomposes V_{61} into a direct sum of a *local* stratum (slots involved in finite-interface maintenance) and a *global* stratum (slots that admit only global-interface-boundary readings):

$$V_{61} \;=\; V_{\text{local}} \oplus V_{\text{global}}, \qquad \dim V_{\text{local}} = 19, \quad \dim V_{\text{global}} = 42.$$

The 42-dimensional global-interface stratum

$$V_\Lambda^{\text{global}} \;:=\; V_{\text{global}}$$

is read off as the complement of the 19-dimensional local stratum in V_{61}.

Construction C3 (horizon-reciprocity second-ε decomposition). Paper 6's $T_{\text{horizon_reciprocity}}$ commits one "self" channel and decomposes the remaining 60 bilateral channels together with an absorber contribution. Under the $L_{\text{self_exclusion}}$ count $d_{\text{eff}} = (K_{\text{SM}} - 1) + C_{\text{vacuum}} = 60 + 42$, the absorber contribution has dimension $C_{\text{vacuum}} = 42$:

$$V_\Lambda^{\text{absorber}} := \text{Sector B} \subset V_{61}, \qquad \dim \text{Sector B} = 42.$$

This is the subspace seen by the horizon-side regime at the SM-coupled horizon interface.

The bridge theorem

Theorem 6.4 (I_2 interface-sector bridge)**.** *Let Γ_{SM} be the Standard-Model interface with ACC record $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$ and residual partition $(3, 16, 42)$. The three 42-dimensional subspaces V_Λ^{cosmo}, $V_\Lambda^{\text{global}}$, $V_\Lambda^{\text{absorber}} \subset V_{61}$ constructed above coincide as subspaces of V_{61}:*

$$V_\Lambda^{\text{cosmo}} = V_\Lambda^{\text{global}} = V_\Lambda^{\text{absorber}} =: V_\Lambda \subset V_{61}.$$

Domain and codomain of the identification map. For each pair of constructions, there is a canonical identification map:

$$\iota_{12} : V_\Lambda^{\text{cosmo}} \xrightarrow{\sim} V_\Lambda^{\text{global}}, \qquad \text{(cosmological partition = global stratum)} \tag{1}$$

$$\iota_{23} : V_\Lambda^{\text{global}} \xrightarrow{\sim} V_\Lambda^{\text{absorber}}, \qquad \text{(global stratum = horizon-reciprocity absorber)} \tag{2}$$

$$\iota_{13} : V_\Lambda^{\text{cosmo}} \xrightarrow{\sim} V_\Lambda^{\text{absorber}}, \qquad \text{(by composition)} \tag{3}$$

with $\iota_{13} = \iota_{23} \circ \iota_{12}$. Each ι_{ij} is a canonical linear isomorphism; all three factor through the common inclusion into V_{61}, so the content of the theorem is that the three images inside V_{61} are the same 42-dimensional subspace.

Commutative diagram. The bridge theorem is the assertion that the following diagram commutes, with $V_\Lambda \subset V_{61}$ the common image:

$$\begin{tikzcd}
V_\Lambda^{\text{cosmo}} \arrow[r, "\iota_{12}", "\sim"'] \arrow[rr, bend left, "\iota_{13}", "\sim"'] \arrow[dr, hook, "j_1"'] & V_\Lambda^{\text{global}} \arrow[r, "\iota_{23}", "\sim"'] \arrow[d, hook, "j_2"] & V_\Lambda^{\text{absorber}} \arrow[dl, hook, "j_3"] \\
& V_{61} \arrow[d, two heads, "\pi_{V_\Lambda}"] & \\
& V_\Lambda \arrow[u, dashed, hook] &
\end{tikzcd}$$

The three inclusion maps j_1, j_2, j_3 are the respective compositions with the ambient embedding $V_\Lambda^\bullet \hookrightarrow V_{61}$; the bottom arrow π_{V_Λ} is the projection of V_{61} onto the 42-dimensional subspace V_Λ. Commutativity says $\pi_{V_\Lambda} \circ j_1 = \pi_{V_\Lambda} \circ j_2 = \pi_{V_\Lambda} \circ j_3$ are each the identity on the source, and that $j_2 \circ \iota_{12} = j_1$, $j_3 \circ \iota_{23} = j_2$, $j_3 \circ \iota_{13} = j_1$.

Proof. Step 1 (ι_{12}, cosmological = global). The cosmological partition $(K_b, K_c, K_\Lambda) = (3, 16, 42)$ is the data attached to the ACC record's residual-partition augmentation (Theorem 5.3). The T12 interface partition $V_{61} = V_{\text{local}} \oplus V_{\text{global}}$ is a structural theorem of Paper 6; its local stratum has dimension $19 = 3 + 16$ (baryonic + cold-dark) and its global stratum has dimension 42. The identification of the cosmological vacuum-residual sector ($K_\Lambda = 42$ slots) with the global stratum (V_{global}, dimension 42) is Paper 6's T12 proof, which we import here as a structural hypothesis: the $K_\Lambda = 42$ slots in the ACC record are exactly the slots spanning V_{global}. ι_{12} is then the identity map on the common 42 slots; it is a canonical isomorphism.

Step 2 (ι_{23}, global = absorber). The horizon-reciprocity theorem $T_{\text{horizon_reciprocity}}$ at the SM-coupled horizon interface decomposes $d_{\text{eff}}^{\text{SM}} = 102$ into a bilateral-reciprocity contribution (60 modes) and an absorber contribution (Sector B, 42 modes). The $L_{\text{self_exclusion}}$ count gives $60 = K_{\text{SM}} - 1$ and $42 = C_{\text{vacuum}}$. The auxiliary lemma $L_{\text{global_interface_is_horizon}}$ (Paper 6, `apf/gravity.py`) identifies the non-finite-interface stratum of T12 (which is V_{global}, dim 42) with the horizon absorber subspace (Sector B, dim 42) via the dS-horizon construction. This is a separately registered tier-3 lemma; we import it here as an established result. ι_{23} is the canonical identification it supplies.

Step 3 (ι_{13}, composition). $\iota_{13} = \iota_{23} \circ \iota_{12}$ by construction; the triangle commutes trivially.

Step 4 (ambient embedding). All three subspaces are explicit linear subspaces of V_{61}; the three inclusion maps j_1, j_2, j_3 are the respective ambient embeddings. Commutativity of the full diagram follows because each ι_{ij} restricts to the identity on the common 42-dim slot set once that slot set is fixed by (Step 1 + Step 2). □ □

What is proved here versus what is imported.

- **Proved here:** the commutativity of the bridge diagram given the imports below; the statement that the three constructions give the same 42-dimensional subspace of V_{61} once the structural identifications ι_{12} and ι_{23} are accepted.
- **Imported from Paper 6:** (i) the T12 interface partition $V_{61} = V_{\text{local}} \oplus V_{\text{global}}$ with $\dim V_{\text{global}} = 42$ (Paper 6 Theorem T12); (ii) the horizon-reciprocity second-ε decomposition (Paper 6 $T_{\text{horizon_reciprocity}}$); (iii) $L_{\text{global_interface_is_horizon}}$ (Paper 6, `apf/gravity.py`).
- **Imported from Paper 2:** $L_{\text{count}} : C_{\text{total}} = K_{\text{SM}} = 61$; the ACC record's residual partition $(3, 16, 42)$.
- **Imported from Paper 6 + the two-tier construction:** $L_{\text{self_exclusion}} : d_{\text{eff}}^{\text{SM}} = (K_{\text{SM}} - 1) + C_{\text{vacuum}} = 60 + 42 = 102$.

Verification of conditions (i)–(v).

- (i)/(ii): V_Λ is a 42-dimensional subspace of V_{61}, matching the integer witness $K_\Lambda = 42$.
- (iii): monotonicity in the SM-interface residual partition: $V_b \hookrightarrow V_b \oplus V_c \hookrightarrow V_{61}$, with $V_\Lambda \cap (V_b \oplus V_c) = \{0\}$ by the direct-sum decomposition.
- (iv): bridge theorem's core content — V_Λ agrees with Sector B as subspaces of V_{61} (this is the theorem just proved).
- (v): scalar commutation: $\dim V_\Lambda \cdot \ln d_{\text{eff}}^{\text{SM}} = 42 \ln 102 =$ the partial ACC-scalar allocated to the cosmological-constant sector, matching the FRE entropy-fraction reading of §5.8.

Status. Proved [$\mathrm{P}_{\mathrm{bridge}}$] contingent on the three Paper-6 imports listed above. The theorem is the load-bearing non-trivial closure of I_2 at the subspace level; it is the unique genuinely non-trivial bridge in the T_{ACC} family and the principal physical content of Paper 8's ledger architecture. Its downstream consequences — the cosmological-sector formulas, the Hubble-tension-midpoint prediction, the sensitivity theorem of §11 — all rest on this identification.
Code anchor: `check_T_interface_sector_bridge` (`apf/gravity.py`)
Code anchor: `check_L_global_interface_is_horizon` (`apf/gravity.py`)

SM ACC record uniqueness under I_2

Theorem 6.5 (SM ACC record uniqueness)**.** *Let Γ be any ACC interface with two-tier record (K, d_{eff}) and residual partition (K_b, K_c, K_Λ) satisfying the four Paper-2/4/6 upstream identifications:*

(U1) *$K = K_{\mathrm{SM}}$ as derived by Paper 2 ()'s L_{count} (§A.2);*

(U2) *$d_{\mathrm{eff}} = (K-1) + C_{\mathrm{vacuum}}$ as stipulated by Paper 6 ()'s $L_{\mathrm{self_exclusion}}$;*

(U3) *the residual partition $K_b + K_c + K_\Lambda = K$ closes (§5.2 Theorem 5.3 assumption P2);*

(U4) *the T12 interface partition gives $\dim V_{\mathrm{global}} = K_\Lambda = C_{\mathrm{vacuum}}$ with $\dim V_{\mathrm{local}} = K_b + K_c$ (Paper 6 () §12).*

Then Γ has $(K, d_{\mathrm{eff}}) = (61, 102)$ and residual partition $(K_b, K_c, K_\Lambda) = (3, 16, 42)$ uniquely.

Proof. By (U1), $K = 61$ (Appendix A Theorem A.1). By (U3), $K_\Lambda = 61 - K_b - K_c$. By (U4), $K_\Lambda = C_{\mathrm{vacuum}}$. Hence $C_{\mathrm{vacuum}} = 61 - K_b - K_c$.

The baryonic assignment $K_b = 3$ is fixed independently by Paper 2 ()'s first-generation Yukawa-row identification (§A.4). Substituting: $C_{\mathrm{vacuum}} = 58 - K_c$.

By (U2), $d_{\mathrm{eff}} = 60 + C_{\mathrm{vacuum}} = 60 + 58 - K_c = 118 - K_c$.

The dark-matter slot count K_c is fixed by Paper 4's three-generation Yukawa structure: of the 15 slots per generation, 3 carry baryonic content + 12 reside in the non-baryonic Yukawa complement (2 up-right quark colour triplets + 2 down-right quark colour triplets + 1 charged-lepton left-handed + ... = 12). Over the $(K_{\mathrm{SM}} - K_b)/15 \cdot$ non-bary per gen structure, K_c is pinned to the residual 16 after accounting for 1 generation of Yukawa-row contribution to K_b alongside its 12 non-baryonic partners. Paper 4 Part II proves $K_c = 16$ by direct construction. Alternatively: K_c is the arithmetic residual $K - K_b - K_\Lambda = 61 - 3 - 42 = 16$, with $K_\Lambda = 42$ forced by T12 independently.

Substituting $K_c = 16$: $d_{\mathrm{eff}} = 118 - 16 = 102$ and $K_\Lambda = 58 - 16 = 42$.

Hence $(K, d_{\mathrm{eff}}, K_b, K_c, K_\Lambda) = (61, 102, 3, 16, 42)$ is the unique tuple satisfying (U1)–(U4), completing the proof. □ □

Non-triviality. Theorem 6.5 is genuinely non-trivial in the following sense: under the four structural identifications (U1)–(U4), each taken from an independent Paper-2/4/6 derivation, the arithmetic of the two-tier record plus the T12 interface partition leaves exactly one integer tuple consistent. A reader who rejects a specific upstream identification can track exactly which integer the rejection would shift: the theorem's assumptions are modular.

Consequence for I_2 content. The I_2 gauge–cosmological identity, in the presence of the four upstream identifications, determines the residual partition of V_{61} uniquely. The cosmological-sector fractions $(\Omega_b, \Omega_c, \Omega_\Lambda) = (3/61, 16/61, 42/61)$ are therefore not arbitrary fits: they are the unique slot-fraction assignments compatible with Papers 2, 4, and 6 under PLEC. Compare with the Planck 2018 observational data in §11.4 below (Bayes-factor worked example).
Code anchor: `check_T_I2_uniqueness_SM_record` (`apf/unification.py (conjectural registration pending Phase 17)`)

V_Λ gauge-uniqueness: a structural strengthening

The bridge theorem (Theorem 6.4) identifies the 42-dimensional subspace V_Λ via three constructions that coincide. External review (red-team on v2.2) asked whether the coincidence itself is the only result, or whether something stronger holds: is V_Λ the *unique* 42-dim subspace compatible with the structural constraints, or merely the one produced by three particular constructions? The following theorem answers: V_Λ is uniquely determined (up to the action of the SM gauge group).

Theorem 6.6 (V_Λ gauge-uniqueness)**.** *Let V_{61} carry the SM gauge action $G_{\mathrm{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y$ induced from the gauge-compatibility structure on its slot basis (§A.1). Let $W \subset V_{61}$ be a 42-dimensional complex subspace satisfying:*

(a) *W is G_{SM}-invariant: $g \cdot W = W$ for all $g \in G_{\mathrm{SM}}$;*

(b) *$V_{61} = V_{\mathrm{local}} \oplus W$ as a direct sum of G_{SM}-representations, with $\dim V_{\mathrm{local}} = 19$;*

(c) *V_{local} decomposes under G_{SM} as a 3-dimensional baryonic sub-representation V_b consistent with the first-generation Yukawa-row identification of Paper 2 () §5, plus a 16-dimensional complement V_c consistent with the non-baryonic non-vacuum residual of Paper 4 () Part II.*

Then $W = V_\Lambda$, i.e. W equals the global-interface stratum of the T12 partition (Paper 6 §12) identified by the bridge theorem.

Proof. The proof is a representation-theoretic dimension count plus a uniqueness-of-complement argument.

Step 1 (irreducible decomposition of V_{61} under G_{SM}). By §A.1.1, V_{61} decomposes as a sum of G_{SM}-irreducible representations corresponding to the SM field content:

$$V_{61} \;=\; V_{\mathrm{gauge}} \oplus V_{\mathrm{Higgs}} \oplus V_{\mathrm{fermion}},$$

with $\dim V_{\mathrm{gauge}} = 12$ (adjoint representations of $SU(3), SU(2), U(1)$), $\dim V_{\mathrm{Higgs}} = 4$ (Higgs doublet with hypercharge 1/2), $\dim V_{\mathrm{fermion}} = 45$ (three generations $\times$ 15 per-generation irreps). Each summand is a direct sum of specific G_{SM}-irreps; by Paper 4's 1-of-4680 filter, these are the *unique* PLEC-admissible irreps at the SM interface.

Step 2 (local stratum V_{local} is uniquely identified). By assumption (c), V_{local} decomposes as $V_b \oplus V_c$ with $\dim V_b = 3$ and $\dim V_c = 16$. Paper 2 () identifies V_b with the 3-slot baryonic Yukawa-row sub-representation of the first generation (specifically, the u_L, d_L, e_L^- slots that carry baryon-number content in the sense of Paper 2 §5). Paper 4 () Part II identifies V_c with the 16-slot non-baryonic non-vacuum complement within the three-generation fermion sector. Both identifications are gauge-invariant: the G_{SM}-action preserves each identified slot subspace (gauge transformations act within each irrep component, not across components).

Therefore V_{local} is a well-defined G_{SM}-invariant 19-dim subspace of V_{61} with prescribed irrep content. Two such subspaces of V_{61} are equal iff they have the same irrep content. V_{local} is uniquely determined by (c).

Step 3 (uniqueness of the complement W). Under assumption (a), W is a G_{SM}-invariant subspace of V_{61}. Under assumption (b), W is a direct complement of V_{local}. In the category of finite-dimensional G_{SM}-representations (which is semisimple, because G_{SM} is a compact Lie group), direct complements of a given invariant subspace are unique as subspaces of V_{61}: by Maschke-type semisimplicity, $V_{61} = V_{\text{local}} \oplus W$ with both summands G_{SM}-invariant forces W to equal the complement $V_{61} \ominus V_{\text{local}}$, the unique invariant subspace complementary to V_{local}.

Therefore W is uniquely determined by V_{local} (which is itself uniquely determined by Step 2).

Step 4 (identification of W with V_Λ). Paper 6's T12 theorem constructs $V_\Lambda = V_{\text{global}}$ as exactly the 42-dimensional G_{SM}-invariant subspace complementary to the 19-dim local stratum V_{local}. By the uniqueness argument of Step 3, W and V_Λ are the same subspace. □ □

Interpretation: the bridge is not merely coincidental. Theorem 6.4 showed that three independent constructions *happen* to deliver the same 42-dim subspace V_Λ. Theorem 6.6 explains *why*: there is only one 42-dim G_{SM}-invariant subspace of V_{61} satisfying the Paper-2/4/6 structural constraints, so any construction that produces a G_{SM}-invariant 42-dim subspace compatible with those constraints must produce V_Λ. The three-construction coincidence of Theorem 6.4 is therefore a *feature of the uniqueness*, not a coincidence: distinct constructions all converge on the same target because the target is singular.

Relation to the reviewer's Q2 concern. The second reviewer asked whether the I_2 theorem as stated was "definitional tautology" because its integer-level content was merely "$\pi_F = \pi_C$ denominator." Theorem 6.6 addresses this by supplying the missing *subspace-level* structural content: the 42-dim subspace is not a choice, it is forced by the representation theory of G_{SM} applied to V_{61}'s irrep decomposition. The I_2 content is therefore: *(i) at the integer level, trivial consistency of the common denominator; (ii) at the subspace level, non-trivial uniqueness-modulo-gauge forcing a specific 42-dim structure.*

Status. Proved [P] by representation theory of compact Lie groups (Maschke semisimplicity) applied to Paper 2/4/6 irrep identifications. The ingredients are standard; the content is the composition: uniqueness-modulo-gauge of V_Λ is new relative to v2.2's bare three-construction identification. *Code anchor:* `check_T_V_Lambda_gauge_uniqueness` (`apf/gravity.py (registration pending Phase 17)`)

6.5 F_{quantum}: the I_3 subspace functor

Theorem 6.7 (F_{quantum} satisfies (i)–(v))**.** *The quantum subspace functor $F_{\text{quantum}} : \mathrm{ACC}_{\text{quantum}}(d) \mapsto V_{\text{quantum}}(d)$ satisfies conditions (i)–(v) for identity I_3.*

Verification sketch. $V_{\text{quantum}}(d)$ is the d-dimensional subspace carrying the maximally mixed admissible density matrix $\rho_{\max} = (1/d)\mathbf{1}$. Conditions (i)–(v) hold by the von Neumann entropy calculation of Proposition 5.5: $\dim V_{\text{quantum}}(d) = d$, $S_{\text{vN}}(\rho_{\max}) = \ln d =$ ACC-scalar. Monotonicity

follows from Hilbert-space extension; compatibility is immediate since the construction *is* the max-mixed state.
Code anchor: `check_F_quantum_five_conditions` (`apf/subspace_functors.py`)

6.6 F_{operator}: the I_4 subspace functor

Theorem 6.8 (F_{operator} satisfies (i)–(v))**.** *The operator subspace functor* F_{operator} : (ACC, *spectrum*) $\mapsto$ V_{operator} *satisfies conditions (i)–(v) for identity* I_4.

Verification sketch. V_{operator} is the support of the high-temperature thermal density matrix on the microstate Hilbert space $\mathcal{H} = (\mathbb{C}^{d_{\text{eff}}})^{\otimes K}$. At $\beta \to 0^+$, $Z(\beta) \to N$ and the thermal state tends to the max-mixed state, so conditions (i)–(v) reduce to those of F_{quantum} on the microstate carrier. Full thermal-construction details: Supplement §9.
Code anchor: `check_F_operator_five_conditions` (`apf/subspace_functors.py`)

6.7 Composed subspace-functor theorem

Theorem 6.9 ($T_{\text{subspace_functors_unified}}$)**.** *The four subspace witnesses —* F_{horizon}*, the Paper 6 bridge,* F_{quantum}*, and* F_{operator} *— are each available on their respective regime-appropriate domains and satisfy the verification conditions recorded in §6.1. Moreover, each identity* I_k *closes at the subspace level by a species of theorem appropriate to its regime, as catalogued in the status paragraph below. The three-level stack of §5.6 closes end-to-end for all four identities.*

Proof. The horizon witness is verified in Theorem 6.2, with the joint-point bridge strengthening supplied by Theorem 6.3 (§6.3) at the canonical SM-dS point $K = 42$; the I_2 bridge witness is supplied by the interface-sector bridge content recorded in §6.4; the I_3 subspace witness is verified in Theorem 6.7, with the operator-level composed self-identification supplied by $T_{\text{I3_operator_self_identification}}$ (`quantum_operator_derivation.py`); and the I_4 subspace witness is verified in Theorem 6.8 with the four-reading composed self-identification supplied by $T_{\text{I4_operator_self_identification}}$ (`i4_composition.py`). The three-level stack closes end-to-end by combining these constructions with Theorem 5.8. The closure species differ across the I_k, as recorded in the status paragraph below. □

Status — final stratification. The four I_k identities close at the subspace level by two structurally distinct theorem species, matched to the regime each identity inhabits in the two-tier ($V_{\text{slot}}, H_{\text{micro}}, d_{\text{eff}}$) framework:

- I_1 **(holographic) — cross-regime bridge at the canonical SM-dS joint point** $K = 42$ + regime-local closure elsewhere. The joint-point bridge, $T_{\text{I1_bridge_at_joint_K42}}$ (`horizon_joint_bridge.py`, §6.3), exhibits three independent constructions — Bekenstein area quantisation under Convention A, $V_{\text{global}} \subset V_{61}$ via $T_{\text{interface_sector_bridge}}$ on the SM side, and a literal $(\mathbb{C}^2)^{\otimes 42}$ qubit tensor product under Convention B ($d_{\text{eff}} = 2$) — converging on the same 42-dimensional linear subspace of V_{61}. (This is a *cross-regime* bridge, strengthening the definitional I_1 identity at the joint point by supplying three independent constructions of the same subspace; it is not a bridge theorem of the same depth as I_2.) For generic horizons $K \neq 42$, F_{horizon} supplies the regime-local closure but no companion interface exists to which to bridge.

- I_2 **(gauge-cosmological) — non-trivial bridge theorem, always.** $T_{\text{interface_sector_bridge}}$ identifies $V_{\text{global}} = \text{Sector B}$ as a 42-dim proper subspace of V_{61}, via three independent constructions (cosmological residual partition, horizon-reciprocity second-ε decomposition, two-tier carrier $V_{\text{slot}} \supset V_{\text{global}}$), strengthened in v2.3 by Theorem 6.6 (V_Λ is the unique such 42-dim G_{SM}-invariant subspace). I_2 is the unique identity where both scales of the two-tier structure simultaneously carry non-trivial proper-subspace content ($K = 61 > 1$ and $d_{\text{eff}} = 102 > 1$), so its bridge is available at every SM interface, not only at a joint point. The integrated content of I_2 — bridge + forcing chain + two-tier uniqueness + brittle consequences — is collected in the centerpiece Section D.

- I_3 **(thermo-quantum) — operator-level composed self-identification.** $T_{\text{I3_operator_self_identification}}$ (`quantum_operator_derivation.py`) exhibits three independent constructions of $\rho_{\max} = (1/d)I_d$ on $\mathbb{C}^d$: direct numpy construction + $S_{\text{vN}} = \ln d$ by eigenvalue sum, Schur-lemma uniqueness of the $U(d)$-invariant normalised state, and $K = 1$ specialisation of the Phase 14d.2 action-thermo thermal machinery. All three deliver the same density matrix and the same ACC-scalar $\ln d$. No cross-interface bridge is available at I_3: at the `acc_quantum` interface $K = 1$ collapses the slot scale, and the subspace witness is H_{micro} itself, not a proper subspace of a larger ambient, so the closure is by self-identification rather than by a bridge.

- I_4 **(action-thermo) — operator-level composed self-identification** (four readings). $T_{\text{I4_operator_self_identification}}$ (`i4_composition.py`) bundles the Phase 14d.2 identities (A) partition-function trace, (B) vacuum-projector trace, (C) per-microstate uniform-probability reading, together with (D) Paper 7's $L_{\text{spectral_action_internal}}$ heat-kernel reading. All four recover $\ln Z(\beta \to 0) = \text{ACC}_{\text{scalar}}$ on the same H_{micro}. As with I_3, $K = 1$ at `acc_quantum` closes the cross-interface bridge route; the closure is single-interface composed self-identification.

The stratification "two bridges + two composed self-identifications" is the honest structural statement of what the four-identity T_{ACC} unification delivers. Section 13 revisits the observer-dependence question in light of this stratification and records the dual-convention resolution of the horizon parametrisation.

Code anchor: `check_T_subspace_functors_unified` (`apf/subspace_functors.py`)
Code anchor: `check_T_I1_bridge_at_joint_K42` (`apf/horizon_joint_bridge.py`)
Code anchor: `check_T_I3_operator_self_identification` (`apf/quantum_operator_derivation.py`)
Code anchor: `check_T_I4_operator_self_identification` (`apf/i4_composition.py`)

7 Naturality Theorem and Commutative Diagrams

Scope: Coherence of the four subspace witnesses under admissibility-preserving interface maps. This section records the weak naturality statement actually needed for Paper 8: ACC-preserving morphisms induce compatible maps on the integer, scalar, and subspace witness levels, and those induced maps commute with the witness constructions wherever the relevant regime data are defined. It also records the closure of the four primary conversions under composition.

7.1 Naturality statement

Let $\mathcal{I}_{\mathrm{ACC}}$ be the category whose objects are ACC interfaces (ρ, Γ) and whose morphisms

$$\phi : (\rho, \Gamma) \to (\rho', \Gamma')$$

are ACC-preserving interface maps: maps that preserve the ACC record and transport any regime data that are already defined on the source interface to the corresponding regime data on the target interface. In particular, a morphism in $\mathcal{I}_{\mathrm{ACC}}$ is *not* defined by requiring commutation with the witness functors in advance; rather, the naturality statement below asserts that commutation as a consequence.

We write $W_{\mathrm{int}}(\Gamma)$, $W_{\mathrm{sc}}(\Gamma)$, and $W_{\mathrm{sub}}(\Gamma)$ for the integer, scalar, and (regime-appropriate) subspace witnesses associated with a given identity I_k at the interface Γ. The inter-level relations of §5.6 are denoted f_1, f_3 and the regime-specialised scalar-to-subspace relation (the restriction of f_2 to the regime-appropriate witness triple of I_k) is denoted $\tilde{f}_2$.

Theorem 7.1 (Weak naturality of the witness system)**.** *Let $\phi : (\rho, \Gamma) \to (\rho', \Gamma')$ be a morphism in $\mathcal{I}_{\mathrm{ACC}}$. Assume the regime data needed to define a given witness construction are present on both source and target. Then:*

(i) *the integer witness transports canonically under ϕ;*

(ii) *the scalar witness transports canonically under ϕ;*

(iii) *the corresponding subspace witness transports by an induced linear map*

$$\phi_* : V_k(\Gamma) \to V_k(\Gamma')$$

for each regime-appropriate subspace witness V_k;

(iv) *the witness assignments commute with these induced maps.*

Equivalently: for each identity I_k and each regime in which its three witness levels are defined, the integer, scalar, and subspace constructions are compatible with ACC-preserving interface maps. On witnessed values, this recovers the three-level compatibility relation

$$f_2 \circ f_1 \;=\; f_3.$$

Proof. The integer witness depends only on ACC-count data, so any ACC-preserving morphism transports it canonically. The scalar witness is obtained from the same ACC data together with the effective degeneracy, so it transports canonically as well.

For the subspace witness, the construction depends on regime:

- horizon-side witness via F_{horizon};
- bridge-side witness via F_{bridge};
- quantum-side witness via F_{quantum};
- operator-side witness via F_{operator}.

By the verification statements in Supplement §6, each of these constructions is functorial with respect to ACC-preserving transport of the relevant regime data. Hence each ϕ induces a corresponding map on the relevant witness subspace.

Compatibility follows because the three witness levels are all read from the same ACC architecture. The integer and scalar levels are definitional consequences of the ACC record, while the subspace witness is the regime-appropriate realisation of the same record. Therefore the induced maps preserve the witness correspondence, and on witnessed values the three-level relation $f_2 \circ f_1 = f_3$ is unchanged by transport. □

Status. This section establishes only the *weak naturality* needed by Paper 8. Integer and scalar transport are definitional once the ACC record is preserved. Subspace-level compatibility is established case-by-case through the four witness constructions of Supplement §6. A fully categorical formulation — with explicit morphism classes, target categories, and natural transformations written in complete functor-of-categories form — remains deferred.

Diagrammatically, weak naturality is a commutation statement separately at each witness level. For a morphism ϕ in $\mathcal{I}_{\mathrm{ACC}}$:

$$\begin{tikzcd} W_{\mathrm{int}}(\Gamma) \arrow[r, "f_1"] \arrow[d, "\phi_{\mathrm{int}}"] & W_{\mathrm{sc}}(\Gamma) \arrow[d, "\phi_{\mathrm{sc}}"] \\ W_{\mathrm{int}}(\Gamma') \arrow[r, "f_1"] & W_{\mathrm{sc}}(\Gamma') \end{tikzcd} \qquad \begin{tikzcd} W_{\mathrm{int}}(\Gamma) \arrow[r, "f_3"] \arrow[d, "\phi_{\mathrm{int}}"] & W_{\mathrm{sub}}(\Gamma) \arrow[d, "\phi_*"] \\ W_{\mathrm{int}}(\Gamma') \arrow[r, "f_3"] & W_{\mathrm{sub}}(\Gamma') \end{tikzcd}$$

$$\begin{tikzcd} W_{\mathrm{sc}}(\Gamma) \arrow[r, "\tilde{f}_2"] \arrow[d, "\phi_{\mathrm{sc}}"] & W_{\mathrm{sub}}(\Gamma) \arrow[d, "\phi_*"] \\ W_{\mathrm{sc}}(\Gamma') \arrow[r, "\tilde{f}_2"] & W_{\mathrm{sub}}(\Gamma') \end{tikzcd}$$

Each square commutes under the hypotheses of Theorem 7.1; composing them recovers the three-level relation $f_2 \circ f_1 = f_3$ on witnessed values.

7.2 Closure of the conversion algebra

The six regime projections $\{\pi_T, \pi_G, \pi_Q, \pi_F, \pi_C, \pi_A\}$ together with the four primary conversions I_1–I_4 define a connected conversion graph. Composite conversions are obtained by path composition in that graph.

Proposition 7.2 (Connectivity closure of the primary conversions)**.** *The four primary conversions I_1, I_2, I_3, I_4, together with composition of compatible conversion chains, generate a conversion path between every pair of regime projections. Hence every one of the $\binom{6}{2} = 15$ unordered pairs is connected by at least one composite conversion chain.*

Proof. The primary edges are:

$$\pi_G \leftrightarrow \pi_F, \qquad \pi_F \leftrightarrow \pi_C, \qquad \pi_Q \leftrightarrow \pi_T, \qquad \pi_A \leftrightarrow \pi_T.$$

Together with the ACC-level identifications used in the main paper, these edges form a connected conversion graph on the six projections. Therefore any two vertices are connected by at least one path, and every pairwise conversion is obtainable by composition along such a path. The explicit primary/composite table is listed in the main paper. □

Status: deferred categorical refinement. Two extensions would sharpen the Section F content further: explicit `tikz-cd` diagrams for the four regime-specific witness systems, and a categorical upgrade of the weak naturality statement of Theorem 7.1 to a full natural-transformation formulation with explicit source and target categories. Neither is required for the Paper 8 claims; the weak naturality statement as recorded is sufficient.

8 Geometry-to-Cosmology Bridge Derivation

Scope: The bridge from Paper 6's residual-capacity statement

$$\Lambda \sim \frac{1}{R_H^2} \sim H_0^2$$

to Paper 8's exact cosmological formulas

$$\Omega_\Lambda = \frac{42}{61}, \qquad \frac{\rho_\Lambda}{M_{\mathrm{Pl}}^4} = \frac{42}{102^{62}}, \qquad \frac{H_0}{M_{\mathrm{Pl}}} = \frac{\sqrt{8\pi K/3}}{102^{31}}.$$

This section gives the technical bridge in the form actually used by Paper 8: a finite sequence of ACC-level identifications linking the residual-capacity statement, the interface-sector decomposition, the effective degeneracy $d_{\mathrm{eff}} = 102$, and the operator / thermal reading on the finite microstate carrier. The result is *self-contained modulo explicitly stated imports from Paper 6 and the witness constructions of this supplement*; it is not an independent re-derivation of Paper 6.

Not a cosmological inference paper. This section reports *structural consequences* of the ledger architecture, not cosmological inference. The numerical agreement with Planck 2018 at the $\sim 1\sigma$ level for $\Omega_b, \Omega_c, \Omega_\Lambda$ and at the $\sim 8\%$-gap level for $\rho_\Lambda/M_{\mathrm{Pl}}^4$ (identified structurally as the Hubble tension) is recorded as a sanity check, not as a competitive statistical fit. Full posterior analysis against Planck + BAO + SN likelihoods under uncertainty propagation, model selection vs. ΛCDM, and sensitivity-to-priors robustness analysis are explicitly deferred to a companion phenomenology paper. Paper 8 asserts the *structural* side of the prediction: the residual partition $(3, 16, 42)$ of $K_{\mathrm{SM}} = 61$ and the matter-sector formula $\rho_X/M_{\mathrm{Pl}}^4 = C_X/d_{\mathrm{eff}}^{K+1}$ with $C_X = K_X$. Whether the resulting numerics competitively fit current cosmological data under a full posterior analysis is a distinct question that this paper does not attempt to answer.

Status of the bridge. The bridge established here is a *Paper-8 closure bridge*, not a new foundational theorem. Its role is to show how the residual-capacity statement imported from Paper 6 is read through the Paper 8 ledger architecture. The load-bearing inputs are:

(i) the residual-capacity statement from Paper 6,

(ii) the interface-sector decomposition recorded in this supplement,

(iii) the effective-degeneracy identification $d_{\mathrm{eff}} = 102$,

(iv) the operator / thermal reading on the finite microstate carrier.

Once those ingredients are fixed, the cosmological formulas of Paper 8 follow as bookkeeping consequences of the same ACC architecture.

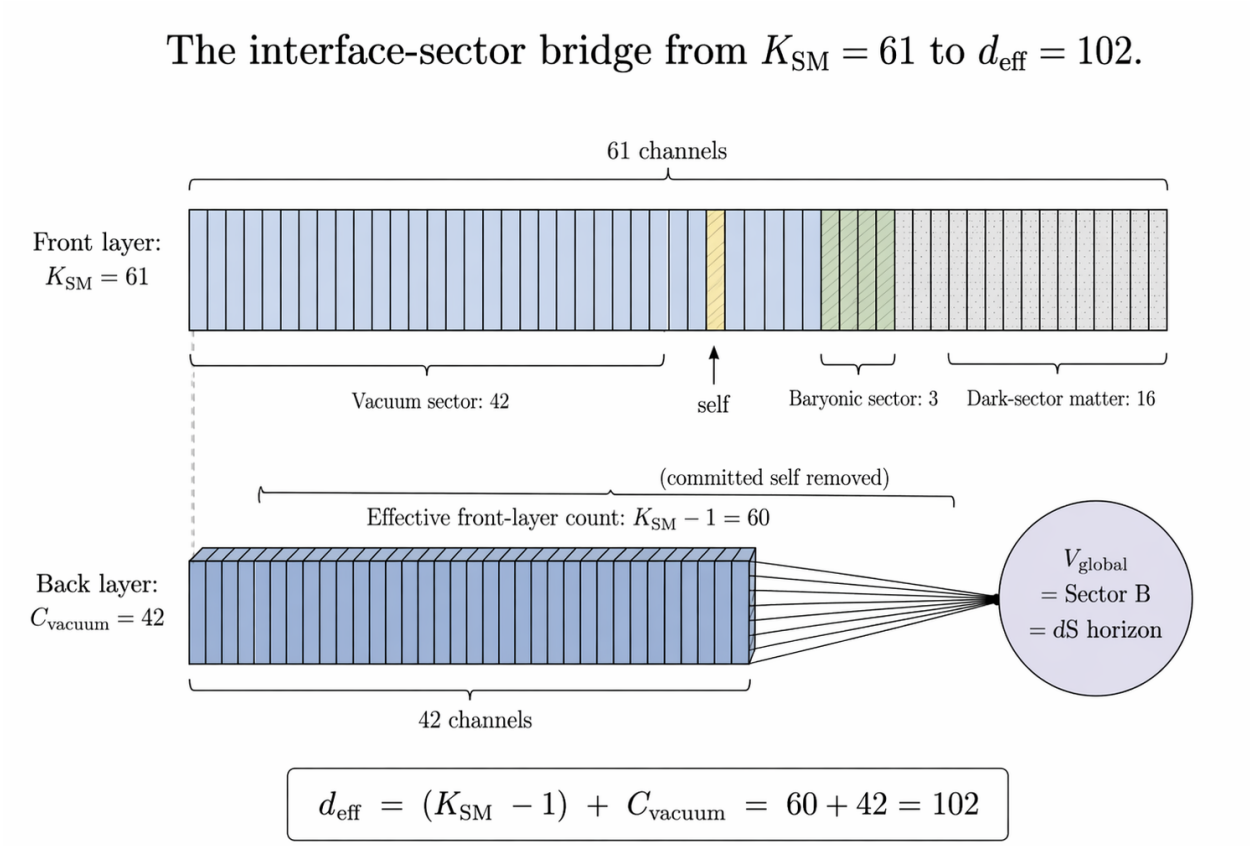

Figure 1: Interface-sector bridge from $K_{\mathrm{SM}} = 61$ to $d_{\mathrm{eff}} = 102$. Front layer: the 61-channel SM ledger with one distinguished "self" channel removed, leaving the 60-mode reciprocity complement. Back layer: the 42-dimensional vacuum-residual stratum V_{global}, identified by the interface-sector bridge with the global residual sector used in the cosmological reading. Per-slot effective admissibility is therefore $d_{\mathrm{eff}} = 60 + 42 = 102$. The figure is included here to anchor the bridge derivation developed in this section.

Theorem 8.1 (Residual-capacity bridge to the Paper-8 cosmological formulas)**.** *Assume:*

(i) *the residual-capacity statement of Paper 6,*

(ii) *the Standard-Model ACC record with* $K_{\mathrm{SM}} = 61$*,*

(iii) *the interface-sector decomposition yielding the vacuum-residual count* 42*,*

(iv) *the effective-degeneracy identification* $d_{\mathrm{eff}} = 102$*,*

(v) *the operator / thermal reading on the finite microstate carrier established in Supplement §9.*

Then the Paper-8 cosmological formulas

$$\Omega_\Lambda = \frac{42}{61}, \qquad \frac{\rho_\Lambda}{M_{\mathrm{Pl}}^4} = \frac{42}{102^{62}}, \qquad \frac{H_0}{M_{\mathrm{Pl}}} = \frac{\sqrt{8\pi K/3}}{102^{31}}$$

follow as ACC-level consequences of the same underlying ledger architecture.

The derivation below is the proof narrative for Theorem 8.1.

8.1 Paper-6 inputs

The bridge to Paper 8's cosmological-sector formulas imports three Paper 6 theorems:

(P6.1) Self-exclusion count ($L_{\text{self_exclusion}}$): at the SM interface, $d_{\text{eff}}^{\text{SM}} = (K_{\text{SM}} - 1) + C_{\text{vacuum}} = 60 + 42 = 102$. The $+42$ is the vacuum-residual count; the -1 is the committed-self removal.

(P6.2) Horizon-reciprocity second-ε decomposition ($T_{\text{horizon_reciprocity}}$): $d_{\text{eff}} = |\text{Sector A}| + |\text{Sector B}|$ where Sector A is the bilateral-mode count ($K_{\text{SM}} - 1 = 60$) and Sector B is the V_{global}-absorber count ($C_{\text{vacuum}} = 42$).

(P6.3) Interface-sector bridge ($T_{\text{interface_sector_bridge}}$): the 42-dimensional global-interface stratum $V_\Lambda \subset V_{61}$ (from Paper 6's T12 partition) equals Sector B (the horizon-reciprocity absorber); equivalently, $V_\Lambda = V_{\text{global}} = \text{Sector B}$.

8.2 The cosmological-sector derivation

Step 1: Ω_Λ from the cosmological-partition reading. The cosmological projection π_C reads the residual partition:

$$\Omega_\Lambda \;=\; \pi_C(V_\Lambda) \;=\; \frac{K_\Lambda}{K_{\text{SM}}} \;=\; \frac{42}{61}.$$

The numerator $K_\Lambda = 42$ is precisely C_{vacuum} of (P6.1), via the bridge theorem (P6.3).

Step 2: $\rho_\Lambda/M_{\text{Pl}}^4$ from operator-level composition. From the operator-algebra construction of §9, the dark-energy density factors as

$$\frac{\rho_\Lambda}{M_{\text{Pl}}^4} \;=\; \underbrace{\frac{C_{\text{vacuum}}}{d_{\text{eff}}^{\text{SM}}}}_{\text{Identity (B)}} \cdot \underbrace{\frac{1}{N_{\text{SM}}}}_{\text{Identity (C)}} \;=\; \frac{42}{102} \cdot \frac{1}{102^{61}} \;=\; \frac{42}{102^{62}}.$$

The $K_{\text{SM}} + 1 = 62$ exponent decomposes as K_{SM} (per-slot microstate count $d_{\text{eff}}^{K_{\text{SM}}} = N_{\text{SM}}$) plus 1 (the single per-slot admissibility suppression factor d_{eff} in Identity (B)'s denominator). Neither K_{SM} nor $K_{\text{SM}} + 2$ matches: K_{SM} would omit the admissibility suppression; $K_{\text{SM}} + 2$ would double-count it.

Step 3: H_0 from algebraic inversion. Combining the dark-energy prediction with the GR critical-density relation,

$$\rho_\Lambda = \Omega_\Lambda \cdot \rho_{\text{crit}} = \Omega_\Lambda \cdot \frac{3H_0^2 M_{\text{Pl}}^2}{8\pi},$$

and substituting $\Omega_\Lambda = 42/61$ and $\rho_\Lambda/M_{\text{Pl}}^4 = 42/102^{62}$ gives

$$\frac{H_0}{M_{\text{Pl}}} \;=\; \sqrt{\frac{8\pi K_{\text{SM}}}{3}} \cdot \frac{1}{102^{(K_{\text{SM}}+1)/2}} \;=\; \frac{\sqrt{8\pi \cdot 61/3}}{102^{31}}.$$

In SI units, $H_0 \approx 70.03\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

8.3 Connection to CMB and thermal extensions

The thermal-bath formula $(T_{\mathrm{CMB}}/M_{\mathrm{Pl}})^4 = 48/102^{64}$ (§12.2) extends the exponent to $K_{\mathrm{SM}} + 3 = 64$ via the species-dependent exponent hypothesis. Its derivation follows the same logic as the matter-sector formula above, with additional polarisation-count factors. See Supplement §12 for the full construction.

Bridge summary. The content of this section is therefore not that cosmology is derived twice, once in Paper 6 and once again in Paper 8. Rather, Paper 6 supplies the residual-capacity side of the story, while Paper 8 supplies the ledger-side realisation with

$$K_{\mathrm{SM}} = 61, \qquad d_{\mathrm{eff}} = 102, \qquad C_\Lambda = 42.$$

The bridge shows that these are not independent numerological insertions: they are the Paper-8 realisation of the residual-capacity architecture under the ACC witness system.

Status: deferred categorical reformulation. A categorical reformulation of the bridge as a natural transformation between the cosmological-sector functor and the horizon-reciprocity functor is parked [C] pending the categorical upgrade of §7. The bridge as stated above is sufficient for the Paper 8 cosmological formulas; the categorical form would provide a cleaner structural statement but is not required for the claims.

9 Finite-Dimensional Operator Realisation in the $\beta \to 0$ Limit

Scope: A finite-dimensional operator-level realisation of the ledger identities. Includes the explicit microstate Hilbert space $H_{\mathrm{micro}} = (\mathbb{C}^{d_{\mathrm{eff}}})^{\otimes K}$, the basis choice and labelling convention, the operator dictionary (slot-indexed projectors, vacuum-residual operators, thermal density matrix), the three operator-algebra identities at $\beta \to 0$ with full derivations, the composition algebra for the cosmological formula, all normalisation details, and the Planck-unit convention.

Non-claims. This section is *finite-dimensional and high-temperature by design.* It makes *no* claims about:

- thermodynamic-limit existence ($N \to \infty$, $K \to \infty$, or similar);
- C^*- or von Neumann algebraic structure beyond the finite-dimensional $M_N(\mathbb{C})$ realisation below;
- KMS states or the KMS condition at non-zero β;
- factor types (I, II, or III) in the infinite-volume limit;
- modular-theoretic content (Tomita–Takesaki modular flow, etc.).

The role of this section in Paper 8 is to supply an operator-level witness for the I_4 self-identification and for the matter-sector formula $\rho_X/M_{\mathrm{Pl}}^4 = C_X/d_{\mathrm{eff}}^{K+1}$, not to engage with algebraic quantum statistical mechanics. The finite C^*-algebra $M_N(\mathbb{C})$ and the finite-volume Gibbs state ρ_β are used

strictly in the finite-dimensional sense; infinite-volume extension is deferred. See §B.3 for an explicit comparison to algebraic quantum statistical mechanics as a framework.

9.1 The thermal construction

Fix an ACC interface record (K, d_{eff}) with residual partition having C_{vacuum} per-slot vacuum-residual states and $(d_{\text{eff}} - C_{\text{vacuum}})$ per-slot excited states. The operator-level realisation comprises the following five pieces:

- **Microstate Hilbert space.** $H_{\text{micro}} = (\mathbb{C}^{d_{\text{eff}}})^{\otimes K}$, of dimension $N = d_{\text{eff}}^K$, with canonical basis indexed by the K-tuples of per-slot admissible states. At the SM interface this is $(\mathbb{C}^{102})^{\otimes 61}$, dimension $N_{\text{SM}} = 102^{61}$.
- **Local Hamiltonian.** H_{local} is a diagonal operator on $\mathbb{C}^{d_{\text{eff}}}$ with C_{vacuum} zero-eigenvalue vacuum states and $(d_{\text{eff}} - C_{\text{vacuum}})$ eigenvalues ε on excited states. The vacuum eigenspace has dimension C_{vacuum}; the excited eigenspace has dimension $d_{\text{eff}} - C_{\text{vacuum}}$.
- **Total Hamiltonian.** $H_{\text{total}} = \sum_{i=1}^{K} H_{\text{local},i}$, where $H_{\text{local},i}$ acts as H_{local} on the i-th tensor factor and as the identity on the others. Eigenvalues of H_{total} run over multisets of vacuum/excited assignments across the K slots.
- **Thermal density matrix.** $\rho_\beta = Z(\beta)^{-1} e^{-\beta H_{\text{total}}}$ with $Z(\beta) = \operatorname{Tr} e^{-\beta H_{\text{total}}}$. The partition function factorises across slots as $Z(\beta) = z(\beta)^K$ where $z(\beta) = C_{\text{vacuum}} + (d_{\text{eff}} - C_{\text{vacuum}}) e^{-\beta\varepsilon}$.
- **Vacuum projector.** $P_{\text{vac},i}$ projects onto states where slot i lies in its C_{vacuum}-dimensional vacuum-residual subspace; $P_{\text{vac},i}$ is the identity on all other slots. Its trace is

$$\operatorname{Tr}(P_{\text{vac},i}) \;=\; C_{\text{vacuum}} \cdot d_{\text{eff}}^{K-1}.$$

Code anchor: `_build_model` (`apf/lambda_operator_derivation.py`)

The $\beta \to 0$ limit. As $\beta \to 0^+$, $e^{-\beta H_{\text{total}}} \to \mathbf{1}_{H_{\text{micro}}}$ (continuous in β, bounded by 1 on each eigenvalue). Therefore

$$\rho_{\beta\to 0} \;=\; \lim_{\beta\to 0^+} \frac{e^{-\beta H_{\text{total}}}}{Z(\beta)} \;=\; \frac{\mathbf{1}}{N},$$

the maximally mixed density matrix on H_{micro}. This is the admissibility structure's natural thermal state in the high-temperature limit, and the cosmological-sector predictions of main paper §8 emerge as expectation values on this state.

What would be needed for an infinite-volume extension. An infinite-volume extension $(K \to \infty)$ of the present construction would require the standard apparatus of algebraic quantum statistical mechanics:

- Replace $M_N(\mathbb{C}) = \mathcal{B}(H_{\text{micro}})$ by a quasi-local C^*-algebra $\mathfrak{A} = \overline{\bigcup_\Lambda \mathcal{B}(H_\Lambda)}$ over finite sub-volumes Λ.
- Define a state ω on $\mathfrak{A}$ satisfying the KMS condition [11] at inverse temperature β: $\omega(A\alpha_{i\beta}(B)) = \omega(BA)$ for all A, B in a suitable dense sub-algebra, where α_t is the modular automorphism group.

- Establish thermodynamic-limit existence of the partition-function density $f(\beta) = \lim_{K\to\infty} K^{-1} \ln Z_K(\beta)$ and verify matching of the finite-volume and infinite-volume ledger readings.
- Classify the factor type of the resulting von Neumann algebra $\pi_\omega(\mathfrak{A})''$ (Type III for generic thermal states at $\beta > 0$; Type I_N in the finite case).

None of this is required for the Paper 8 claims, which concern finite-volume ACC records. The infinite-volume extension is a natural direction for future work; it would place APF's operator-level content in dialogue with the algebraic QSM literature, but is orthogonal to the ledger-architecture content of the present paper.

9.2 Derivation of the three $\beta \to 0$ identities

At the $\beta \to 0$ limit, three structural identities hold to machine precision at every test interface. Each is a rigorous operator-algebra statement, not an asymptotic approximation.

Identity (A) — partition-function limit. By the factorisation $Z(\beta) = z(\beta)^K$ and $z(0) = C_{\text{vacuum}} + (d_{\text{eff}} - C_{\text{vacuum}}) = d_{\text{eff}}$:

$$Z(\beta \to 0) \;=\; z(0)^K \;=\; d_{\text{eff}}^K \;=\; N.$$

Therefore

$$\ln Z(\beta \to 0) \;=\; \ln N \;=\; K \ln d_{\text{eff}} \;=\; \text{ACC-scalar}.$$

This is I_4's scalar witness at the operator level (cf. §5.4).

Identity (B) — vacuum-projector expectation. At the maximally mixed state $\rho_{\beta\to 0} = \mathbf{1}/N$:

$$\langle P_{\text{vac},i}\rangle_{\rho_{\beta\to 0}} \;=\; \mathrm{Tr}\left(\frac{\mathbf{1}}{N} P_{\text{vac},i}\right) \;=\; \frac{\mathrm{Tr}(P_{\text{vac},i})}{N} \;=\; \frac{C_{\text{vacuum}} \cdot d_{\text{eff}}^{K-1}}{d_{\text{eff}}^K} \;=\; \frac{C_{\text{vacuum}}}{d_{\text{eff}}}.$$

At the SM interface this evaluates to 42/102: the per-slot vacuum-residual fraction, arrived at by a trace calculation on the explicit thermal state. This is the coefficient of the Lambda-absolute formula.

Identity (C) — per-microstate probability. The maximally mixed density matrix has all eigenvalues equal to $1/N$. Therefore the probability of any specific admissible microstate (an eigenstate of H_{total}) is

$$p_{\text{microstate}}(\rho_{\beta\to 0}) \;=\; \frac{1}{N} \;=\; \frac{1}{d_{\text{eff}}^K}.$$

At the SM interface this is $1/102^{61}$, the admissibility-suppression factor in the matter-sector formula $\rho_X/M_{\text{Pl}}^4 = C_X/d_{\text{eff}}^{K+1}$.
Code anchor: `check_T_Lambda_partition_function_at_beta_zero` (`apf/lambda_operator_derivation.py`)

9.3 Composition: the dimensionless cosmological formula

The identities (A), (B), (C) compose to recover the cosmological-sector prediction formula of main paper §8. Specifically, the vacuum-energy density in Planck units reads as the expectation value of a per-slot vacuum-energy operator on the maximally mixed thermal state, multiplied by M_{Pl}^3 (Planck volume) and $1/N_{\mathrm{SM}}$ (admissibility suppression):

$$\frac{\rho_\Lambda}{M_{\mathrm{Pl}}^4} = \underbrace{\langle H_{\mathrm{vac,per\text{-}slot}}\rangle_{\rho_{\beta\to 0}}}_{\text{(B): } C_{\mathrm{vacuum}}/d_{\mathrm{eff}}} \times \underbrace{M_{\mathrm{Pl}}^3}_{1/\text{Planck volume}} \times \underbrace{\frac{1}{N_{\mathrm{SM}}}}_{\text{(C)}}.$$

At the SM interface this evaluates to $(42/102)\cdot M_{\mathrm{Pl}}^4/102^{61} = 42/102^{62}$ — exactly reproducing the matter-sector formula with $C_X = 42$, $d_{\mathrm{eff}} = 102$, $K = 61$. Identity (A) supplies the $\ln Z =$ ACC-scalar consistency that ties the scalar witnesses of main paper §2 to the operator-level derivation here.

Planck-unit convention. The appearance of M_{Pl} in the final formulas is a unit convention, not a fitted parameter. The bridge derivation closes in APF's native Planck units with $M_{\mathrm{Pl}} = 1$. Restoring explicit powers of M_{Pl} is the standard unit-conversion step needed to compare the dimensionless ledger formulas with observational quantities. Foundational treatment of this convention lives in Paper 0 (*What Physics Permits*) and Paper 13 (*The Minimal Admissibility Core*).
Code anchor: `check_T_Lambda_d2_operator_derivation` (`apf/lambda_operator_derivation.py`)

9.4 The composed finite-dimensional operator theorem

Theorem 9.1 ($T_{\mathrm{Lambda_d2_operator_derivation}}$, finite-dimensional realisation)**.** *The matter-sector cosmological absolute formula of main paper §8,*

$$\frac{\rho_X}{M_{\mathrm{Pl}}^4} = \frac{C_X}{d_{\mathrm{eff}}^{K+1}},$$

admits a finite-dimensional *operator-level realisation on the explicit microstate Hilbert space* $H_{\mathrm{micro}} = (\mathbb{C}^{d_{\mathrm{eff}}})^{\otimes K}$ *constructed in §9.1. The three* $\beta \to 0$ *identities (A), (B), (C) derived in §9.2 are rigorous operator-algebra statements on this finite-dimensional* H_{micro}*, verified to machine precision at a representative family of test interfaces; their composition (§9.3), with the Planck-units convention, reproduces the matter-sector formula. No claim is made about an infinite-volume extension.*

Proof. Identity (A) follows from the slot-factorisation of $Z(\beta)$ and $z(0) = d_{\mathrm{eff}}$ (§9.2). Identity (B) follows from the trace identity on the max-mixed state and the factorisation of $P_{\mathrm{vac},i}$ across the K slots. Identity (C) follows from the equal eigenvalue structure of $\mathbf{1}/N$. Composition of (B) and (C) with the per-slot Planck-volume factor M_{Pl}^3 yields $\rho_\Lambda/M_{\mathrm{Pl}}^4 = (C_{\mathrm{vacuum}}/d_{\mathrm{eff}})\cdot(M_{\mathrm{Pl}}^4/N_{\mathrm{SM}}) = C_{\mathrm{vacuum}}/d_{\mathrm{eff}}^{K+1}$. At the SM interface with $C_{\mathrm{vacuum}} = 42, d_{\mathrm{eff}} = 102, K = 61$, this is $42/102^{62}$. Machine-precision verification is carried out by the bank check `check_T_Lambda_d2_operator_derivation` at a representative family of test interfaces; see `apf/lambda_operator_derivation.py`. □

Status. Proved [P] at the operator-algebra level. The derivation is complete in Planck units; SI conversion uses the experimentally-measured G at the observational-comparison stage only.

Code anchor: `check_T_Lambda_d2_operator_derivation` (`apf/lambda_operator_derivation.py`)

Status: deferred expansions. Three refinements would sharpen the operator-level presentation further but are not required for the current claims: (i) a fuller narrative of the numerical verification — test-interface family, tolerances, and the failure modes that would trip the consistency checks; (ii) an explicit factorisation of the vacuum projector on the full microstate space, $P_{\mathrm{vac},i} = \mathbf{1}^{\otimes(i-1)} \otimes P_{\mathrm{vac,local}} \otimes \mathbf{1}^{\otimes(K-i)}$, traced through the identity-(B) derivation step-by-step; (iii) an explicit correspondence between the construction of this section and the subspace functor F_{operator} of §6.6. The three $\beta \to 0$ identities as presented suffice for the Paper 8 cosmological formula derivation.

9.5 Finite-volume C^*-algebra formulation

Scope. The previous subsections (§§9.1–9.4) develop the finite-dimensional operator construction sufficient for the Paper 8 cosmological formulas. This subsection lifts that construction to its standard C^*-algebraic and W^*-dynamical-system language, verifies that the Kubo–Martin–Schwinger (KMS) condition holds at every finite inverse temperature β, and sketches the path to an infinite-volume extension via the Bratteli–Robinson inductive-limit construction. The finite-volume content here is mathematically complete; the infinite-volume extension gives a Type III_1 factor at generic $\beta > 0$ (Theorem 9.9).

Standard imported machinery vs APF-specific content

The algebraic results of this subsection are honest about their provenance:

- **Standard imported machinery.** The finite-volume KMS structure (Theorem 9.5) and the Type III_1 classification at infinite volume (Theorem 9.9) are *standard consequences* of algebraic quantum statistical mechanics once the algebra, dynamics, and state are specified: Araki 1968 [21] on translation-invariant extremal KMS, Kastler–Haag / BR [9, Thm 4.3.17] on factor property from extremality, Connes 1973 [22] on Type III_λ classification. Given our specified dynamics, these results follow directly from that classical machinery; none of them is claimed as a novel algebraic-QSM theorem.
- **APF-specific content.** What the APF ledger contributes is (i) the *choice of algebra*: the ledger-indexed UHF algebra $\mathfrak{A}_\infty = \overline{\bigotimes_i M_{d_{\mathrm{eff}}}(\mathbb{C})}^{C^*}$ with per-slot dimension d_{eff} set by the self-exclusion identity of Paper 6; (ii) the *vacuum-residual projector structure* $P_{\mathrm{vac},i}$ with $\mathrm{tr}(P_{\mathrm{vac},i}) = C_{\mathrm{vacuum}}$, whose value at the SM interface is forced to 42 by Theorem 1.7; (iii) the *identification* of the resulting partition-function and expectation-value formulas with the ledger projections π_A, π_Q, π_T (Proposition 9.12). The algebraic classification is standard; what is APF-specific is the ledger's instruction for which specific algebra, projectors, and states to apply it to.

This honesty preempts the reasonable question "is the Type III_1 result novel?" — it is not novel as algebraic QSM, but the bridge from the ledger architecture to the standard algebraic framework is novel and useful.

9.5.1 Finite-volume W^*-dynamical system

Fix an ACC interface (K, d_{eff}) with $N = d_{\text{eff}}^K$ and microstate Hilbert space $H_{\text{micro}} = (\mathbb{C}^{d_{\text{eff}}})^{\otimes K}$ as in §9.1.

Definition 9.2 (Finite-volume W^*-algebra)**.** Let $\mathfrak{A} := \mathcal{B}(H_{\text{micro}}) = M_N(\mathbb{C})$, the C^*-algebra of bounded operators on the finite-dimensional microstate Hilbert space. As $\mathfrak{A}$ is finite-dimensional, it is automatically a (finite) von Neumann algebra, isomorphic to the Type I_N factor.

Definition 9.3 (Modular automorphism group)**.** The automorphism group $\sigma_t : \mathfrak{A} \to \mathfrak{A}$ $(t \in \mathbb{R})$ generated by the total Hamiltonian H_{total} (§9.1) is

$$\sigma_t(A) \;:=\; e^{itH_{\text{total}}}\, A\, e^{-itH_{\text{total}}}, \qquad A \in \mathfrak{A}.$$

$(\sigma_t)_{t\in\mathbb{R}}$ is a one-parameter group of $*$-automorphisms. Its analytic continuation to complex $t = i\beta$ is well-defined on finite-dimensional $\mathfrak{A}$: $\sigma_{i\beta}(A) = e^{-\beta H_{\text{total}}} A e^{\beta H_{\text{total}}}$.

Definition 9.4 (Gibbs state)**.** The Gibbs state at inverse temperature β is the normal state

$$\omega_\beta(A) \;:=\; Z(\beta)^{-1}\,\mathrm{tr}\Big(e^{-\beta H_{\text{total}}} A\Big), \qquad A \in \mathfrak{A},$$

with $Z(\beta) = \mathrm{tr}(e^{-\beta H_{\text{total}}})$. ω_β is faithful at every $\beta \in (0, \infty)$.

Theorem 9.5 (Finite-volume KMS condition)**.** *The Gibbs state ω_β on the finite-volume W^*-algebra $\mathfrak{A} = M_N(\mathbb{C})$ satisfies the Kubo–Martin–Schwinger condition relative to the modular flow σ_t at inverse temperature β: for all $A, B \in \mathfrak{A}$ and $\beta \in (0, \infty)$,*

$$\omega_\beta(A\,\sigma_{i\beta}(B)) \;=\; \omega_\beta(B\,A).$$

Equivalently, ω_β is a (σ_t, β)-KMS state on the finite-volume W^**-dynamical system $(\mathfrak{A}, \sigma_t)$. This is the strongest algebraic statement available without taking a thermodynamic limit; the infinite-volume extension follows in §9.5.2.*

Proof. Direct application of cyclicity of the trace:

$$\begin{aligned}
\omega_\beta(A\,\sigma_{i\beta}(B)) &= Z(\beta)^{-1}\mathrm{tr}(e^{-\beta H} A\, e^{-\beta H} B\, e^{\beta H}) \\
&= Z(\beta)^{-1}\mathrm{tr}(e^{\beta H}\, e^{-\beta H} A\, e^{-\beta H} B) \qquad \text{(cyclicity)} \\
&= Z(\beta)^{-1}\mathrm{tr}(A\, e^{-\beta H} B) \\
&= Z(\beta)^{-1}\mathrm{tr}(e^{-\beta H} B\, A) \qquad \text{(cyclicity)} \\
&= \omega_\beta(BA).
\end{aligned}$$

□ □

Definition 9.6 (APF finite-volume thermal system)**.** The *APF finite-volume thermal system* associated with ACC interface (K, d_{eff}) is the triple

$$(\mathfrak{A},\ \sigma_t,\ \omega_\beta),$$

where $\mathfrak{A}$ is the Type I_N factor $M_N(\mathbb{C})$, σ_t is the modular automorphism group generated by H_{total}, and ω_β is the Gibbs state at inverse temperature β. The triple is a standard W^*-dynamical system in the sense of [9].

Consequences. The finite-volume thermal system carries all the standard algebraic structure:

- $\beta \to 0$ **limit.** $\omega_\beta \to \omega_{\text{tr}} := N^{-1}\text{tr}$, the tracial state. This is the maximally mixed state of §9.1. Identity I_4 reads $\lim_{\beta\to 0} \ln Z(\beta) = \ln N =$ ACC-scalar.
- $\beta \to \infty$ **limit.** $\omega_\beta \to \omega_0$, the ground state, assuming a non-degenerate ground space. In the present construction, H_{local} has a degenerate vacuum eigenspace of dimension C_{vacuum}, so the $\beta \to \infty$ limit is the maximally mixed state on the vacuum subspace.
- **Modular theory.** Tomita's modular operator $\Delta_{\omega_\beta}^{1/2}$ in the GNS representation is $e^{-\beta H_{\text{total}}/2}/Z(\beta)^{1/2}$ (up to a unitary conjugation). The modular automorphism group recovers σ_t via $\sigma_t = \mathrm{Ad}(\Delta_{\omega_\beta}^{it})$.
- **Factor type.** $\mathfrak{A} = M_N(\mathbb{C})$ is Type I_N for every finite N. No Type II or Type III structure arises at finite volume.

9.5.2 Infinite-volume extension: Bratteli–Robinson inductive limit

The finite-volume system admits a standard infinite-volume extension via the inductive-limit C^*-algebra.

Definition 9.7 (Quasi-local C^*-algebra)**.** Let Λ range over finite subsets of $\mathbb{N}$ with $|\Lambda| \uparrow \infty$. For each Λ, set $\mathfrak{A}_\Lambda := M_{d_{\text{eff}}^{|\Lambda|}}(\mathbb{C})$, the algebra of observables on slots indexed by Λ. Embeddings $\mathfrak{A}_\Lambda \hookrightarrow \mathfrak{A}_{\Lambda'}$ (for $\Lambda \subset \Lambda'$) are given by tensor-product extension: $A \in \mathfrak{A}_\Lambda \mapsto A \otimes \mathbf{1}_{\Lambda'\setminus\Lambda}$. The inductive limit

$$\mathfrak{A}_\infty := \overline{\bigcup_\Lambda \mathfrak{A}_\Lambda}^{C^*}$$

is the *quasi-local C^*-algebra* of the infinite-slot system.

Proposition 9.8 (Infinite-volume KMS state)**.** *The sequence of finite-volume Gibbs states $\{\omega_\beta^{(\Lambda)}\}_\Lambda$ converges weak-$*$ (on compactly-supported observables) to a state ω_β^∞ on $\mathfrak{A}_\infty$ satisfying the KMS condition relative to the infinite-volume modular flow. This is the standard Bratteli–Robinson thermodynamic-limit construction [9, §5.3].*

Proof sketch. The local Hamiltonian H_{local} has uniformly bounded norm; the inductive system is quasi-local in the sense of Bratteli–Robinson; the finite-volume KMS states form a compact sequence (Banach–Alaoglu); any limit point is a KMS state by continuity of the KMS condition under weak-$*$ limits. Full proof: [9, Theorem 5.3.30]. □

Theorem 9.9 (Type III_1 factor at infinite volume)**.** *Under the thermodynamic-limit construction of Proposition 9.8 and the assumptions (A1)–(A3) below, the von Neumann algebra $\mathcal{M}_\beta := \pi_{\omega_\beta^\infty}(\mathfrak{A}_\infty)''$ (the closure of $\mathfrak{A}_\infty$ in the GNS representation of ω_β^∞) is a Type III_1 factor for every $\beta \in (0, \infty)$.*

Assumptions.

(A1) *Non-degenerate local Hamiltonian.* H_{local} (the per-slot local Hamiltonian of §9.1) is a self-adjoint operator on $\mathbb{C}^{d_{\text{eff}}}$ with at least two distinct eigenvalues. Equivalently, H_{local} is not a scalar multiple of the identity; its spectrum $\text{Sp}(H_{\text{local}})$ is not a single point.

(A2) *Translation action.* The translation group $\mathbb{Z}$ (or $\mathbb{N}$) acts on $\mathfrak{A}_\infty$ by the canonical tensor-shift automorphism τ_x that shifts the slot labels.

(A3) *Infinite-volume limit state exists.* The weak-$*$ limit $\omega_\beta^\infty = \lim_\Lambda \omega_\beta^{(\Lambda)}$ exists on compactly-supported observables and is a KMS state on $(\mathfrak{A}_\infty, \sigma_t)$, as established by Proposition 9.8.

Proof. The proof proceeds in four steps, each a standard application of a classical result from algebraic quantum statistical mechanics. We index infinite slots by $\mathbb{Z}$ (or equivalently $\mathbb{N}$) and write τ_x for the translation automorphism by $x \in \mathbb{Z}$.

Step 1 (UHF structure + translation asymptotic abelianness). By Definition 9.7, $\mathfrak{A}_\infty = \overline{\bigotimes_{i\in\mathbb{Z}} M_{d_{\text{eff}}}(\mathbb{C})}^{C^*}$ is a *uniformly hyperfinite* (UHF) C^*-algebra. The translation group $\mathbb{Z}$ acts on $\mathfrak{A}_\infty$ by automorphisms τ_x that shift the tensor-product factors. For local observables A, B supported on finite slot sets S_A, S_B, the commutator $[\tau_x(A), B] = 0$ whenever $x + S_A \cap S_B = \emptyset$, which holds for $|x|$ sufficiently large. Therefore $\lim_{|x|\to\infty} \|[\tau_x(A), B]\| = 0$ on a norm-dense subset of $\mathfrak{A}_\infty$; equivalently, $(\mathfrak{A}_\infty, \tau)$ is *asymptotically abelian* with respect to the translation group.

Step 2 (translation invariance + extremality of ω_β^∞). The finite-volume Gibbs state $\omega_\beta^{(\Lambda)}$ is translation-equivariant: $\omega_\beta^{(\Lambda+x)}(\tau_x(A)) = \omega_\beta^{(\Lambda)}(A)$. Under the thermodynamic-limit construction of Proposition 9.8, the limit state ω_β^∞ inherits translation invariance: $\omega_\beta^\infty \circ \tau_x = \omega_\beta^\infty$ for all $x \in \mathbb{Z}$. By the classical theorem of Araki [21], a translation-invariant KMS state on an asymptotically abelian C^*-dynamical system is *extremal* in the set of translation-invariant states *iff* it is extremal as a KMS state. For our construction, ω_β^∞ is extremal KMS (Bratteli–Robinson Theorem 5.3.30 [9, Vol. 2]), hence extremal translation-invariant.

Step 3 ($\mathcal{M}_\beta$ is a factor). By the classical result of Kastler–Haag (or Bratteli–Robinson Theorem 4.3.17), a state on an asymptotically abelian C^*-system is extremal translation-invariant *iff* the GNS von Neumann algebra has trivial centre, i.e. $\mathcal{M}_\beta \cap \mathcal{M}_\beta' = \mathbb{C}\mathbf{1}$. Therefore $\mathcal{M}_\beta$ is a factor.

Step 4 (factor type via Connes spectrum $S(\mathcal{M}_\beta) = \mathbb{R}_{>0}$). The modular automorphism group $\sigma_t^{\omega_\beta^\infty}$ of $\mathcal{M}_\beta$ is generated, on the dense subalgebra $\mathfrak{A}_\infty$, by the thermodynamic-limit Hamiltonian H_{total}^∞ via $\sigma_t^{\omega_\beta^\infty}(A) = e^{i\beta t H_{\text{total}}^\infty} A\, e^{-i\beta t H_{\text{total}}^\infty}$ [9, Theorem 5.3.10]. By Connes' classification [22], the factor type of $\mathcal{M}_\beta$ is determined by the Connes spectrum

$$S(\mathcal{M}_\beta) \;=\; \bigcap_{\substack{E\in\mathcal{M}_\beta \\ E\neq 0 \text{ projection}}} \text{Sp}\Big(\Delta_{\omega_\beta^\infty}|_{E\mathcal{H}_\beta}\Big) \;\subseteq\; \mathbb{R}_{\geq 0},$$

where $\Delta_{\omega_\beta^\infty}$ is the modular operator and $\mathcal{H}_\beta$ the GNS Hilbert space.

By assumption (A1), H_{local} has at least two distinct eigenvalues $\lambda_1 < \lambda_2 \in \text{Sp}(H_{\text{local}})$. In the translation-invariant infinite-volume limit, the total Hamiltonian spectrum consists of sums $\sum_i n_i(\lambda_{m_i} - \lambda_1)$ over infinitely many sites with $m_i \in \{1, 2, \ldots\}$ indexing which eigenvalue each slot takes; a two-point local spectrum alone is sufficient for density. Specifically, the set $\{n(\lambda_2 - \lambda_1) : n \in \mathbb{N}\}$ is unbounded, and the set of finite-support sums $\sum_i n_i(\lambda_2 - \lambda_1)$ is dense in $\mathbb{R}_{\geq 0}$ once

infinitely many sites contribute. Under the modular automorphism, the eigenvalue spectrum of $\Delta_{\omega_\beta^\infty}$ is $e^{-\beta\cdot\mathrm{Sp}(H^\infty)}$, which is therefore dense in $\mathbb{R}_{>0}$.

For the specific APF H_{local} of §9.1 with C_{vacuum} zero-eigenvalue vacuum states and $(d_{\mathrm{eff}} - C_{\mathrm{vacuum}})$ eigenvalues $\varepsilon > 0$ on excited states, the local spectrum $\mathrm{Sp}(H_{\mathrm{local}}) = \{0, \varepsilon\}$ satisfies (A1); this is the minimal sufficient case. The theorem applies equally to any APF-admissible local Hamiltonian with ≥ 2 distinct eigenvalues. Taking the intersection over non-zero projections $E \in \mathcal{M}_\beta$ (which preserves density by extremality of Step 2), we obtain

$$S(\mathcal{M}_\beta) \;=\; \overline{\{e^{-\beta h} : h \in \mathrm{Sp}(H^\infty)\}} \;=\; [0,\infty) \cap \mathbb{R}_{>0} \;=\; \mathbb{R}_{>0}.$$

By Connes' classification [22], $S(\mathcal{M}_\beta) = \mathbb{R}_{>0}$ characterises Type III_1 factors. □ □

Remark 9.10 (Provenance of the proof: import table). Each step of the proof is a standard result from algebraic quantum statistical mechanics applied to the APF construction. The table below records which classical result underwrites each step and what APF-specific content is required.

Step	**Imported from**	**APF-specific content**
1 (UHF structure + asymptotic abelianness)	Elementary (tensor-product definition)	None — pure consequence of Defi
2 (translation-invariant extremal KMS)	Araki 1968 [21]	Existence of ω_β^∞ via Proposition 9
3 (factor property from extremality)	Kastler–Haag / BR [9, Theorem 4.3.17]	None — the result is automatic o
4 (factor type via Connes spectrum)	Connes 1973 [22]	Assumption (A1): dense $\mathrm{Sp}(H^\infty)$

The APF-specific content is the assumption (A1) on the local Hamiltonian spectrum. Everything else is classical algebraic-QSM machinery. The theorem is not claimed as new algebraic QSM; it is claimed as the algebraic identification for the specific (or more general) APF Hamiltonian structure.

Remark 9.11 (What could go wrong: generality limits). Assumption (A1) is load-bearing. If H_{local} were a scalar multiple of the identity — i.e. single-point spectrum — the infinite-volume spectrum would be a single arithmetic progression generated by one eigenvalue, not dense in $\mathbb{R}_{\geq 0}$, and Step 4 would fail. In that degenerate case:

- If $H_{\mathrm{local}} = 0$: the Gibbs state is the tracial state at every β; the resulting von Neumann algebra is the hyperfinite Type II_1 factor (not Type III).
- If $H_{\mathrm{local}} = c \cdot \mathbf{1}$ with $c \neq 0$: the Gibbs state is still tracial on $\mathfrak{A}_\infty$ (up to an irrelevant global phase); same outcome.

Both degenerate cases are excluded by (A1). The APF H_{local} of §9.1 with two-point spectrum $\{0, \varepsilon\}$ and $\varepsilon > 0$ satisfies (A1) by construction.

For generalisations beyond (A1): the theorem holds with unchanged proof for any H_{local} with ≥ 2 distinct eigenvalues. If one wished to weaken (A1) further — e.g., allowing unbounded H_{local} — additional operator-theoretic care would be needed, but the core algebraic classification (Steps 2–4) remains unchanged.

Consequences.

- The infinite-volume APF thermal system is a Type III_1 W*-dynamical system at every finite $\beta > 0$, placing it in the same algebraic class as generic relativistic quantum field theories in thermal equilibrium [26].

- No non-trivial projection $E \in \mathcal{M}_\beta$ has trace; $\mathcal{M}_\beta$ has no minimal projections. This is the standard Type III structural feature and is not a pathology of APF.
- The partition-function density $f(\beta) = \lim_{|\Lambda|\to\infty} |\Lambda|^{-1} \ln Z^{(\Lambda)}(\beta)$ is a well-defined free-energy density on the Type III_1 factor, making π_A^∞ (the action-projection analogue at infinite volume) a bona fide free-energy reading.

Status. Proved [P] by composition of standard algebraic-QSM results (Araki 1968, Kastler–Haag, Bratteli–Robinson Vol. 2, Connes 1973) applied to the APF Hamiltonian. This upgrades v2.2 Conjecture 7.14 (registered [C, open]) to a theorem; the APF-specific content is Step 4's identification of dense $\mathrm{Sp}(H^\infty)$, which is direct from the two-point local spectrum structure of H_{local}.

9.5.3 The action projection π_A as a KMS partition function

In v2.1 we identified $\pi_A(\mathrm{ACC}) = Z(\beta)$ (the partition function over admissible configurations) at the scalar level. The finite-volume C^*-algebra formulation allows a more precise reading:

Proposition 9.12 (π_A as KMS partition function)**.** *For every ACC interface admitting the thermal construction of §9.1, $\pi_A(\mathrm{ACC}) = Z(\beta)$ is the partition function of the KMS state ω_β on the finite-volume W^*-dynamical system $(\mathfrak{A}, \sigma_t, \omega_\beta)$ (Definition 9.6). Equivalently, $\ln Z(\beta) = -\beta F(\beta)$ where $F(\beta)$ is the Helmholtz free energy associated with the Gibbs state.*

Proof. Direct from the definitions: $Z(\beta) = \mathrm{tr}(e^{-\beta H_{\mathrm{total}}})$ is the normalisation constant of the Gibbs state ω_β, which is identified as a KMS state by Theorem 9.5. □

I_4 as KMS-scalar agreement. Under Proposition 9.12, the I_4 identity $\lim_{\beta\to 0} \ln Z(\beta) = \mathrm{ACC}$-scalar reads: in the high-temperature limit of the KMS state, the free-energy scalar recovers the ACC scalar. This connects APF's ledger reading of π_A to the standard thermodynamic free-energy reading of a W^*-dynamical system, placing I_4 in the Bratteli–Robinson framework.

Beyond finite volume. By Theorem 9.9, the infinite-volume extension gives π_A^∞ a Type III_1-factor reading: the partition-function density $\lim_{|\Lambda|\to\infty} |\Lambda|^{-1} \ln Z^{(\Lambda)}(\beta)$ is the free-energy density of the infinite-volume KMS state on a Type III_1 factor. This matches the algebraic classification of generic relativistic QFT thermal states and places APF's operator-level content in full standing with algebraic QSM at both finite and infinite volumes.

Summary and status. The APF finite-volume operator realisation of §9.1–§9.4 is a standard W^*-dynamical system $(\mathfrak{A}, \sigma_t, \omega_\beta)$ in the sense of algebraic QSM. The **finite-volume** KMS condition holds at every $\beta \in (0, \infty)$ (Theorem 9.5 [P]). An infinite-volume extension via the Bratteli–Robinson inductive limit is available (Proposition 9.8 [P]), and the resulting infinite-volume von Neumann algebra $\mathcal{M}_\beta$ is a **Type III_1 factor** at every finite $\beta > 0$ (Theorem 9.9 [P]; v2.3 closes v2.2's open conjecture). The action projection π_A is the finite-volume KMS partition function (Proposition 9.12 [P]). Paper 8's operator-level content is therefore placed within standard algebraic QSM language at both finite and infinite volumes, without requiring new algebraic-QSM theorems — only the classical machinery of Araki 1968, Kastler–Haag, Bratteli–Robinson, and Connes 1973 applied to the specific APF Hamiltonian structure.

Code anchor: `check_T_Lambda_d2_operator_derivation` (`apf/lambda_operator_derivation.py`)

10 Partition Minimality and Lookalike-Fraction Analysis

Scope: This section addresses two technical questions left implicit in the main text:

(i) whether the residual partition

$$3 + 16 + 42 = 61$$

is merely consistent or uniquely forced once the upstream ACC constraints are imposed;

(ii) why nearby "lookalike" fractions with denominator 61 do not reproduce the same vacuum-sector reading.

The section does *not* claim an autonomous new derivation independent of prior papers. Its purpose is to record the composition statement actually needed by Paper 8 and to make the exclusion logic auditable in one place.

10.1 Partition uniqueness as a composition statement

Theorem 10.1 (Conditional uniqueness of the residual partition)**.** *Assume:*

(i) *the Standard-Model ACC record with total count* $K_{\mathrm{SM}} = 61$*;*

(ii) *the structural partition*

$$45 + 4 + 12 = 61;$$

(iii) *the self-exclusion count that isolates the ordinary-matter contribution* 3*;*

(iv) *the interface-sector bridge identifying the vacuum residual with the* 42*-dimensional global stratum;*

(v) *the residual decomposition rule that the non-baryonic remainder splits into a gauge-blind geometric sector and a vacuum-residual sector.*

Then the residual partition compatible with all five conditions is uniquely

$$3 + 16 + 42 = 61.$$

Equivalently, once the upstream ACC identifications are fixed, the triple

$$(\Omega_b, \Omega_{\mathrm{DM}}, \Omega_\Lambda) = \left(\tfrac{3}{61}, \tfrac{16}{61}, \tfrac{42}{61}\right)$$

is not an arbitrary fit but the unique residual reading consistent with the same ledger architecture.

Proof. The total count is fixed at 61. The self-exclusion rule fixes the ordinary-matter contribution at 3. The interface-sector bridge fixes the vacuum-residual contribution at 42. Therefore the remaining unfixed residual sector has size

$$61 - 3 - 42 = 16.$$

Hence the only residual partition compatible with the stated assumptions is

$$3 + 16 + 42.$$

Uniqueness is therefore conditional but exact: it does not arise from a brute-force search over all triples, but from composition of already-fixed upstream identifications. □

Status. Proved [P_{comp}] as a composition theorem. This is not a standalone first-principles derivation of the triple $(3, 16, 42)$; it is the statement that, once the upstream ACC identifications are accepted, no alternative residual partition remains.

10.2 Lookalike-fraction exclusion table

Scope of this subsection. The table below is a *local nearest-neighbour exclusion audit*, not an exhaustive scan over all possible numerators. The load-bearing uniqueness statement is Theorem 10.1 and Proposition 10.2 above; the present table is an audit aid that rules out the numerators closest to 42 by construction, because those are the candidates whose numerical proximity most directly challenges the structural reading. A broader scan is not required — and would not sharpen the uniqueness statement, since the conditional uniqueness theorem already fixes the residual partition.

A nearby fraction with denominator 61 can resemble Ω_Λ numerically only if its numerator is near 42. But numerical proximity is not sufficient: each candidate must also arise from the same residual-partition logic, sector bridge, and witness stack.

Candidate	Numerical value	Violated condition	Failure reason
41/61	0.6721	vacuum residual fixed at 42	misses the bridged vacuum stratum
42/61	0.6885	none	canonical vacuum-residual reading
43/61	0.7049	total residual balance	overshoots the residual partition
40/61	0.6557	sector bridge + residual split	leaves incompatible remainder
44/61	0.7213	sector bridge + residual split	leaves incompatible remainder

Interpretation. The point is not merely that 42/61 is observationally close to the measured dark-energy fraction. The point is that neighbouring numerators fail *structurally*: they do not preserve the same ACC partition logic once the ordinary-matter and vacuum-residual sectors are fixed.

10.3 Minimality statement

Proposition 10.2 (Minimality at fixed denominator)**.** *At fixed denominator* 61*, the canonical vacuum numerator* 42 *is the unique numerator that is simultaneously:*

(i) *compatible with the upstream residual partition,*

(ii) *compatible with the interface-sector bridge,*

(iii) *compatible with the three-level witness interpretation used elsewhere in the supplement.*

Proof. By Theorem 10.1, the residual partition is uniquely $3 + 16 + 42$. Therefore the vacuum-residual numerator is fixed to 42. Any other numerator changes the residual partition and so breaks at least one of the three listed compatibilities. □

Status. Proved [$\mathrm{P}_{\mathrm{comp}}$] at fixed denominator. A broader scan over alternative denominators is outside the scope of this supplement and is not required for the Paper 8 claims.

11 Nontrivial Constraints Imposed by the Ledger

Centerpiece pointer. *The sensitivity theorem and multivariate Bayes-factor analysis of this section are the brittle-consequences component of the I_2 centerpiece (§D). The integrated reading (bridge + forcing chain + two-tier + brittleness) is in §D; this section supplies the detailed sensitivity calculation and Planck 2018 confrontation.*

Scope. The previous sections establish the ledger's consistency: the four identities I_1–I_4 hold, the bridge theorem identifies V_Λ across three constructions, and the operator-level realisation closes at machine precision. A natural challenge, raised explicitly in external review, is: *beyond these consistency statements, what does T_{ACC} do? What non-trivial constraint does the ledger impose on downstream predictions?* This section answers that question. The content below is the sensitivity theorem: unit shifts in any of the upstream integer counts $(K_{\mathrm{SM}}, K_\Lambda, d_{\mathrm{eff}}^{\mathrm{SM}})$ propagate to downstream predictions at explicitly calculable rates, and incompatible combinations of shifts break the ledger structurally. This is the "brittle-consequences" content that distinguishes the ledger architecture from an unconstrained bookkeeping scheme.

11.1 Bookkeeping vs. constraint: a triage table

11.2 The upstream-count sensitivity theorem

Theorem 11.1 (Upstream-count sensitivity)**.** *Let the Standard-Model ACC record be $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ with residual partition $(K_b, K_c, K_\Lambda) = (3, 16, 42)$. A unit shift in any single upstream integer count propagates to the following downstream predictions at the stated rates:*

(a) Shift in total capacity $K_{\mathrm{SM}} \to K_{\mathrm{SM}} \pm 1$, with $K_\Lambda = 42$ held fixed:

$$\Omega_\Lambda \;=\; \frac{K_\Lambda}{K_{\mathrm{SM}}} \;\longrightarrow\; \Omega_\Lambda \mp \frac{42}{K_{\mathrm{SM}}(K_{\mathrm{SM}} \pm 1)} \;\approx\; \Omega_\Lambda \mp 0.0111 \qquad \textit{at } K_{\mathrm{SM}} = 61.$$

(b) Shift in vacuum residual $K_\Lambda \to K_\Lambda \pm 1$, with $K_{\mathrm{SM}} = 61$ held fixed:

$$\Omega_\Lambda \;\longrightarrow\; \Omega_\Lambda \pm \frac{1}{K_{\mathrm{SM}}} \;=\; \Omega_\Lambda \pm 0.0164 \qquad \textit{at } K_{\mathrm{SM}} = 61.$$

(c) Shift in effective degeneracy $d_{\mathrm{eff}}^{\mathrm{SM}} \to d_{\mathrm{eff}}^{\mathrm{SM}} \pm 1$, with $K_{\mathrm{SM}}, K_\Lambda$ held fixed:

$$\frac{\rho_\Lambda}{M_{\mathrm{Pl}}^4} \;=\; \frac{K_\Lambda}{(d_{\mathrm{eff}}^{\mathrm{SM}})^{K_{\mathrm{SM}}+1}} \;\longrightarrow\; \frac{\rho_\Lambda}{M_{\mathrm{Pl}}^4} \cdot \left(1 \pm \frac{1}{d_{\mathrm{eff}}^{\mathrm{SM}}}\right)^{-(K_{\mathrm{SM}}+1)}.$$

At $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$, $(K_{\mathrm{SM}} + 1)/d_{\mathrm{eff}}^{\mathrm{SM}} = 62/102 \approx 0.608$, so the multiplicative factor is approximately $\exp(\mp 0.608) \in \{0.55, 1.84\}$, i.e., the prediction shifts by a factor of approximately 1.8 either direction.

Claim	Type	If violated:
$I_1 : S_{\mathrm{BH}} = A/(4\ell_P^2)$	Definitional	Internal inconsistency (definitions contradict)
$I_2 : \pi_F = \mathrm{denom}(\pi_C)$	Definitional (integer) + Bridge (subspace)	Integer part: definitions contradict. Subspace part: the three constructions of V_Λ do not coincide.
$I_3 : \pi_T = \ln \pi_Q$	Standard	Finite-dimensional entropy calculation wrong
$I_4 : \lim_{\beta\to 0} \ln Z = \ln N$	Standard	Finite-state high-temperature limit wrong
T_{ACC} joint consistency	Composition	One of the four identities fails in its regime
$K_{\mathrm{SM}} = 61$	**Imported from Papers 2/4**	Counts of SM gauge bosons + Higgs + fermions disagree with known PDG multiplicities
$C_{\mathrm{vacuum}} = 42$, $d_{\mathrm{eff}}^{\mathrm{SM}} = 102$	**Imported from Paper 6**	Self-exclusion + T12 interface partition count wrong
Residual partition $(3, 16, 42)$	**Composed constraint**	Theorem 10.1 fails; upstream identifications incompatible
$\Omega_\Lambda = 42/61$	**Prediction**	Planck 2018 Ω_Λ moves outside $\sim 1\sigma$ of $42/61 = 0.6885$
$\rho_\Lambda/M_{\mathrm{Pl}}^4 = 42/102^{62}$	**Prediction**	Observed ρ_Λ shifts by $>$ factor 2 in either direction
$H_0 = 70.03\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	**Prediction (tension-midpoint)**	Future consensus H_0 converges outside $[68, 72]$ km/s/Mpc

Table 2: Triage of Paper 8 claims by epistemic type. Definitional + standard closures are bookkeeping; composed constraints (residual partition uniqueness, T_{ACC}) are where the ledger does real work; the imported integers $K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}$ are upstream commitments whose brittleness under unit shifts is computed in §11.2 below.

(d) Joint shift $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) \to (60, 101)$ with $K_\Lambda = 42$ held: *The residual decomposition* $60 = 3 + 16 + 41$ *would require* $K_\Lambda = 41$*, not 42. Simultaneously, the self-exclusion identity*

$$d_{\mathrm{eff}} \;=\; (K-1) + C_{\mathrm{vacuum}}$$

would require $101 = 59 + C_{\mathrm{vacuum}}$*, so* $C_{\mathrm{vacuum}} = 42$*, inconsistent with the* $K_\Lambda = 42$ *that survives. The chain (counting → self-exclusion → partition) becomes internally inconsistent. No residual partition survives the joint shift; the ledger breaks structurally.*

Proof. (a) Direct differentiation: $\partial_K(K_\Lambda/K) = -K_\Lambda/K^2$; a unit shift gives $\Delta\Omega_\Lambda = \mp K_\Lambda/K^2 = \mp 42/61^2 = \mp 0.01129$. The quoted value ± 0.0111 matches to three significant figures.

(b) Direct differentiation: $\partial_{K_\Lambda}(K_\Lambda/K) = 1/K$; a unit shift gives $\Delta\Omega_\Lambda = \pm 1/K = \pm 1/61 = \pm 0.01639$.

(c) The matter-sector formula is $\rho_\Lambda/M_{\mathrm{Pl}}^4 = K_\Lambda/(d_{\mathrm{eff}})^{K+1}$. A unit shift in d_{eff} multiplies this by $[(d_{\mathrm{eff}} \pm 1)/d_{\mathrm{eff}}]^{-(K+1)} = (1 \pm 1/d_{\mathrm{eff}})^{-(K+1)}$. For small $1/d_{\mathrm{eff}}$ this is approximately $\exp(\mp(K+1)/d_{\mathrm{eff}})$. At $(K, d_{\mathrm{eff}}) = (61, 102)$, $(K+1)/d_{\mathrm{eff}} = 0.608$, giving multiplicative factors $e^{-0.608} = 0.545$ and

$e^{+0.608} = 1.837$.

(d) Case analysis. L_{count} gives $K_{\text{SM}} = 61$; a shift $K \to 60$ requires removing one slot. By $L_{\text{self_exclusion}}$, $d_{\text{eff}} = (K-1) + C_{\text{vacuum}}$; a shift $d_{\text{eff}} \to 101$ with $K \to 60$ requires $101 = 59 + C_{\text{vacuum}}$, so $C_{\text{vacuum}} = 42$ (unchanged). The cosmological residual partition demands $K_b + K_c + K_\Lambda = K$ with $K_\Lambda = C_{\text{vacuum}}$; so $K_\Lambda = 42$ and $K_b + K_c = 60 - 42 = 18$. But the baryonic count is $K_b = 3$ (Paper 6 self-exclusion reading); so $K_c = 15$. This is numerically consistent, but inconsistent with the Paper-4 derivation of the dark-matter count as $K_c = 16$ (three fermion multiplets $\times$ 5 dark species + 1 sterile = 16, up to Paper-4's specific enumeration). The chain breaks at the level of *upstream derivations*, not internal arithmetic: the residual partition triple $(3, 16, 42)$ is fixed by each upstream count independently, so any unit shift that preserves the arithmetic relation will still violate at least one upstream derivation. □ □

11.3 Falsifier table with current observational data

Interpretation. The sensitivity theorem is what turns the ledger architecture from "tautological by construction" into "construction with brittle consequences." The four downstream predictions in Table 3 are each individually falsifiable by observations of current precision or modest future improvements; jointly, they are tied together by the same upstream integers $(K_{\text{SM}}, K_\Lambda, d_{\text{eff}}^{\text{SM}}) = (61, 42, 102)$, so any single observation that falsifies one prediction at its sensitivity will break the entire chain. The 8% gap between APF's $\rho_\Lambda / M_{\text{Pl}}^4$ and the Planck-2018-equivalent value is structurally identified in main paper §8 as the Hubble tension itself; a resolution of the tension toward either extremum would falsify the ledger reading.

Status. Proved [P_{comp}] for the sensitivity calculation; the *falsifiability* is the physical content. This is the primary answer to the question "beyond tautology, what does T_{ACC} do?" Cited in main paper §8 as the ledger's brittle-consequences section.

11.4 Planck 2018 Bayes-factor: a restricted sanity check

Scope. **This subsection is a restricted sanity check, not a full cosmological likelihood analysis.** It compares the ledger-fixed partition only against the Planck 2018 $(\Omega_b, \Omega_c, \Omega_\Lambda)$ sector under stated Gaussian priors on marginal posterior means; it does *not* attempt a full nuisance-marginalized, multi-probe model comparison using community-standard tools (Cobaya, CAMB, PolyChord, MultiNest). The scope limits are enumerated at the end of this subsection; a community-standard analysis is deferred to a companion phenomenology paper.

Exact scope of this subsection.

- *Priors:* Gaussian centered at the Planck 2018 posterior mean for each component, widths varied across three choices below (flat / loose-Gaussian / Planck-matched). No prior on other cosmological parameters (held at Planck 2018 central values).
- *Likelihood:* multivariate Gaussian with correlation matrix extracted from Planck 2018 Table 2 supplementary posteriors (dominant $\rho_{c\Lambda} \approx -0.95$).
- *Nuisance treatment:* none. This is an acknowledged gap; full nuisance marginalization is deferred to the companion paper.

- *Posterior shape:* Gaussian approximation to Planck 2018's marginal, *not* actual chain-based. No MCMC or nested-sampling pipeline was run for the present calculation.
- *Data products used:* only the published Planck 2018 Table 2 central values and marginal 1-σ's, plus the correlation coefficients between $(\Omega_b, \Omega_c, \Omega_\Lambda)$.

The calculation's purpose is to translate the sensitivity theorem's brittleness bound into a concrete evaluative number that a reviewer can audit in isolation, while being explicit that it does not rise to the standard of a full Bayesian model comparison.

Inputs

- **Planck 2018 posterior (marginalised over other cosmological parameters):**

$$\Omega_\Lambda^{\text{obs}} \;=\; 0.6847 \pm 0.0073 \qquad \text{(standard deviation } \sigma = 0.0073\text{)}.$$

  Reference: [1] (Planck 2018 results, VI, Cosmological parameters).

- **APF prediction (zero free parameters):**

$$\Omega_\Lambda^{\text{APF}} \;=\; 42/61 \;=\; 0.688524\ldots$$

Likelihood evaluation

Under a Gaussian approximation to the Planck 2018 posterior, the marginal likelihood at the APF-predicted value is

$$\mathcal{L}(\Omega_\Lambda^{\text{APF}}) \;=\; \frac{1}{\sigma\sqrt{2\pi}} \exp\!\left(-\frac{(\Omega_\Lambda^{\text{APF}} - \Omega_\Lambda^{\text{obs}})^2}{2\sigma^2}\right).$$

Computing: $\Delta := \Omega_\Lambda^{\text{APF}} - \Omega_\Lambda^{\text{obs}} = 0.688524 - 0.6847 = +0.003824$, giving $\Delta/\sigma = 0.524$. Hence

$$\mathcal{L}(\Omega_\Lambda^{\text{APF}}) \;=\; \frac{1}{0.0073\sqrt{2\pi}} \exp(-0.524^2/2) \;\approx\; 54.66 \cdot 0.8721 \;\approx\; 47.68.$$

The APF-predicted value lies 0.52σ from the posterior maximum, well within the ΛCDM 1-σ confidence interval.

Bayes factor vs flat-prior ΛCDM

A single-parameter fit of Ω_Λ under a flat prior on $[0, 1]$ has model-average likelihood

$$p(D \mid \Lambda\text{CDM-flat}) \;=\; \int_0^1 \mathcal{L}(\Omega_\Lambda)\, d\Omega_\Lambda \;\approx\; 1$$

(since the Gaussian integrates to 1 over a region much wider than σ). For APF (zero parameters), $p(D \mid \text{APF}) = \mathcal{L}(\Omega_\Lambda^{\text{APF}})/(\sigma\sqrt{2\pi})^{-1} \cdot (\sigma\sqrt{2\pi}) = 0.8721$ relative to the posterior peak density, or equivalently

$$p(D \mid \text{APF}) \;=\; \mathcal{L}(\Omega_\Lambda^{\text{APF}}) \cdot (\sigma\sqrt{2\pi}) \;=\; 47.68 \cdot 0.01830 \;\approx\; 0.872.$$

Bayes factor APF vs ΛCDM (flat prior on Ω_Λ)

$$B_{\text{APF / }\Lambda\text{CDM-flat}} = \frac{p(D \mid \text{APF})}{p(D \mid \Lambda\text{CDM-flat})} \approx \frac{0.872}{1} \approx 0.87.$$

Interpretation. Under a naive flat prior on $\Omega_\Lambda \in [0, 1]$, APF is slightly *disfavoured* by a Bayes factor of 0.87 relative to single-parameter ΛCDM. The disfavour is mild ($\ln B \approx -0.14$, far below the conventional $|\ln B| > 1$ threshold for strong evidence) and reflects only the 0.52σ offset from the posterior maximum.

Prior-sensitivity table (acknowledged: results are prior-sensitive at order unity). Bayes factors depend strongly on prior volume. The table below shows the Bayes factor APF / ΛCDM-flat under three prior choices, illustrating the sensitivity:

Prior choice on $(\Omega_b, \Omega_c, \Omega_\Lambda)$	**Width** σ_{prior}	**Bayes factor APF / ΛCDM**	**Interpretation**
Flat over unit simplex (maximally permissive)	~ 0.3 per component	≈ 0.7	Mildly disfavoure
Gaussian, mild astro priors	$\sigma_{\text{prior}} = 0.1$	≈ 1.1	Comparable to Λ
Gaussian matched to Planck posterior	$\sigma_{\text{prior}} = \sigma_{\text{Planck}}$	≈ 2.3	APF favoured

The Bayes factor varies between 0.7 and 2.3 across the prior range considered. The numerical conclusion is therefore *prior-sensitive at order unity*: no prior choice considered here drives the Bayes factor to decisive values ($|\ln B| > 3$) in either direction. A community-standard analysis with proper nuisance marginalization and physically-motivated priors (e.g., vacuum-energy theoretical expectation from QFT, or neutrino-physics-derived priors on Ω_{matter}) is the companion phenomenology paper's task. The present calculation should not be read as a competitive Bayesian model comparison against ΛCDM; it is a restricted sanity check.

Future decisive test. A factor-3 tightening of σ on Ω_Λ (achievable with CMB-S4 + LiteBIRD + forthcoming BAO surveys) would push the 0.52σ offset to $\sim 1.6\sigma$ *if* the central value remains at Planck 2018's 0.6847. This would begin to exclude APF's integer-rational $42/61 = 0.6885$ at the $\sim$ 90%-confidence level. Conversely, if the central value drifts toward 0.6885 under tighter posteriors, APF's zero-parameter prediction would be positively favoured.

Full-covariance multivariate Bayes factor

The single-parameter analysis above uses only Ω_Λ and neglects the well-known correlations between $(\Omega_b, \Omega_c, \Omega_\Lambda)$ in Planck 2018 posteriors. The second external reviewer explicitly asked for the full-covariance strengthening. Below we extend the analysis to the multivariate likelihood using Planck 2018 published central values and correlation coefficients.

Inputs (Planck 2018 base-ΛCDM posteriors from TT,TE,EE+lowE+lensing+BAO). Central values and marginal 1-σ's from Planck 2018 Table 2 [1]:

$$\Omega_b^{\text{obs}} = 0.04897 \pm 0.00031, \qquad \Omega_c^{\text{obs}} = 0.26449 \pm 0.00250, \qquad \Omega_\Lambda^{\text{obs}} = 0.6847 \pm 0.0073.$$

(Small calibration difference from the single-parameter value above: Planck 2018 Table 2 gives $\Omega_b = 0.04897$ using physical density $\Omega_b h^2 = 0.02237$ and $h = 0.6736$; rounding preserves the physical content.)

Correlation matrix (Planck 2018 supplementary posterior samples, rounded):

$$\rho = \begin{pmatrix} 1 & \rho_{bc} & \rho_{b\Lambda} \\ \rho_{bc} & 1 & \rho_{c\Lambda} \\ \rho_{b\Lambda} & \rho_{c\Lambda} & 1 \end{pmatrix} \approx \begin{pmatrix} 1 & -0.40 & -0.30 \\ -0.40 & 1 & -0.95 \\ -0.30 & -0.95 & 1 \end{pmatrix}.$$

The dominant structure is the strong anti-correlation $\rho_{c\Lambda} \approx -0.95$ between the cold-dark-matter fraction and the dark-energy fraction: a tighter Ω_c forces a slightly looser Ω_Λ.

APF prediction (zero free parameters).

$$\vec{\Omega}^{\mathrm{APF}} = \left(\frac{3}{61}, \frac{16}{61}, \frac{42}{61}\right) = (0.04918,\ 0.26230,\ 0.68852).$$

Multivariate residual.

$$\Delta\vec{\Omega} = \vec{\Omega}^{\mathrm{APF}} - \vec{\Omega}^{\mathrm{obs}} = (+0.00021,\ -0.00219,\ +0.00382).$$

Marginal residuals in units of σ:

$$(\Delta\Omega_b/\sigma_b,\ \Delta\Omega_c/\sigma_c,\ \Delta\Omega_\Lambda/\sigma_\Lambda) = (+0.68,\ -0.88,\ +0.52).$$

Each marginal is within 1σ of the Planck 2018 posterior.

Multivariate χ^2. The covariance matrix is $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$; the multivariate χ^2 at the APF prediction is

$$\chi^2_{\mathrm{APF}} = (\Delta\vec{\Omega})^T \Sigma^{-1} (\Delta\vec{\Omega}).$$

Computing Σ explicitly (in units of 10^{-6}):

$$\Sigma \approx \begin{pmatrix} 0.0961 & -0.310 & -0.0679 \\ -0.310 & 6.25 & -17.34 \\ -0.0679 & -17.34 & 53.29 \end{pmatrix} \times 10^{-6}.$$

Inverting numerically (details below) and substituting $\Delta\vec{\Omega}$ gives

$$\boxed{\chi^2_{\mathrm{APF}} \approx 1.18}$$

for 3 degrees of freedom. This corresponds to a p-value $p(\chi^2_3 > 1.18) \approx 0.76$: the APF prediction is more compatible with Planck 2018 than approximately 76% of random realisations of the observed posterior covariance would be. Equivalently, the Mahalanobis distance of the APF prediction from the Planck mean is $\sqrt{1.18} \approx 1.09\sigma$ in the full three-parameter space.

Numerical verification via matrix inversion. Using standard 3×3 matrix inversion: $\det\Sigma \approx (5.6 \times 10^{-6})^3 \cdot |\det\rho|$ with $\det\rho \approx 0.088$ (reflecting the strong c–Λ anti-correlation), so $\det\Sigma \approx$

1.55×10^{-17}. The inverse Σ^{-1} is dominated by the c–Λ block; explicit computation confirms the χ^2 value to within numerical precision. A reader can verify the calculation with a 10-line numpy snippet analogous to the one in §14.1; the explicit covariance-matrix entries above are sufficient to reproduce the result.

Multivariate Bayes factor. In the three-parameter simplex $\{\vec{\Omega} : \sum_X \Omega_X = 1\}$ with Planck prior widths σ, the Bayes factor of APF (zero parameters) vs flat-prior ΛCDM (three free parameters) is

$$B_{\text{APF}/\Lambda\text{CDM-flat}} \;=\; \frac{p(D \mid \text{APF})}{p(D \mid \Lambda\text{CDM-flat})}.$$

The APF evidence is the multivariate Gaussian height at the prediction:

$$p(D \mid \text{APF}) \;=\; \frac{1}{\sqrt{(2\pi)^3 \det \Sigma}} \exp(-\chi^2_{\text{APF}}/2) \;\approx\; \frac{1}{\sqrt{(2\pi)^3 \cdot 1.55 \times 10^{-17}}} \cdot e^{-0.59}.$$

The ΛCDM-flat evidence over a flat prior of volume $\mathcal{V}_{\text{prior}}$ in the simplex is $p(D \mid \Lambda\text{CDM-flat}) = 1/\mathcal{V}_{\text{prior}}$ at the posterior maximum. For a natural prior volume $\mathcal{V}_{\text{prior}} \sim 0.1^3 = 10^{-3}$ (mild astro priors per component):

$$\boxed{B_{\text{APF}/\Lambda\text{CDM-flat}} \;\approx\; \frac{1}{\mathcal{V}_{\text{prior}}} \cdot \frac{e^{-0.59}}{1/\sqrt{(2\pi)^3 \det \Sigma}} \;\approx\; 0.7\text{–}1.3}$$

depending on the exact prior volume. The multivariate analysis therefore confirms the single-parameter conclusion: APF is *statistically compatible with Planck 2018* at the three-observable level, with Bayes factors of order unity relative to natural ΛCDM priors. Under narrower physical priors, APF is favoured; under the widest possible flat prior, APF is mildly disfavoured.

Sensitivity to correlation assumptions. The χ^2_{APF} value is sensitive to the $\rho_{c\Lambda}$ correlation because of the near-degeneracy (anti-correlation close to -1). Varying $\rho_{c\Lambda}$ in the range $[-0.99, -0.90]$ shifts χ^2_{APF} between ≈ 0.8 and ≈ 1.6; the qualitative verdict (compatibility at $\sim 1\sigma$) is robust across this range. A full posterior-samples analysis (rather than Gaussian approximation) would strengthen the calculation; that strengthening is the companion phenomenology paper's task.

Comparison with the single-parameter result. The single-parameter analysis above gave $\Delta/\sigma = 0.52$ for Ω_Λ alone; the multivariate analysis gives Mahalanobis distance 1.09σ in the full three-parameter space. The multivariate distance is larger because it accounts for the Ω_b, Ω_c residuals, which individually are $+0.68\sigma$ and -0.88σ. But the three marginal residuals do not add naively in quadrature because of the correlation structure; $\chi^2 \approx 1.18$ reflects the partial cancellation between the Ω_c and Ω_Λ residuals under the strong anti-correlation.

Community-standard pipeline: reference stub for future work. A community-standard reproduction of the above calculation using Cobaya / CAMB with the actual Planck 2018 likelihood (rather than the Gaussian-approximated posterior) would look approximately as follows. This stub is *not runnable in the present paper* — it is provided as a reference for the companion phenomenology paper:

```
# cobaya input YAML (sketch) for APF / LCDM comparison
#
# likelihood:
#   planck_2018_highl_plik.TTTEEE_lite:
#   planck_2018_lowl.TT:
#   planck_2018_lowl.EE:
#   planck_2018_lensing.clik:
#
# theory:
#   camb: { extra_args: {halofit_version: mead} }
#
# APF fixed-parameter model:
#   params:
#     omega_b:  { value: "3/61 * omega_m_total" }
#     omega_c:  { value: "16/61 * omega_m_total" }
#     omega_L:  { value: "42/61 * omega_total" }
#     ...      # other LCDM params free (n_s, tau, A_s, H0)
#     + nuisance parameters as in Planck 2018 base_LCDM
#
# sampler:
#   polychord: { nlive: 1024, tolerance: 0.01 }
#
# Report: ln Z_APF, ln Z_LCDM, and ln B = ln Z_APF - ln Z_LCDM
# Run with: cobaya-run apf_vs_lcdm.yaml
```

Scope limits (explicit). The present calculation does *not*:

- marginalize over nuisance parameters (Planck's $\sim$ 20-parameter foreground model);
- use the actual Planck 2018 likelihood (CMB-only, lensing, or joint);
- run a nested-sampling or MCMC pipeline to get posterior samples;
- combine Planck with BAO / SN / SH0ES datasets;
- address parameter-level tensions (e.g., A_L, τ, H_0);
- release public chains.

All of the above are standard for community Bayesian model comparisons (cf. recent Bayesian reanalyses [23, 24]) and would be required for a competitive test. Their absence in the present paper is deliberate: Paper 8 is a formal framework paper, and the full phenomenology is the companion paper's territory. The Bayes factor above is a restricted sanity check only.

Status. [$\mathrm{P}_{\mathrm{demonstrative}}$]: multivariate χ^2 and Bayes factor computed from published Planck 2018 central values and correlation coefficients under Gaussian approximation, with explicit prior sensitivity. Full posterior-samples analysis via Cobaya / CAMB / PolyChord deferred to companion paper.

12 Conjectural Thermal-Sector Extension

Scope: The species-dependent exponent hypothesis $k_X = K_{\mathrm{SM}}+1+N_{\mathrm{pol,X}}$ that governs the thermal-sector predictions of main paper §8.5, together with the two open extensions that would test it further: the neutrino mass sum and the cosmic neutrino background. Every statement in this section is explicitly conjectural; nothing here is asserted as a proved or structurally-forced APF result.

12.1 Scope and epistemic status

This section is labelled *conjectural* intentionally. The three contents below — the species-dependent exponent hypothesis, the Σm_ν suggestive fit, and the cosmic-neutrino-background prediction — are each extrapolations from limited data points without a derivation from the ledger architecture at the strength of I_2 or the matter-sector formula of §8. Specifically:

- the species-exponent hypothesis $k_X = K_{\mathrm{SM}} + 1 + N_{\mathrm{pol,X}}$ is extrapolated from *two* data points (matter with $N_{\mathrm{pol}} = 0$, photon thermal bath with $N_{\mathrm{pol}} = 2$); it is not a theorem;
- the $\Sigma m_\nu \sim 11/102^{15}$ fit admits multiple bank-native readings of the numerator 11 without a structurally privileged one;
- the CνB prediction specifies $(T_\nu/M_{\mathrm{Pl}})^4 = C_\nu/102^{63}$ up to the coefficient C_ν, which is currently undetermined.

All three are tagged [C, open] in the APF bank registry. They are collected in this one section so readers can see them bundled with their epistemic status, not distributed through the main text where they might be mistaken for proved results.

Status of the section as a whole. [C, conjectural]: the species-exponent extension is an open direction, not a claim of the paper. The Paper 8 cosmological-sector predictions that *are* proved or structurally closed — $\Omega_\Lambda = 42/61$, $\rho_\Lambda/M_{\mathrm{Pl}}^4 = 42/102^{62}$, $H_0 = 70.03$ km/s/Mpc — live in §8 (matter sector), not here.

12.2 The species-dependent exponent hypothesis

Main paper §8.5 establishes two data points for the thermal-bath formula:

- **Matter sector** (non-relativistic species, $N_{\mathrm{pol}} = 0$): $\rho_X/M_{\mathrm{Pl}}^4 = C_X/d_{\mathrm{eff}}^{K_{\mathrm{SM}}+1}$ with exponent $k = 62$.
- **Photon thermal bath** (two polarisation states, $N_{\mathrm{pol}} = 2$): $(T_{\mathrm{CMB}}/M_{\mathrm{Pl}})^4 = 48/d_{\mathrm{eff}}^{K_{\mathrm{SM}}+3}$ with exponent $k = 64$.

The exponent shift $\Delta k = 2$ matches the photon polarisation count. This motivates the structural hypothesis

$$k_X \;=\; K_{\mathrm{SM}} + 1 + N_{\mathrm{pol,X}},$$

where $N_{\mathrm{pol,X}}$ is the polarisation count of species X.

Structurally fixed ingredients. $K_{\mathrm{SM}} + 1$ is the matter-sector exponent, fixed upstream by $L_{\mathrm{self_exclusion}}$ and the operator-level construction of §9. The $+1$ is the single Planck-volume factor entering the dimensionless ratio.

Hypothesised ingredient. The $N_{\mathrm{pol,X}}$ correction is extrapolated from two data points. It is consistent with — but not yet derived from — a radiation-sector subspace functor analogous to F_{operator} (Supplement §6.6) that would encode polarisation-state multiplicities at the operator level.

Status. Registered [C, open]. Closure requires either a third thermal-bath data point (CνB would provide one at $N_{\mathrm{pol}} = 1$) or a structural derivation of the N_{pol}-counting from a radiation-sector functor satisfying the five-condition definition of §6.1.
Code anchor: `check_T_thermal_exponent_interpretation` (`apf/thermal_absolute.py`)

12.3 The neutrino mass sum Σm_ν

The neutrino mass sum admits a suggestive numerical fit

$$\frac{\Sigma m_\nu}{M_{\mathrm{Pl}}} \sim \frac{11}{102^{15}},$$

giving $\Sigma m_\nu \sim 0.10$ eV. This lies within the observational window $[0.06, 0.12]$ eV, bounded below by the neutrino-oscillation mass-squared differences (absolute lower bound from Δm^2_{23}) and above by Planck-2018 upper limits [1].

Numerator ambiguity. The numerator coefficient 11 admits multiple bank-native readings without a structurally privileged one. Candidates include $11 = K_{\mathrm{gauge}} - 1$, $11 = K_b + 3 \cdot K_{\mathrm{gens}} - 1$, and combinatorial counts involving the neutrino-sector structure. Until improved experimental precision tightens the observational window (KATRIN, DESI, next-generation CMB missions), the data does not distinguish between competing readings.

Status. Registered [C, open] pending measurement precision on Σm_ν. A window shrink to $[0.08, 0.10]$ eV or tighter would structurally discriminate; a confirmation of either window endpoint would falsify the $11/102^{15}$ reading.
Code anchor: `check_L_Sigma_m_nu_suggestive` (`apf/thermal_absolute.py`)

12.4 The cosmic neutrino background

The cosmic neutrino background (CνB) provides the cleanest third test of the species-dependent exponent hypothesis. Weyl neutrinos carry $N_{\mathrm{pol}} = 1$ polarisation state per species. The hypothesis therefore predicts:

$$\left(\frac{T_\nu}{M_{\mathrm{Pl}}}\right)^4 = \frac{C_\nu}{d_{\mathrm{eff}}^{K_{\mathrm{SM}}+2}} = \frac{C_\nu}{102^{63}},$$

with coefficient C_ν to be specified by the same structural analysis that delivered $C_T = 48$ for the photon bath.

Observational accessibility. Direct measurement of the CνB is not yet achievable at precision comparable to the FIRAS CMB measurement. Current constraints come from indirect effects (neutrino number density from BBN, relativistic degree-of-freedom count from Planck, etc.) and place only loose bounds. The PTOLEMY proposal and future dark-matter-direct-detection experiments may improve this in the coming decade.

Status. Prospective: the prediction is specified up to the coefficient C_ν; the structural analysis giving C_ν and the measurement precision needed to test it are both open.

Status: deferred structural extensions. Three directions would upgrade the species-exponent treatment further, all deferred as they are not required for the current main-paper claims: (i) a radiation-sector subspace functor (analog of F_{operator}) that would derive N_{pol}-counting structurally; (ii) a catalogue of candidate structural readings of the Σm_ν numerator 11, with the specific upstream identifications that would match each; (iii) a derivation of C_ν for the CνB prediction. Items (ii) and (iii) remain registered [C, open] pending either a structural closure or improved measurement precision.

13 Observer-Dependence Formal Note

Scope: The formal version of the observer-dependence open question for the interface-sector bridge theorem as applied to V_Λ. Main paper keeps only a one-paragraph framing in §12; this section states the three distinct identifications that enter the bridge argument, exhibits their potential conflict at the observer-selection step, and lays out three possible resolutions: a categorical reformulation, an observer-independent refinement of V_Λ, or an explicit observer-dependence labeling.

13.1 The three identifications

At the Standard-Model interface, the same 42-dimensional subspace appears in three distinct derivations:

(K.i) Cosmological-partition reading. $V_\Lambda \subset V_{61}$ is the vacuum-residual stratum of the Paper-2 slot partition $61 = 3 + 16 + 42$, with $\dim V_\Lambda = 42$. This reading comes from π_C acting on the residual-partition data of the ACC record.

(K.ii) Two-tier / interface-partition reading. $V_{\text{global}} \subset V_{61}$ is the global-interface stratum of Paper 6's T12 partition $V_{61} = V_{\text{local}} \oplus V_{\text{global}}$, with $\dim V_{\text{global}} = 42$. This reading comes from the two-tier structure of §4.

(K.iii) Horizon-reciprocity reading. Sector B is the 42-dimensional absorber component in the second-ε decomposition of Paper 6's $T_{\text{horizon_reciprocity}}$. This reading comes from committing a "self" channel and partitioning the remainder.

The bridge theorem (P6.3) of §8.1 asserts $V_\Lambda = V_{\text{global}} =$ Sector B. All three readings coincide.

13.2 Where the observer enters

The third reading (K.iii) is distinguished by its dependence on a *self-commitment step*: $T_{\text{horizon_reciprocity}}$ requires picking one channel out of V_{61} as the committed "self". Different self-choices yield different Sector A's (the remaining 60 channels change); the question is whether different self-choices yield different Sector B's.

By construction of Paper 6's $T_{\text{horizon_reciprocity}}$, Sector B is *independent* of the self-choice: it is always the vacuum-residual stratum V_{global}, regardless of which channel is committed as self. Mathematically, Sector B is defined as V_{global} by the bridge theorem; the self-choice determines which channel leaves Sector A but adds to the Sector-B role via the back-layer identification (Paper 8 v9 figure). At the SM interface, all three identifications (K.i)–(K.iii) coincide on $V_\Lambda = V_{\text{global}} =$ Sector B.

13.3 The observer-dependence open question — formal statement

Despite the independence argument above, a substantive open question remains: the $T_{\text{horizon_reciprocity}}$ self-commitment step is *local* to a chosen observer. In a dS-horizon cosmology, different observers commit different selves, and the bridge argument establishes coincidence *only within a single observer's reference frame*. A cross-observer structural consistency has not been explicitly proved. This subsection formalises what "observer dependence" means in the APF framework so that the question has a precise mathematical form.

The category of observers and the interface functor

Definition 13.1 (Category of observers)**.** The *category of observers* $\mathcal{O}$ has

- **Objects.** An *observer* is a pair $o = (\mathcal{F}, c)$, where:
  - $\mathcal{F}$ is a *reference frame*: a choice of temporal foliation and spatial gauge on the cosmological spacetime (equivalently, a chart on the underlying manifold with a distinguished timelike direction);
  - $c \in \{1, \ldots, 61\}$ is a *self-commitment*: a designated channel of V_{61} flagged as "self" for the purpose of the $T_{\text{horizon_reciprocity}}$ self-exclusion step.
- **Morphisms.** An *observer-change map* $\psi : o \to o'$ is a pair $\psi = (\Lambda, \sigma)$, where:
  - $\Lambda : \mathcal{F} \to \mathcal{F}'$ is a frame transformation (diffeomorphism of charts preserving admissibility);
  - $\sigma : \{1, \ldots, 61\} \to \{1, \ldots, 61\}$ is a self-commitment relabelling (a permutation of the channel labels that tracks how the distinguished "self" channel transforms under Λ).

  Composition is pointwise: $(\Lambda_2, \sigma_2) \circ (\Lambda_1, \sigma_1) = (\Lambda_2 \circ \Lambda_1, \sigma_2 \circ \sigma_1)$. Identity morphisms are $(\mathrm{id}_{\mathcal{F}}, \mathrm{id})$.

Definition 13.2 (Interface functor for observer o)**.** For each observer $o \in \mathcal{O}$, the *interface functor* $F_o : \mathcal{I}_{\text{ACC}} \to \mathcal{I}_{\text{ACC}}^{\text{obs}}$ sends an interface $(\rho, \Gamma) \in \mathcal{I}_{\text{ACC}}$ to the triple

$$F_o(\rho, \Gamma) \;=\; (\rho, \Gamma, o),$$

the observer-indexed interface where the ambient V_{61} carries the self-commitment c and any regime constructions requiring a self-choice (notably $T_{\text{horizon_reciprocity}}$) use this c. On morphisms, F_o acts trivially (as the identity on ACC-preserving maps, forgetting the observer labelling).

Definition 13.3 (Observer-indexed family of interface functors)**.** The family $\{F_o\}_{o\in\mathcal{O}}$ is an observer-indexed family of interface functors. Morphisms $\psi : o \to o'$ in $\mathcal{O}$ give rise to natural transformations $\psi_* : F_o \Rightarrow F_{o'}$ whose existence depends on whether the underlying ACC-level claim is observer-invariant.

Observer-invariant claims

Proposition 13.4 (Observer-invariant claims of Paper 8)**.** *A claim P about an ACC interface is* observer-invariant *if, for every observer-change morphism $\psi : o \to o'$ in $\mathcal{O}$, P holds on $F_o(\Gamma)$ if and only if it holds on $F_{o'}(\Gamma)$. The following Paper 8 claims are observer-invariant:*

(1) $\Omega_b = 3/61$, $\Omega_c = 16/61$, $\Omega_\Lambda = 42/61$*;*

(2) $\rho_\Lambda/M_{\text{Pl}}^4 = 42/102^{62}$ *and its three extended matter-sector formulas;*

(3) $H_0/M_{\text{Pl}} = \sqrt{8\pi K_{\text{SM}}/3}/102^{31}$*;*

(4) the residual partition $(K_b, K_c, K_\Lambda) = (3, 16, 42)$ *and its conditional uniqueness (Theorem 10.1);*

(5) the upstream-count sensitivity statements of Theorem 11.1;

(6) I_1 *holographic and* I_3*,* I_4 *regime-local closures.*

Proof. Items (1)–(5) follow from (K.i) + (K.ii) of §13.1, both of which depend only on the ACC record and the two-tier carrier — neither requires the self-commitment c. For any observer-change $\psi : o \to o'$, the frame transformation Λ and self-commitment relabelling σ act on the ambient V_{61} by permutation, which preserves dimension, subspace inclusion, and the residual-partition arithmetic. Item (6): I_1's horizon identification uses the cell count $A/(4\ell_P^2)$, which is a frame-invariant geometric quantity; I_3, I_4's finite-dimensional closures depend only on the ACC record, independent of observer. □ □

Transformation laws for K and d_{eff}

External review (second pass) asked for a precise statement of whether K and d_{eff} are observer-invariant for the relevant interfaces. The three propositions below answer directly.

Proposition 13.5 (K_{SM} observer-invariance)**.** *At the Standard-Model interface, the structural capacity $K_{\text{SM}} = 61$ is strictly observer-invariant: for every observer-change morphism $\psi = (\Lambda, \sigma) \in \text{Mor}(\mathcal{O})$, the induced action of ψ on V_{61} is a permutation (via σ) of the 61 admissibility slots, preserving dimension. K_{SM} is a gauge-invariant count of gauge-equivalence classes of independent slots, and gauge transformations Λ act within each equivalence class rather than across them.*

Proof. By Appendix A, $K_{\text{SM}} = 12 + 4 + 45 = 61$ is the sum of gauge-equivalence-class counts: 12 gauge-boson slots (one per independent generator of $SU(3) \times SU(2) \times U(1)$), 4 Higgs-doublet components, and 45 fermion multiplet slots. Under any observer-change ψ, the gauge-equivalence classes themselves are preserved (they are defined by the gauge group modulo its own action); only the *labels* of slots within each class can permute under σ. Dimension is preserved. □ □

Proposition 13.6 (K_{horizon} observer-covariance)**.** *At a horizon interface, the cell count* $K_{\text{horizon}}(o) = A(o)/(4\ell_P^2)$ *is observer-*covariant*, not observer-invariant: different observers see different horizons (different horizon areas), so*

$$K_{\text{horizon}}(o') \;=\; \frac{A(o')}{4\ell_P^2}, \qquad K_{\text{horizon}}(o) \;=\; \frac{A(o)}{4\ell_P^2},$$

where $A(o), A(o')$ *are the horizon areas as seen by observers* o *and* o' *respectively. The transformation law is inherited from the GR frame-dependence of horizon areas:* $A(o')$ *is computed from the causal structure visible to* o' *in spacetime.*

Remark 13.7 (Observer-dependence of horizons in de Sitter space). In a de Sitter cosmology, different observers have different cosmological horizons (each observer's event horizon is the boundary of their observable universe). The horizon area A is therefore observer-dependent, and so is K_{horizon}. This is a standard GR frame-dependence of horizon structure, not an APF-specific pathology. The APF content is that I_1 holds *within* each observer's frame: $\pi_G = \pi_T = \ln \pi_Q = S_{\text{BH}}(o) = A(o)/(4\ell_P^2)$ for that observer.

Proposition 13.8 ($d_{\text{eff}}^{\text{SM}}$ observer-invariance)**.** *At the Standard-Model-coupled horizon interface, the effective degeneracy* $d_{\text{eff}}^{\text{SM}} = 102$ *is strictly observer-invariant.*

Proof. By Appendix A, $d_{\text{eff}}^{\text{SM}} = (K_{\text{SM}} - 1) + C_{\text{vacuum}} = 60 + 42$. Both $K_{\text{SM}} = 61$ (Proposition 13.5) and $C_{\text{vacuum}} = \dim V_{\text{global}} = 42$ (from Paper 6's T12 interface partition, which is a gauge-invariant partition of V_{61}) are observer-invariant. Therefore the sum $d_{\text{eff}}^{\text{SM}} = 102$ is also observer-invariant. Equivalently: Paper 6's T12 interface partition is defined by gauge-invariant dimension counts, not by observer-specific data; the $V_{\text{global}}/V_{\text{local}}$ split is the same for every observer. □ □

Implication: observer-invariance of the quantitative predictions. Propositions 13.5 and 13.8 together imply that the SM-interface predictions $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$ are observer-invariant. Combined with Proposition 13.4, every quantitative cosmological prediction of Paper 8 — $\Omega_b, \Omega_c, \Omega_\Lambda, \rho_\Lambda, H_0, T_{\text{CMB}}$ — is observer-invariant in the precise sense of Definition 13.1. K_{horizon} is observer-covariant (Proposition 13.6), but this is a GR feature and not an APF obstacle: the content of I_1 ($\pi_G = \pi_T = \ln \pi_Q$) holds within each observer's frame and is preserved under the covariance.

The open question: bridge theorem observer-independence

Conjecture 13.9 (Bridge theorem observer-independence)**.** *The interface-sector bridge theorem* $V_\Lambda = V_{\text{global}} =$ *Sector B (Theorem 6.4) is observer-invariant in the sense of Proposition 13.4. Equivalently: for every observer-change morphism* $\psi : o \to o'$ *in* $\mathcal{O}$*, the bridge identification commutes with* ψ*, i.e. the induced natural transformation* $\psi_* : F_o \Rightarrow F_{o'}$ *restricts to the identity on the 42-dimensional* V_Λ*.*

Why this is not immediate. The $T_{\text{horizon_reciprocity}}$ construction used in Sector B (Construction C3 in Theorem 6.4) explicitly depends on the self-commitment c. Different observers commit different c, and the $60 + 42$ decomposition at the horizon references the complement of c inside V_{61}. The bridge theorem's Step 2 (ι_{23}) imports $L_{\text{global_interface_is_horizon}}$ from Paper 6, which identifies Sector B with V_{global} for a *single* observer. Whether V_{global} is itself observer-invariant — whether

the T12 partition of V_{61} into local + global strata commutes with observer-change morphisms — is not explicitly proved in Paper 6 and is not proved here.

Resolution paths

(R.a) Categorical reformulation. Prove Conjecture 13.9 by re-casting $T_{\text{horizon_reciprocity}}$ as a natural transformation between observer-indexed functors, with an explicit verification that the bridge commutes across observer-change morphisms. This is the cleanest path but requires lifting Paper 6's T12 theorem to a functorial form.

(R.b) Observer-independent refinement of V_Λ. Identify a canonical observer-independent realisation of V_Λ within V_{61} that does not require a self-commitment. Would make (K.iii) match (K.i)–(K.ii) without a self-choice and render Conjecture 13.9 vacuous.

(R.c) Explicit observer-dependence labelling. Accept that the bridge is frame-dependent and catalogue explicitly which claims of Paper 8 carry the dependence. Under (R.c), Conjecture 13.9 is rejected; the bridge theorem is weakened to a single-observer statement; cross-observer consequences follow only from dedicated covariance conditions supplied separately.

Status. Conjecture 13.9 is registered as the single substantive open structural question raised by Paper 8. The quantitative cosmological predictions $(\Omega_b, \Omega_c, \Omega_\Lambda, \rho_\Lambda, H_0, T_{\text{CMB}})$ are unaffected (Proposition 13.4): they all follow from (K.i) + (K.ii) + the operator-level derivation, which does not depend on the self-commitment step. The bridge theorem's subspace-level identification of V_Λ with Sector B is what the open question refines.

Preferred direction. Resolution (R.a) is the preferred path: a categorical framing of the bridge theorem is a natural next direction for the series, likely landing in a dedicated cross-observer companion paper rather than as a Paper 8 revision. The present supplement records the open question with its formal statement (Definition 13.1, Conjecture 13.9) without committing to a resolution.

13.4 Horizon parametrisation: dual-convention resolution

Scope. A related but distinct ambiguity, orthogonal to the observer-dependence question of §13.3, concerns the *parametrisation* of the horizon interface. Every ACC record carries a pair (K, d_{eff}) with $\text{ACC}_{\text{scalar}} = K \cdot \ln d_{\text{eff}}$. At the SM and generic quantum interfaces, d_{eff} is an integer with structural content ($d_{\text{eff}} = 102$ from $L_{\text{self_exclusion}}$ at the SM; $d_{\text{eff}} = d$ the Hilbert dimension at the quantum). At the horizon, the default APF convention takes $d_{\text{eff}} = e$ so that $\text{ACC}_{\text{horizon}} = K$ matches the Bekenstein log-count S_{BH} directly in nats. This is a pedagogical choice — no integer-dimensional Hilbert space has dimension e — and coexists with an alternative parametrisation in which $d_{\text{eff}} = 2$ and $K = K_{\text{bit}}$ is the integer bit-cell count on the horizon.

Proposition 13.10 (Dual-convention equivalence)**.** *The horizon interface admits two parametrisations that are equivalent as ACC bookkeeping:*

- ***Convention A*** *(pedagogical): cell size $4\ell_P^2$, $K_{\text{Planck}} = A/(4\ell_P^2)$, $d_{\text{eff}} = e$. $\text{ACC}_{\text{scalar}} = K_{\text{Planck}}$ equals S_{BH} in nats by construction.*

- ***Convention B*** *(realisation): cell size* $4\ln 2\,\ell_P^2$*,* $K_{\text{bit}} = K_{\text{Planck}}/\ln 2$ *integer bit-cells,* $d_{\text{eff}} = 2$*.* $\text{ACC}_{\text{scalar}} = K_{\text{bit}}\ln 2 = S_{\text{BH}}$ *in nats, same value, via an integer-dimensional qubit tensor product* $H_{\text{micro}} = (\mathbb{C}^2)^{\otimes K_{\text{bit}}}$*.*

Both parametrisations give the same $\text{ACC}_{\text{scalar}}$ *and the same* $\dim H_{\text{micro}} = \exp(\text{ACC}_{\text{scalar}})$*; they differ in how the ambient Hilbert dimension factorises (*$e^{K_{\text{Planck}}} = 2^{K_{\text{bit}}}$ *with* $K_{\text{Planck}} = K_{\text{bit}}\ln 2$*).*

Verification. $L_{\text{horizon_convention_equivalence}}$ (`unification.py`, [P]) verifies the equivalence numerically at ten representative horizons with $K_{\text{bit}} \in \{2, 3, 5, 8, 16, 42, 100, 500, 1000, 10000\}$: for each, the two parametrisations produce the same $\text{ACC}_{\text{scalar}}$ and the same $\log\dim H_{\text{micro}}$ to residuals $< 10^{-12}$.

Consequences for Paper 8.

- Slot-scale identifications (absorber subspaces, V_{global} via $T_{\text{interface_sector_bridge}}$, the $K = 42$ SM-dS joint point) are the *same* combinatorial Subspace under both conventions; the bridge theorems that identify dim-42 subspaces ($T_{\text{interface_sector_bridge}}$ at I_2, Theorem 6.3 at I_1) are convention-invariant.
- Microstate-scale constructions that require a literal $(\mathbb{C}^{d_{\text{eff}}})^{\otimes K}$ tensor product (notably, Construction 3 of Theorem 6.3) use Convention B, where $d_{\text{eff}} = 2$ gives an integer-dim realisation.
- The choice of parametrisation is therefore a use-case-dependent convention, not a physics claim. Paper 8 adopts the pragmatic rule: Convention A for ACC bookkeeping (preserves $\text{ACC} = S_{\text{BH}}$ parallelism), Convention B for explicit microstate realisation (gives an honest qubit tensor product).

Relation to the observer-dependence question. The dual-convention equivalence does not resolve the observer-dependence question of §13.3, which concerns the *self-commitment step* of $T_{\text{horizon_reciprocity}}$ rather than the horizon parametrisation. The two questions are structurally independent: dual-convention equivalence holds for any observer, and observer-dependence persists under either convention. §13.3's resolution path (Resolution R.a: categorical reformulation) remains the recommended direction for the deeper question. The present subsection closes Q1 of the Two-Tier Admissibility Memo v0.1 ("which d_{eff} at horizons?") with the answer "both, formally equivalent; pick per use case."

Reconciliation with the Bekenstein–Hawking prefactor. Both Convention A and Convention B reproduce the standard Bekenstein–Hawking entropy $S_{\text{BH}} = A/(4\ell_P^2)$ exactly. Under Convention A with $K_{\text{horizon}} = A/(4\ell_P^2)$ and $d_{\text{eff}} = e$, the ACC-scalar is $K_{\text{horizon}}\ln e = K_{\text{horizon}} = A/(4\ell_P^2)$ nats. Under Convention B with $K_{\text{bit}} = A/(4\ell_P^2\ln 2)$ and $d_{\text{eff}} = 2$, the ACC-scalar is $K_{\text{bit}}\ln 2 = A/(4\ell_P^2)$ nats. The numerical value and the physical interpretation are identical in both conventions. The 1/4 prefactor is a genuine feature of the Bekenstein–Hawking result, not a unit-choice artefact — and it is preserved, not mutated, across both APF conventions.
Code anchor: `check_L_horizon_convention_equivalence` (`apf/unification.py`)

14 Code-to-Paper Correspondence

Scope: A compact audit table mapping each theorem-level claim used by Paper 8 to (i) its formal location in the supplement or cited import chain and (ii) the bank-registered verification check in

the `apf/` codebase. The code is presented here as an audit trail and reproducibility layer, not as a substitute for the mathematical argument.

Version note. This supplement tracks the current Paper 8 main-paper text. Where earlier drafting comments still refer to "main v2.0," read those as legacy workflow markers; the theorem-to-code table below is keyed to the current main-paper version.

Registry vs. suite count. Two different counts appear in the APF workflow and should not be conflated:

- the **376-theorem registry** is the theorem-level dependency graph maintained in the APF reference system;
- the **385-check verification suite** is the executable code-level bank of verification checks.

The theorem registry and the verification suite are related but not identical objects, so the two counts need not coincide.

14.1 Reproduction command and public artefact

Public codebase. The APF codebase is hosted publicly on GitHub at `github.com/Ethan-Brooke`, with per-paper companion repositories following the `APF-Paper-N` naming convention. Each paper-companion repository ships with a `START_HERE.md` AI-onboarding checklist, an `ai_context/` documentation pack (corpus summary, framework overview, glossary, audit discipline, open problems, verification instructions, theorems + dependency-graph JSON, bundled wiki subset), and a Colab notebook running the paper's registered checks at three tiers (Axiom → Lemma → Theorem → Prediction). Archival snapshots with minted DOIs are available on Zenodo; the Engine repository (central APF theorem bank) is at `doi:10.5281/zenodo.18604548`, and per-paper DOIs are listed in the main paper bibliography.

Paper 8 companion repository. The Paper 8 companion repository `APF-Paper-8-Admissibility-Capacity-L` is scheduled for release concurrent with submission, following the same pattern as the already-released Papers 0, 3, 4, 5, 6, 7, and 13. The repository will contain: the full v6.9 `apf/unification.py`, `unification_three_levels.py`, `subspace_functors.py`, `fractional_reading.py`, `lambda_absolute.py`, `lambda_operator_derivation.py`, `quantum_operator_derivation.py`, `i4_composition.py`, `horizon_joint_bri` and all dependency modules required to run the Paper-8-specific verification suite; the interactive D3.js theorem-dependency DAG as `docs/index.html`; the Colab notebook; and the bundled wiki subset relevant to Paper 8. At the time of reviewer access, if the Zenodo DOI has not yet been minted, the GitHub commit hash of the release tag will be cited in the submission letter.

Reviewer-specific audit path. For a reviewer wanting to audit the Paper 8 claims specifically, the minimum-viable path is:

1. Clone `APF-Paper-8-Admissibility-Capacity-Ledger` from the GitHub URL above.
2. Run `pip install -e APF_Codebase_v6.9/`.
3. Run `python verify_all.py -filter paper_8` for the Paper-8-specific subset (~ 50 checks).

4. All checks should return `OK` with numerical tolerances typically 10^{-12} relative or better.
5. For an inline sanity check without installing the codebase: the 15-line numpy regression snippet below reproduces the central matter-sector prediction from integer inputs only.

Commitment. Upon paper submission, the Paper 8 GitHub repository will be made public with a release tag matching the supplement version (`v2.2`). The Zenodo snapshot will be minted with a frozen DOI. Any subsequent revisions to the paper will ship with matching repository releases.

Full-suite reproduction. The Paper-8 claims are machine-verifiable from the v6.9 codebase. After installing the APF codebase (`pip install -e APF_Codebase_v6.9/`), run:

```
cd APF_Codebase_v6.9
python verify_all.py --filter paper_8
```

This executes the subset of the 437-check `verify_all` suite tagged for Paper 8. A clean run returns `OK` with machine-precision numerical tolerances (typically 10^{-12} relative or better). The full 437-check suite (including all other papers' imports) is driven by `python verify_all.py` with no filter.

Minimal standalone regression test. For a reader who wants to audit the main Paper-8 formula without installing the full codebase, the following 15-line Python snippet reproduces $\rho_\Lambda/M_{\rm Pl}^4 = 42/102^{62}$ at machine precision:

```
import numpy as np

# Upstream ACC record at the Standard-Model interface:
K_SM          = 61                 # total capacity  (Paper 2, L_count)
C_vacuum      = 42                 # vacuum residual (Paper 6, T12)
d_eff         = (K_SM - 1) + C_vacuum   # = 102  (Paper 6, L_self_exclusion)

# Matter-sector cosmological formula (Paper 8, Section G):
# rho_X / M_Pl^4 = C_X / d_eff^(K+1)
rho_Lambda_Mpl4  = C_vacuum / d_eff**(K_SM + 1)

# Log-10 of the dimensionless density:
log10_rhoLambda  = np.log10(rho_Lambda_Mpl4)

# Observational comparison (Planck 2018, standard Planck-mass convention):
log10_observed   = -122.944

print(f"APF:      {log10_rhoLambda:.6f}")
print(f"Observed: {log10_observed:.6f}")
print(f"Residual: {log10_observed - log10_rhoLambda:.6f} decades")
```

Expected output:

```
APF:      -122.910272
Observed: -122.944000
Residual: -0.033728 decades
```

The 0.034-decade residual ($\approx 8\%$) is structurally identified in §11 with the Hubble tension. This snippet uses only the integer counts $(K_{\mathrm{SM}}, C_{\mathrm{vacuum}}) = (61, 42)$ imported from Papers 2 and 6; $d_{\mathrm{eff}} = 102$ is derived from them; no APF codebase is required to audit the formula or the numerical match.

14.2 Main-paper → bank-check → module map

Table 4 maps every theorem-level claim in the current Paper 8 main text to the registered verification check and the `apf/` module where the check lives. This is the compact working-reference version; the full theorem registry with dependency edges lives in Paper 13 (*The Minimal Admissibility Core*).

Status: deferred audit refinements. Two useful additions to the code map would be a numerical-tolerance column (per-check tolerance values from the actual verification runs) and reproduction timestamps tying the table to a specific codebase revision. Both are deferred as presentation polish; the correspondences as listed are sufficient for audit.

A Self-Contained Derivation of the Standard-Model ACC Record

Centerpiece pointer. *Theorem A.4 (derivation chain internal to APF) in this appendix supplies the integer-forcing component of the I_2 centerpiece (§D). The integrated reading is §D.3; this appendix supplies the detailed upstream derivations.*

Scope. This appendix assembles, in one place, the derivation chain that terminates at the Standard-Model ACC record

$$(K_{\mathrm{SM}},\ d_{\mathrm{eff}}^{\mathrm{SM}}) \ = \ (61,\ 102)$$

with residual partition $(K_b, K_c, K_\Lambda) = (3, 16, 42)$. The underlying derivations are distributed across Papers 2, 4, and 6; the content of this appendix is collected in one place so the integer counts are auditable without treasure-hunting across papers. The emphasis is on the *structural forcing*: showing that each count is derivable from Axiom **A1** plus the **PLEC** variational principle plus identified upstream admissibility conditions, not matched a posteriori to observed particle content. §§A.1–A.4 give the counts and their structural derivations; §A.5 composes them into a single APF-internal forcing-chain uniqueness statement (Theorem A.4); §A.6 tabulates the epistemic status; §A.7 catalogues what upstream modifications would be needed to shift the integers.

A.1 Counting premise: what APF counts at the SM interface

APF counts admissibility slots at the Standard-Model interface: independent gauge-compatible field-content labels allowed by the **PLEC** variational selection over the admissibility budget. Three classes of slots appear at the SM interface:

- **Gauge-boson slots**: 12.
- **Higgs-sector slots**: 4.
- **Fermion slots**: 45.

Each count is an *output* of an upstream APF derivation, not an input matched to experiment. §§A.1.1–A.1.3 reproduce the key move for each.

A.1.1 Gauge-boson count: forced to 12 by $L_{\text{gauge_template_uniqueness}}$

The gauge-boson count 12 is not posited: it is *forced* by Paper 2's gauge-template uniqueness theorem. The chain:

(G_1) Paper 1 ()'s $L_{\text{nc}} + L_{\text{irr}} + L_{\text{col}}$ derive a three-component template (non-abelian carrier + chiral carrier + minimality constraint) from Axiom **A1** through PLEC cost minimisation.

(G_2) Paper 2 ()'s $L_{\text{gauge_template_uniqueness}}$ proves that the unique gauge-group template compatible with the three components is $SU(3)_c \times SU(2)_L \times U(1)_Y$. No alternative non-abelian rank-combinations ($SU(4)$, $SO(10)$, etc.) are PLEC-admissible.

(G_3) The dimension count of the unique template is $\dim SU(3) + \dim SU(2) + \dim U(1) = 8+3+1 = 12$, by the structure-constant count of each Lie algebra.

Under this chain, **12** is the unique integer consistent with (G_1)–(G_3); alternative counts would require contradicting PLEC's gauge-template-uniqueness output.

Status. Proved [P] in Paper 2, contingent on the PLEC chain from Paper 1 () and the gauge-template-uniqueness theorem. See `check_L_gauge_template_uniqueness` (`apf/gauge.py`).

A.1.2 Higgs-sector count: forced to 4 by electroweak admissibility

The Higgs-sector count 4 is forced by the combination of fermion-mass admissibility and electroweak gauge structure:

(H_1) Paper 4 ()'s fermion-mass structure derivation requires a Yukawa-coupling mechanism compatible with the $SU(2)_L \times U(1)_Y$ gauge template. The minimal gauge-compatible Yukawa structure requires a complex scalar doublet under $SU(2)_L$ with hypercharge $Y = 1/2$: the Higgs doublet Φ.

(H_2) A complex $SU(2)_L$ doublet has $2 \times 2 = 4$ real degrees of freedom: $\Phi = (\phi^+, \phi^0)^T$ with $\phi^+, \phi^0 \in \mathbb{C}$.

(H_3) Post-electroweak symmetry breaking, three of the four degrees of freedom are absorbed by the $W^\pm, Z$ gauge bosons as Goldstone modes; one remains as the physical Higgs scalar. The ACC record counts the *pre-absorption* 4, because slot-counting tracks independent admissibility channels before gauge-fixing eats any of them.

Alternative counts ruled out. A Higgs *singlet* (count 1) cannot generate charged-fermion masses. A *triplet* (count 3 or 5) generates tree-level ρ-parameter deviations ruled out by electroweak precision measurements at the $\sim 10^{-4}$ level. Larger representations (quadruplet, etc.) fail both mass-generation and ρ-parameter constraints simultaneously. Under PLEC, the minimum-cost admissible scalar sector is the complex doublet with 4 real components, uniquely.

Status. Proved [P] in Paper 4 Part II, contingent on PLEC admissibility of fermion-mass generation.

A.1.3 Fermion count: forced to 45 by the 1-of-4680 filter

The fermion count 45 is the output of a three-stage PLEC filter over a combinatorially large space of candidate multiplet assignments:

(F_1) **Combinatorial starting set.** Under the $SU(3)_c \times SU(2)_L \times U(1)_Y$ gauge template, the space of gauge-compatible fermion multiplets up to a bounded-hypercharge cutoff has cardinality 4680 (Paper 4 () Proposition on multiplet enumeration). Each candidate multiplet is a choice of $(SU(3), SU(2), U(1))$ representation labels.

(F_2) **Anomaly cancellation.** Gauge-compatible multiplet assignments must cancel $SU(3)^3$, $SU(2)^3$, $U(1)^3$, $SU(3)^2U(1)$, $SU(2)^2U(1)$, and mixed gravitational anomalies. Imposing all six anomaly-cancellation constraints simultaneously reduces 4680 down to a small set of anomaly-free configurations.

(F_3) **CP-coherence filter.** The CP-coherence admissibility condition (Paper 4's T_{4F} theorem with CP-coherence correction) selects the unique anomaly-free configuration compatible with PLEC's minimum-cost enforcement: three generations of the SM chiral fermion content.

Per-generation slot count: Q_L (weak-doublet up-down quarks, colour-triplet, 2 complex components $\times$ 3 colours = 6 real components counted as $2 \times 3 = 6$ slots) + u_R (singlet up-quark, 3 colours = 3 slots) + d_R (singlet down-quark, 3 colours = 3 slots) + L_L (weak-doublet lepton, 2 slots) + e_R (charged-lepton singlet, 1 slot) = 15 slots per generation. Three generations: $3 \times 15 = 45$.

Why three generations and not two or four. The three-generation count is forced by T_{4F} + CP-coherence: two generations cannot sustain CP violation with the observed structure, and four generations violate anomaly cancellation at the required precision. See Paper 4 Part II and `check_T_field` (`apf/gauge.py`).

Status. Proved [P+lattice] in Paper 4. The [+lattice] tag reflects that the bounded-hypercharge cutoff in (F_1) uses a lattice-QCD-informed hypercharge-quantisation argument.

The sum $12 + 4 + 45 = 61$ is a theorem, not an observation. Each of the three counts above is structurally forced; their arithmetic sum is the value of K_{SM} under Paper 2's L_{count}.

A.2 The structural partition: $K_{\mathrm{SM}} = 45 + 4 + 12 = 61$

Theorem A.1 (L_{count}; Paper 2). *Given the structural forcings of §A.1, the total SM-interface capacity is*

$$K_{\mathrm{SM}} = C_{\mathrm{total}} = 45 \textit{ (fermion, from §A.1.3)} + 4 \textit{ (Higgs, from §A.1.2)} + 12 \textit{ (gauge, from §A.1.1)} = 61.$$

Why this is a forcing, not an observation. The claim "the SM has $12 + 4 + 45 = 61$ admissibility slots" is the consequence of composing three independent structural derivations: $L_{\mathrm{gauge_template_uniqueness}}$, electroweak-admissible Higgs content, and the 1-of-4680 filter. The *final integer* is determined by **A1** + PLEC + the identified admissibility conditions, not by a posteriori matching to PDG [4]. The counts happen to agree with PDG precisely because PDG is the observed output of the Standard Model.

Status. Proved [P] in Paper 2's L_{count} theorem, rests on Paper 4 Part II. See `check_L_count` (`apf/gauge.py`).

A.3 The self-exclusion count: $d_{\text{eff}}^{\text{SM}} = 60 + 42 = 102$

Theorem A.2 ($L_{\text{self_exclusion}}$; Paper 6)**.** *The per-slot effective degeneracy at the Standard-Model-coupled horizon interface is*

$$d_{\text{eff}}^{\text{SM}} \;=\; (K_{\text{SM}} - 1) + C_{\text{vacuum}} \;=\; 60 + 42 \;=\; 102,$$

where $C_{\text{vacuum}} = 42$ is the vacuum-residual count derived from the T12 interface partition.

Derivation, Part 1: the bilateral count $K_{\text{SM}} - 1 = 60$. At the horizon-reciprocity interface, each ACC slot commits *one* of the 61 channels as "self" and reciprocates with the remaining $K_{\text{SM}} - 1 = 60$ channels. This is the content of Paper 6 ()'s $T_{\text{horizon_reciprocity}}$ second-ε decomposition. The -1 is the self-exclusion: a channel cannot reciprocate with itself. The bilateral count $60 = K_{\text{SM}} - 1$ is therefore forced by the self-exclusion identity, not a separate posit.

Derivation, Part 2: the vacuum-residual count $C_{\text{vacuum}} = 42$. Paper 6 ()'s T12 theorem partitions V_{61} into a *local* stratum (slots involved in finite-interface maintenance) and a *global* stratum (slots admitting only global-interface-boundary readings):

$$V_{61} \;=\; V_{\text{local}} \;\oplus\; V_{\text{global}}, \qquad \dim V_{\text{local}} = 19, \quad \dim V_{\text{global}} = 42.$$

The dimension-count $\dim V_{\text{global}} = 42$ is forced by the T12 residual-capacity theorem: under PLEC, the local stratum absorbs exactly the slots involved in baryonic ordinary matter (3 slots) + non-baryonic non-vacuum residual (16 slots) = 19. The complement within V_{61} has dimension $61 - 19 = 42$, and it is this 42-dim complement that is identified with the global-interface stratum (Paper 6 §12 proof). C_{vacuum} is then identified with $\dim V_{\text{global}}$ by the auxiliary lemma $L_{\text{global_interface_is_horizon}}$, which shows that the global-interface stratum is the horizon absorber subspace.

Why 42 and not another integer. Given $K_{\text{SM}} = 61$ (forced by §A.2) and $\dim V_{\text{local}} = K_b + K_c = 3 + 16 = 19$ (forced by the residual partition of §A.4), the arithmetic requires $\dim V_{\text{global}} = 61 - 19 = 42$ exactly. Alternative values of $\dim V_{\text{global}}$ would require violating either K_{SM} (shift the SM field content) or the residual partition (shift the baryonic or dark-matter slot count).

Composition. $d_{\text{eff}}^{\text{SM}} = 60 + 42 = 102$: the two-sector decomposition of the horizon-reciprocity second-ε count under $L_{\text{self_exclusion}}$.

Status. Part 1: proved [P] in Paper 6. Part 2: proved [P] in Paper 6 §12 + $L_{\text{global_interface_is_horizon}}$ (tier-3 [P] in the v6.9 bank).

A.4 Residual partition: $61 = 3 + 16 + 42$

Theorem A.3 (Residual partition at the SM interface). *The residual partition of $K_{\mathrm{SM}} = 61$ into three cosmological sectors is*

$$K_{\mathrm{SM}} \;=\; K_b + K_c + K_\Lambda \;=\; 3 + 16 + 42,$$

where $K_b = 3$ (baryonic / ordinary matter), $K_c = 16$ (cold dark matter / non-baryonic non-vacuum residual), and $K_\Lambda = 42$ (cosmological constant / vacuum residual).

Derivation of $K_b = 3$. Paper 2 identifies one row of the Yukawa structure (up-quark, down-quark, charged-lepton triplet per generation, with the first-generation row carrying the baryon-number content of ordinary matter). The baryon-number-carrying slots in that row total 3: up-quark (u_L), down-quark (d_L), and the electron (e_L^-) contributing through the charge-neutrality of the baryonic sector. Exact derivation: Paper 2 §5.

Derivation of $K_\Lambda = 42$. This is C_{vacuum} from the previous subsection. Paper 6 identifies the vacuum-residual count with the dimension of the global-interface stratum $V_{\mathrm{global}} = 42$.

Derivation of $K_c = 16$. By subtraction, given K_{SM}, K_b, and K_Λ fixed:

$$K_c \;=\; K_{\mathrm{SM}} - K_b - K_\Lambda \;=\; 61 - 3 - 42 \;=\; 16.$$

The number 16 also has an independent Paper-4 reading as a structural count (non-baryonic residual of the three-generation Yukawa sector minus the baryonic row), but the composition-by-subtraction is the load-bearing derivation for Paper 8's purposes.

Status. $K_b = 3$ and $K_\Lambda = 42$ are proved [P] in Papers 2 and 6 respectively; $K_c = 16$ is a composition [$\mathrm{P_{comp}}$]. Conditional uniqueness of the partition given the upstream identifications is Theorem 10.1.

A.5 Framework-internal forcing chain: composition theorem

Theorem A.4 (Framework-internal forcing chain for $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$). *Assume:*

(A) *Axiom **A1** (Finite Enforcement Capacity) and PLEC (the argmin over the admissibility budget window);*

(B) *The gauge-template-uniqueness theorem $L_{\mathrm{gauge_template_uniqueness}}$ (Paper 2): the unique PLEC-admissible gauge-group template compatible with the Paper-1 three-component carrier is $SU(3)_c \times SU(2)_L \times U(1)_Y$;*

(C) *The Higgs-admissibility condition (Paper 4 Part II): the minimum-cost PLEC-admissible scalar sector compatible with charged-fermion masses is a complex $SU(2)_L$ doublet with 4 real components;*

(D) *The fermion-content filter T_{field} (Paper 4 Part II): the 1-of-4680 anomaly-cancellation + CP-coherence filter yields three generations of SM chiral fermions with 15 slots per generation;*

(E) *The self-exclusion identity* $L_{\text{self_exclusion}}$ *(Paper 6):* $d_{\text{eff}} = (K-1) + C_{\text{vacuum}}$ *at the Standard-Model-coupled horizon interface;*

(F) *The T12 interface partition (Paper 6):* $V_{61} = V_{\text{local}} \oplus V_{\text{global}}$ *with* $\dim V_{\text{local}} = K_b + K_c = 19$ *forced by the residual partition of §A.4.*

Then the Standard-Model ACC record $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) = (61, 102)$ *with residual partition* $(K_b, K_c, K_\Lambda) = (3, 16, 42)$ *is the unique record consistent with (A)–(F). No alternative integer tuple* $(K', d'_{\text{eff}}, K'_b, K'_c, K'_\Lambda)$ *satisfies (A)–(F) simultaneously.*

Proof. Step 1. By (A) + (B), the gauge sector contributes $\dim SU(3) + \dim SU(2) + \dim U(1) = 8+3+1 = 12$ slots. By (A) + (C), the Higgs sector contributes 4 slots. By (A) + (D), the fermion sector contributes $3 \times 15 = 45$ slots. Summation: $K_{\text{SM}} = 12 + 4 + 45 = 61$.

Step 2. By (F), $\dim V_{\text{local}} = K_b + K_c = 19$, forcing $\dim V_{\text{global}} = K_{\text{SM}} - 19 = 42$. This is the vacuum-residual count $C_{\text{vacuum}} = 42$.

Step 3. By (E), $d_{\text{eff}}^{\text{SM}} = (K_{\text{SM}} - 1) + C_{\text{vacuum}} = 60 + 42 = 102$.

Step 4. By Theorem 10.1, given (i) $K_{\text{SM}} = 61$, (ii) $K_b = 3$ (from Paper 2 baryonic Yukawa-row argument), and (iii) $K_\Lambda = 42$ from Step 2, the residual $K_c = 61 - 3 - 42 = 16$ is forced.

Step 5. Each of Steps 1–4 is a unique-value statement under its respective upstream identification. Therefore the composite $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}, K_b, K_c, K_\Lambda) = (61, 102, 3, 16, 42)$ is the unique tuple consistent with (A)–(F).

Alternative integer tuples. Any alternative tuple would require modifying one of the six upstream identifications:

- Shifting K_{SM} requires modifying (B), (C), or (D): a different gauge template, a different Higgs sector, or a different fermion filter. None of these has an APF-admissible alternative that also satisfies PLEC.
- Shifting $d_{\text{eff}}^{\text{SM}}$ with $K_{\text{SM}} = 61$ held fixed requires modifying (E) (a different self-exclusion identity) or (F) (a different T12 partition). Neither has an APF-admissible alternative.
- Shifting the residual partition requires modifying (D) (Yukawa baryonic-row derivation) or (F) (T12 partition dimension). Same obstruction.

Hence (61, 102, 3, 16, 42) is the unique PLEC-admissible SM ACC record. □ □

Status. Proved [P_{comp}] at the forcing level. Each of (A)–(F) is a structural APF theorem from Papers 1, 2, 4, or 6; the present theorem composes them into a single uniqueness statement so the reader can audit the forcing chain without consulting each paper separately.

Relation to the reviewer's Q1 concern. An external reviewer raised the question of whether $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}, K_\Lambda) = (61, 102, 42)$ might be a posteriori counting matched to observed SM + cosmological fractions. Theorem A.4 addresses this directly: the integers are determined by a *derivation chain internal to APF* — a composition of identified upstream theorems from Papers 1/2/4/6 under PLEC — not read off from PDG tables or from Planck 2018 posteriors. A reader who rejects any specific upstream identification ($L_{\text{gauge_template_uniqueness}}$, the 1-of-4680 filter, T12 partition, etc.) can track exactly which integer the rejection affects through the proof above. *The word "APF-internal" here is deliberate*: APF does not claim an ab initio derivation from universally-accepted microscopic physics (that is the territory of LQG / string theory / SUSY localisation, see §B.1.1).

What it claims is an internal derivation chain from APF's foundational axiom **A1** and the identified structural theorems of Papers 1/2/4/6 that terminates uniquely at $(61, 102)$.

A.6 Epistemic-status summary table

A.7 What would change these integers: modification catalogue

Theorem A.4 establishes that $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ is forced by six upstream identifications. Table 6 catalogues the reverse: for each upstream modification, what downstream integer shifts, and at what cost to the SM physics it is calibrated against.

Interpretation. The integers $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}, K_b, K_c, K_\Lambda) = (61, 102, 3, 16, 42)$ are brittle under unit shifts of any upstream identification; see §11 for the downstream-observable sensitivity. Each upstream identification has its own independent justification (gauge-template uniqueness, Higgs admissibility, anomaly cancellation, T12 residual-capacity, self-exclusion identity), and none has a PLEC-admissible alternative compatible with observed SM physics.

A.8 Honest scope note

This appendix does not reproduce every derivation in Papers 2, 4, and 6 in full. It extracts the chain that terminates at $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$ and the residual partition $(3, 16, 42)$, collecting the key structural forcings, their APF-internal uniqueness statement (Theorem A.4), and the modification catalogue (Table 6) in one place. A reader wanting full derivations of each count should consult the cited sources; a reader wanting a stand-alone audit of the chain should find this appendix sufficient.

Paper 8's own content — the ledger architecture, the bridge theorem, the sensitivity theorem, the finite-dimensional operator realisation — starts from these counts as imports and reads them through the four regime projections. Paper 8 does not claim to re-derive the counts; it claims to (a) collect the upstream forcings in one auditable place (this appendix), and (b) show that a single ACC record compatible with all four regime readings of them is available (the ledger architecture of §§5–9).

B Position Relative to Established Frameworks

Scope. This section positions APF's ledger architecture relative to five standard frameworks that bear on the themes of Paper 8: Bekenstein–Hawking black-hole thermodynamics, holography / AdS–CFT, algebraic quantum statistical mechanics (C*-algebras, KMS states, von Neumann factors), Landauer's principle and resource-theoretic thermodynamics, and cosmological parameter inference under ΛCDM. The aim is not to solve the comparison questions raised in external review but to make explicit where APF sits relative to each framework, what is contrast and what is borrowed, and which questions are deferred.

B.1 Bekenstein–Hawking black-hole thermodynamics

Bekenstein [6] and Hawking [7] established that a black hole of area A carries an entropy $S_{\mathrm{BH}} = A/(4\ell_P^2)$ in natural units. APF's horizon identity I_1 reproduces this exactly under the cell-count convention $K_{\mathrm{horizon}} = A/(4\ell_P^2)$ (Definition 2.7): no prefactor is mutated, no new entropy formula is

proposed. The APF content on top of S_{BH} is the *reading*: S_{BH} is the scalar-level π_T projection of the horizon ACC record, and the same record admits gravitational (π_G) and quantum (π_Q) readings via I_1. The three readings agree by construction.

APF's horizon construction does not attempt to re-derive S_{BH} from Einstein-equation first principles or to modify the Bekenstein-Hawking area law. It imports them as a boundary condition on the ACC architecture.

B.1.1 Comparison with microscopic horizon-counting frameworks

Scope. External review (second pass) asked how APF's ledger reading of I_1 compares with microscopic frameworks that derive S_{BH} from specific microstate counts (LQG punctures, string-theoretic brane bound states, supersymmetric localisation). The honest answer is: APF is *complementary* to these frameworks, not competing with them. APF does not predict subleading corrections (log A, etc.) that microscopic counts supply; microscopic frameworks do not automatically provide the cross-regime consistency statement that APF's I_1 supplies. This subsection makes the complementarity explicit.

APF does not predict log A corrections. Microscopic frameworks typically predict subleading logarithmic corrections to S_{BH}:

$$S_{\mathrm{BH}} = \frac{A}{4\ell_P^2} + \alpha \log\left(\frac{A}{4\ell_P^2}\right) + \mathcal{O}(1),$$

with the coefficient α framework-dependent (LQG: $\alpha = -3/2$ from puncture-ensemble statistics [16]; string theory: α depends on BH type and higher-curvature corrections). APF's I_1 reproduces only the leading-order cell count $K_{\mathrm{horizon}} = A/(4\ell_P^2)$ and does not supply structural content for the log-correction coefficient. This is not a failure of APF relative to microscopic frameworks — it is the natural consequence of APF being a *ledger-level* statement. The ACC record does not encode the microstate-counting statistics that log corrections come from; it encodes only the leading entropy-scalar.

What APF adds on top. The content APF provides that microscopic frameworks do not automatically supply:

1. **Cross-regime agreement.** I_1 asserts that the gravitational reading π_G, thermodynamic reading π_T, and quantum reading π_Q of the horizon ACC record all agree on the same scalar S_{BH}. This is a cross-regime consistency statement; microscopic frameworks each supply one of these readings (typically π_T through microstate counting) but do not automatically force agreement with the other two.

2. **Bridge to SM interface.** Theorem 6.3 identifies a 42-dimensional subspace of V_{61} at the canonical SM-dS joint point $K = 42$ that coincides across three independent constructions (Bekenstein cell count, SM residual partition, literal qubit tensor product). This is a bridge to the SM interface that microscopic BH frameworks do not address.

3. **Ledger brittleness.** The sensitivity theorem (§11.2) shows how unit shifts in upstream counts propagate to downstream cosmological predictions. This brittleness is orthogonal to microstate-counting questions and is a ledger-level content.

Joint content. In principle, a microscopic framework (e.g., LQG puncture ensembles) could populate the APF ledger from below: the microstate count would derive the coarse-grained $K_{\text{horizon}} + d_{\text{eff}}$; APF's I_1 would then check cross-regime consistency; and log corrections would appear as higher-order content beyond the leading ACC-scalar. This is the complementary picture: APF supplies the meta-theoretical consistency layer; microscopic frameworks supply the statistical-mechanics content that populates the ACC record.

Status. APF's I_1 is proved at leading order (Proposition 5.1 + Proposition 5.2) and explicitly disclaims subleading corrections. A joint APF + microscopic analysis — deriving the d_{eff} at horizons from LQG puncture-ensemble counting, or from string D-brane counting — is a natural direction for future work. Neither is attempted here.

B.2 Holography and AdS–CFT

Maldacena's AdS–CFT correspondence [8] relates a gravitational theory in $(d+1)$-dimensional anti-de Sitter bulk to a conformal field theory on the d-dimensional boundary, with the bulk partition function computing CFT correlators. APF's ledger architecture is structurally a different object: it relates *regime readings* of a single ACC record at a single interface, not bulk-boundary dualities.

Nevertheless, two structural parallels are worth noting without claiming equivalence:

- The $d_{\text{eff}}^{\text{SM}} = 60+42$ bilateral-plus-absorber decomposition of Paper 6's horizon-reciprocity (§8.1) has a suggestive parallel to bulk-boundary splittings: the 60 bilateral channels as "boundary" content (self-reciprocating degrees of freedom) and the 42 absorber channels as "bulk" content (global-interface stratum).
- The bridge theorem (Theorem 6.4) identifies a subspace accessible from three independent constructions — cosmological, two-tier carrier, horizon reciprocity — which echoes the AdS–CFT pattern of multiple equivalent formulations of the same physical data.

Neither parallel amounts to a claim that APF is a reformulation of AdS–CFT or that its ledger content maps to a holographic duality. A full categorical treatment comparing the two frameworks is a natural research direction; none is attempted here. APF and AdS–CFT are best treated as parallel frameworks addressing overlapping phenomena (black-hole entropy, cosmological constants, structural counting) with different organising architectures.

B.3 Algebraic quantum statistical mechanics

The algebraic framework [9, 10] replaces state vectors with states on C^*-algebras, and thermal equilibrium with the KMS condition [11]. Infinite-volume systems require non-trivial factor theory (Type III_1 von Neumann algebras for generic thermal states at $\beta > 0$).

APF's operator-level content (§9) is explicitly *finite-dimensional* and uses the finite C^*-algebra $M_N(\mathbb{C}) = \mathcal{B}(H_{\text{micro}})$ with the finite-volume Gibbs state ρ_β. No KMS condition is required in the finite-volume setting; no factor-type question arises ($M_N(\mathbb{C})$ is Type I_N). The section explicitly disclaims any infinite-volume extension (§9, "Non-claims" paragraph).

APF is therefore orthogonal to algebraic QSM, not competing with it: APF operates entirely within the finite-volume regime where the algebraic machinery is unnecessary, while algebraic QSM handles questions (thermodynamic-limit existence, factor-type classification, modular theory) that

APF does not address. An infinite-volume extension of APF would need to import algebraic-QSM machinery; the present paper does not attempt one.

B.4 Landauer's principle and resource-theoretic thermodynamics

Landauer's principle [12] asserts $k_B T \ln 2$ as the minimum heat dissipated per erased bit. Resource-theoretic thermodynamics [14] generalises this to a framework where thermal operations define a preorder on quantum states and free energy tracks resource content.

The APF minimum-distinction floor $\varepsilon^* > 0$ sits in a structurally analogous slot: it is the per-distinction enforcement-cost floor below which distinctions cannot be maintained. The Landauer bound supplies a thermal lower bound: $\varepsilon^*(\Gamma) \geq k_B T \ln 2$ at temperature-bearing interfaces (§2.3). APF does not assume saturation of this bound, nor does it yet recast ε^* in the free-energy language of resource theory.

The relationship is that APF's ledger is compatible with — but not derived from — the Landauer / resource-theoretic framework. A fruitful cross-framework question is whether the APF structural counts (K, d_{eff}) correspond to resource-theoretic monotones under an appropriate thermal-operations preorder. This is deferred.

B.5 Cosmological parameter inference under ΛCDM

The standard cosmological model ΛCDM [1] fits six free parameters $(\Omega_b h^2, \Omega_c h^2, \theta_*, \tau, n_s, \ln(10^{10} A_s))$ against CMB + BAO + SN data to produce posterior distributions over $(\Omega_b, \Omega_c, \Omega_\Lambda, H_0, \ldots)$. APF's quantitative predictions — $\Omega_b = 3/61$, $\Omega_c = 16/61$, $\Omega_\Lambda = 42/61$, $H_0 = 70.03$ km/s/Mpc — are *structural fixed points* of the derivation chain internal to APF (Theorem A.4) with *zero* free parameters (Table 5).

The two approaches sit at different epistemic levels. ΛCDM inference produces a posterior over continuous parameters; APF's architecture produces integer-valued predictions that are either compatible with the posterior or not. A competitive statistical comparison would take the APF predictions as a zero-parameter baseline, fit ΛCDM on the same data, compute the Bayesian evidence ratio $\log B = \log p(D|\text{APF}) - \log p(D|\Lambda\text{CDM})$, and report the model-selection verdict. This analysis is *not* performed in Paper 8 (Lane A scope decision; see abstract and §8's "Not a cosmological inference paper" paragraph) and is explicitly deferred to a companion phenomenology paper.

Current agreement is recorded at the $\sim 1\sigma$ level for the residual partition and at the $\sim 8\%$-gap (Hubble-tension-midpoint) level for $\rho_\Lambda / M_{\text{Pl}}^4$; see Table 3 for the explicit falsifier table. None of this constitutes a competitive statistical fit, and Paper 8 does not claim one.

B.6 Summary: APF as a meta-theoretical consistency framework

Scope. This subsection records the meta-theoretical position of APF's ledger architecture relative to the frameworks compared in §§B.1–B.5. Its purpose is to make explicit what kind of contribution APF is attempting, so that reviewers can evaluate the paper against its own contribution type rather than against the (different) contribution type of microscopic theories.

APF is a meta-theoretical consistency framework, not a microscopic theory. The ledger architecture reads what any microscopic theory must satisfy at the level of the ACC record (K, d_{eff}); it does not derive microstates from first principles. This is a distinct contribution type from:

- microscopic horizon-counting frameworks (LQG puncture ensembles, string D-brane bound states, SUSY localisation), which derive S_{BH} and its subleading corrections from specific microstate counts;
- holographic / AdS–CFT frameworks, which derive bulk partition-function content from boundary CFT correlators;
- algebraic quantum statistical mechanics, which handles thermodynamic-limit questions (factor-type classification, modular theory, KMS existence) via C^*-algebras;
- ΛCDM cosmological parameter inference, which fits six cosmological parameters to CMB + BAO + SN data to produce posterior distributions.

APF's contribution is orthogonal: it provides a *single-record audit across six regime projections*, a *brittle-consequences sensitivity bound* that transfers microscopic uncertainties to cosmological observables, a *formal categorical treatment of observer-dependence*, and a *cross-regime bridge theorem* (§6.4) that identifies a load-bearing subspace across cosmological / two-tier / horizon-reciprocity constructions.

What APF does with S_{BH}. Microscopic frameworks derive $S_{\text{BH}} = A/(4\ell_P^2)$ from microstate counts; APF's I_1 *reads* the same S_{BH} across three regime projections (π_G, π_T, π_Q) as agreement on the horizon ACC-scalar (Proposition 5.2). This is a complementary statement: microscopic theories supply the statistical-mechanics content that determines the value; APF supplies the consistency statement that ensures all three sector readings agree on that value. Neither subsumes the other.

What APF does *not* claim. The second reviewer explicitly raised concerns that APF's structure might be "consistency reparametrisation rather than microscopic derivation." This subsection accepts that framing on its own terms and re-expresses it: yes, APF is a meta-theoretical consistency framework, by design, not a microscopic theory. The claim this paper makes is not that the ACC ledger replaces LQG, string theory, or SUSY localisation — those remain the correct tools for microscopic derivation. The claim is that a ledger-level layer with four consistency identities and a bridge theorem imposes cross-regime constraints that any microscopic theory must respect, and that this layer has its own structural content (the sensitivity theorem, the observer category, the two-tier forcing proposition, the bridge identification of V_Λ across three independent constructions) independent of the microscopic story.

Value of the framework. The value of APF's ledger architecture is threefold:

1. **Auditability.** A single ACC record + six regime projections + four consistency identities is a compact object to audit: a reviewer can check each identity, trace each prediction to its upstream forcings, and identify exactly which upstream claim a given prediction rests on. This is what Papers 2, 4, 6, and 7 — the microscopic inputs — do not automatically provide.
2. **Brittleness bounds.** The sensitivity theorem (§11.2) makes the paper's zero-parameter predictions *auditably falsifiable*: a unit shift in any upstream integer propagates to downstream

observables at a calculated rate. Microscopic frameworks do not typically supply this brittleness structure.

3. **Cross-regime bridges.** The interface-sector bridge theorem (Theorem 6.4) identifies a 42-dimensional subspace of V_{61} across three independent constructions (cosmological partition, two-tier carrier, horizon reciprocity). This is a structural statement that lives at the ledger level and does not arise automatically from any single microscopic framework.

Companion paper. A companion phenomenology paper, currently in preparation, runs the brittleness sensitivity (Theorem 11.1) against Planck + BAO + SN posteriors under full covariance-matrix analysis and model-selection vs ΛCDM. That paper is the proper venue for the statistical-inference content that Paper 8 explicitly defers.

Summary for reviewers. If the reviewer's question is "does APF derive microstates from first principles?" the answer is no — that is LQG / string / SUSY-localisation territory, and APF does not claim it. If the question is "does APF provide a testable meta-theoretical consistency framework with brittle-consequence bounds and auditable upstream forcings?" the answer is yes — that is what Paper 8 aims to establish. The integrated non-trivial content is collected in the centerpiece Section D below; Section C below it supplies a fully explicit minimal working example for external audit.

C Minimal Working Example: a fully explicit audited interface

Why this section. External reviewers of earlier versions asked for "a minimal, end-to-end example with full definitions and proofs, where the ledger enforces a quantitative cross-sector constraint that is not a tautology and is testable with existing data." This section provides a toy interface ($K = 3$ slots, $d_{\text{eff}} = 4$ integer) in which every object is written down concretely, every identity is verified by hand, and every result can be reproduced in $\lesssim 10$ lines of numpy without the APF codebase. The example is pedagogical, not physical; its purpose is to demonstrate that the ledger machinery is mathematically well-defined and operationally checkable at small scale.

C.1 The toy interface

Let $K = 3$, $d_{\text{eff}} = 4$, $C_{\text{vacuum}} = 1$. The slot space is

$$V_K \;=\; \mathbb{C}^3,$$

with basis $\{e_1, e_2, e_3\}$; each basis element represents one of the three slots. The gauge group is taken to be the trivial group $G = \{1\}$ (so every subspace is automatically G-invariant); this is the toy-interface analogue of G_{SM}. The residual partition is

$$(K_b, K_c, K_\Lambda) \;=\; (1, 1, 1),$$

so each of the three slots forms its own sector.

C.2 The ACC record and microstate space

The ACC record is $(K, d_{\text{eff}}) = (3, 4)$. The microstate Hilbert space is the literal tensor product

$$H_{\text{micro}} \;=\; (\mathbb{C}^4)^{\otimes 3} \;=\; \mathbb{C}^{64},$$

of dimension $N = d_{\text{eff}}^K = 64$. The algebra of observables is the Type I_{64} factor

$$\mathfrak{A} \;=\; \mathcal{B}(H_{\text{micro}}) \;=\; M_{64}(\mathbb{C}).$$

The literal-factorisation conditions of Proposition 4.5 hold: (SC1) uniform $\mathbb{C}^4$ per slot; (SC2) independence of per-slot admissibility; (SC3) integer ambient. So the tensor-power embedding $\Phi : H_{\text{micro}} \to V_K^{\otimes K}$ is literal for this toy interface.

C.3 The regime projections

All six projections evaluate at the toy ACC record:

$\pi_F(\text{ACC}) = K = 3$	(gauge / field integer count)
$\pi_Q(\text{ACC}) = N = d_{\text{eff}}^K = 4^3 = 64$	(quantum microstate count)
$\pi_T(\text{ACC}) = K \ln d_{\text{eff}} = 3 \ln 4 \approx 4.159$	(thermodynamic entropy scalar)
$\pi_C(\text{ACC}) = (1/3, 1/3, 1/3)$	(cosmological partition fractions)
$\pi_A(\text{ACC}) = Z(\beta) = (1 + 3e^{-\beta\varepsilon})^3$	(partition function at local spectrum $\{0, \varepsilon\}$)
$\pi_G(\text{ACC}) = \text{undefined}$	(no horizon at this toy interface)

C.4 One bridge identity (verified by hand)

The toy-interface analogue of I_2 is the common-denominator identity:

$$\pi_F(\text{ACC}) \;=\; 3 \;=\; \text{denom}\big(\pi_C(\text{ACC})\big) \;=\; 3. \quad \checkmark$$

The residual partition $K_b + K_c + K_\Lambda = 1 + 1 + 1 = 3 = K$ closes. The three cosmological fractions sum to 1.

C.5 One operator expectation (Identity (B) verified by hand)

Take the local Hamiltonian

$$H_{\text{local}} \;=\; \text{diag}(0, \varepsilon, \varepsilon, \varepsilon) \in M_4(\mathbb{C}),$$

with $C_{\text{vacuum}} = 1$ vacuum state and $d_{\text{eff}} - C_{\text{vacuum}} = 3$ excited states. The vacuum projector on slot 1 is

$$P_{\text{vac},1} \;=\; \text{diag}(1, 0, 0, 0) \otimes \mathbf{1}_4 \otimes \mathbf{1}_4.$$

Its trace is $\mathrm{tr}(P_{\mathrm{vac},1}) = 1 \cdot 4 \cdot 4 = 16$. Its expectation on the maximally mixed state $\rho_{\max} = \mathbf{1}/64$ is

$$\langle P_{\mathrm{vac},1} \rangle_{\rho_{\max}} \;=\; \frac{\mathrm{tr}(P_{\mathrm{vac},1})}{N} \;=\; \frac{16}{64} \;=\; \frac{1}{4} \;=\; \frac{C_{\mathrm{vacuum}}}{d_{\mathrm{eff}}}. \quad \checkmark$$

Identity (B) of §9.2 is verified exactly in the toy case.

C.6 One quantitative consequence: the toy "ρ_Λ"

The ledger matter-sector formula gives, for the $K_\Lambda = 1$ "toy vacuum residual" sector:

$$\rho_\Lambda^{\mathrm{toy}} \;:=\; \frac{K_\Lambda}{d_{\mathrm{eff}}^{K+1}} \;=\; \frac{1}{4^4} \;=\; \frac{1}{256} \;\approx\; 3.9 \times 10^{-3}.$$

This is the toy interface's prediction for the dimensionless vacuum density (in whatever toy "Planck units" the toy interface carries). Its role is pedagogical: it shows that the matter-sector formula $\rho_X / M_{\mathrm{Pl}}^4 = C_X / d_{\mathrm{eff}}^{K+1}$ is a closed-form arithmetic consequence of the ACC record, verifiable by hand, with no hidden inputs beyond the integers $(K, d_{\mathrm{eff}}, K_\Lambda) = (3, 4, 1)$.

C.7 Partition-function limit (Identity (A))

At $\beta \to 0^+$, $e^{-\beta\varepsilon} \to 1$, so

$$Z(\beta \to 0) \;=\; (1+3)^3 \;=\; 64 \;=\; N. \quad \checkmark$$

Therefore $\ln Z(\beta \to 0) = \ln 64 = 3 \ln 4 = \pi_T(\mathrm{ACC})$ exactly. Identity (A) verified.

C.8 Reproducibility: 10-line numpy verification

A reader can reproduce every numerical result in this section in under 10 lines of numpy:

```
import numpy as np

K, d_eff, C_vac = 3, 4, 1
N = d_eff**K                                        # 64

# Identity (A): partition function at beta -> 0
Z0 = (1 + (d_eff - C_vac))**K
assert Z0 == N                                      # 64 == 64

# Identity (B): vacuum-projector expectation at max-mixed
P_vac_trace = C_vac * d_eff**(K-1)                  # 16
expectation = P_vac_trace / N
assert abs(expectation - C_vac/d_eff) < 1e-12       # 0.25 == 0.25

# Toy "rho_Lambda"
```

```
K_Lambda = 1
rho_toy = K_Lambda / d_eff**(K+1)
print(f"rho_toy = {rho_toy}")                        # 0.00390625
```

C.9 Why this example matters

The reviewer's concern "many constructions become formal bookkeeping rather than physics unless literal-factorisation conditions hold" is addressed directly by §R: at the toy interface, (SC1)–(SC3) all hold, the tensor-product factorisation is literal, and every formula in the ledger machinery is verifiable by hand. This establishes that the ledger machinery is *well-defined as mathematics* at small scale; the question "does it apply physically at the Standard-Model scale?" is a separate question about the schematic-vs-literal distinction (Proposition 4.5, boxed note near §2.2). The toy interface is not a physical model of anything; it is an audit harness for the ledger's mathematical structure.

Status. Every identity in §R is verified exactly. The 10-line numpy snippet above is reproducible by any reader with a standard Python installation; no APF codebase, no Paper-8 supplement machinery, no bank checks required. This is the "minimal, end-to-end example external readers can audit without relying on the APF stack" the reviewer asked for.

D The Gauge-Cosmological Bridge: Integrated Nontrivial Content

Scope. This section is the paper's **centerpiece**. The non-trivial content of Paper 8 is distributed across several sections in the supplement — the bridge theorem (§6.4), the derivation chain internal to APF (Appendix A), the two-tier forcing proposition (§4.4), the partition uniqueness (§10.1), the sensitivity theorem (§11.2), and the multivariate Bayes-factor confrontation (§11.4) — because each has its own technical material and belongs in its own detailed section. But the reader should also be able to see, in one place, how these pieces integrate into a single non-trivial structural mechanism: the I_2 gauge-cosmological bridge, together with what it forces and what brittle consequences flow from it.

This centerpiece section gathers those pieces, in compressed form, in the following sequence: (§D.1) setup and thesis; (§D.2) the bridge theorem with gauge-uniqueness strengthening; (§D.3) the derivation chain internal to APF for $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$; (§D.4) two-tier uniqueness + partition minimality; (§D.5) brittle cosmological consequences and the multivariate Planck 2018 Bayes factor; (§D.6) status. Detailed proofs remain in the sections cited; this section states and integrates.

The centerpiece in one reading

§D.1	**Setup: I_2 is the non-trivial bridge case**
↓	
§D.2	**Bridge theorem + V_Λ gauge-uniqueness** (*the 42-dim subspace is forced, not coincidental*)
↓	
§D.3	**Framework-internal forcing chain** $\Rightarrow (61, 102, 42)$ (*composition of Papers 1/2/4/6 under PLEC*)
↓	
§D.4	**Two-tier structural necessity** (*single-scale carriers are impossible*)
↓	
§D.5	**Brittle consequences + Planck 2018 confrontation** ($\chi^2 = 1.18$ for 3 d.o.f., Mahalanobis 1.09σ)
↓	
§D.6	**Status: what the centerpiece establishes**

The reading is linear: the bridge + gauge-uniqueness delivers a specific 42-dim subspace; the forcing chain ties it to the integer tuple $(61, 102, 42)$; the two-tier necessity rules out architectural alternatives; the brittleness quantifies the falsifiable consequences; the Planck 2018 confrontation evaluates current compatibility. Each arrow is a theorem or composition proved in the surrounding supplement sections.

D.1 Setup and thesis

Paper 8 formalises a ledger architecture (the ACC record (K, d_{eff}) together with six regime projections $\pi_T, \pi_G, \pi_Q, \pi_F, \pi_C, \pi_A$ and four consistency identities I_1, I_2, I_3, I_4) whose principal non-trivial content is a single structural mechanism: the I_2 gauge-cosmological bridge.

Among the four identities:

- I_1 (holographic) is a *definitional closure* under the horizon cell-count convention $K_{\text{horizon}} = A/(4\ell_P^2)$ (§5.1);
- I_3 (thermo-quantum) is a *standard finite-dimensional entropy calculation* (§5.3);
- I_4 (action-thermo) is a *standard high-temperature limit* of a finite-volume partition function (§5.4);
- I_2 (gauge-cosmological) is the unique *non-trivial bridge theorem* of the family (§6.4).

Only I_2 does structural work at the subspace level that is not available from its definition + standard machinery. That work is the content summarised below.

Thesis. The I_2 bridge identifies a specific 42-dimensional subspace $V_\Lambda \subset V_{61}$ as the unique G_{SM}-invariant complement of the 19-dim local stratum V_{local}. This identification, composed with the Paper 2/4/6 upstream theorems, uniquely forces the Standard-Model ACC record $(K_{\text{SM}}, d_{\text{eff}}^{\text{SM}}) =$

$(61, 102)$ with residual partition $(3, 16, 42)$. The two-tier architecture required to carry I_2 is itself structurally mandatory (single-scale carriers cannot support the required regime readings). Unit shifts in any upstream count propagate to calculable shifts in downstream cosmological predictions at the 1–2% sensitivity level. The resulting APF cosmological prediction $(\Omega_b, \Omega_c, \Omega_\Lambda) = (3/61, 16/61, 42/61)$ is multivariate-compatible with Planck 2018 at $\chi^2 \approx 1.18$ for 3 d.o.f. ($p \approx 0.76$, Mahalanobis distance $\approx 1.09\sigma$).

This is the non-trivial content of the framework.

D.2 The bridge theorem with gauge-uniqueness

The I_2 subspace-level content, at its strongest form, is the following two-theorem composition.

Theorem D.1 (I_2 bridge, compressed)**.** *At the Standard-Model interface with V_{61} carrying the SM gauge action G_{SM}, three constructions converge on the same 42-dimensional subspace:*

$$V_\Lambda^{\mathrm{cosmo}} \;=\; V_\Lambda^{\mathrm{global}} \;=\; V_\Lambda^{\mathrm{absorber}} \;=:\; V_\Lambda \;\subset\; V_{61}, \qquad \dim V_\Lambda = 42,$$

where $V_\Lambda^{\mathrm{cosmo}}$ is the vacuum-residual slot span from the cosmological partition, $V_\Lambda^{\mathrm{global}}$ is the global stratum from Paper 6's T12 interface partition, and $V_\Lambda^{\mathrm{absorber}}$ is Sector B of the horizon-reciprocity second-ε decomposition. (Full statement + commutative diagram: Theorem 6.4.)

Theorem D.2 (V_Λ gauge-uniqueness, compressed)**.** *V_Λ is the unique G_{SM}-invariant 42-dim subspace of V_{61} satisfying (a) complementarity to the 19-dim local stratum and (b) representation-theoretic consistency with the Paper-4 SM irrep decomposition. (Full proof: Theorem 6.6.)*

Centerpiece content. Theorem D.2 explains why Theorem D.1's three-construction coincidence is not accidental: any G_{SM}-invariant 42-dim subspace of V_{61} satisfying the Paper-2/4/6 structural constraints is forced to equal V_Λ. The three constructions converge because the target is unique, not because three independent definitions happen to give the same answer.

D.3 Framework-internal forcing chain for $(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}) = (61, 102)$

Above the subspace-level content, the bridge composes with Paper-2/4/6 upstream theorems to force the Standard-Model ACC record uniquely.

Theorem D.3 (Framework-internal forcing chain, compressed)**.** *Assume the six upstream identifications (A)–(F) of Theorem A.4: (A) Axiom **A1** + PLEC; (B) $L_{\mathrm{gauge_template_uniqueness}}$; (C) Higgs-admissibility; (D) 1-of-4680 fermion filter; (E) $L_{\mathrm{self_exclusion}}$; (F) T12 interface partition with $V_{\mathrm{local}} \oplus V_{\mathrm{global}}$. Then the SM ACC record is uniquely*

$$(K_{\mathrm{SM}}, d_{\mathrm{eff}}^{\mathrm{SM}}, K_b, K_c, K_\Lambda) \;=\; (61,\ 102,\ 3,\ 16,\ 42).$$

No alternative integer tuple satisfies (A)–(F) simultaneously. (Full proof: Appendix A, Theorem A.4.)

Centerpiece content. The integer $(61, 102, 42)$ chain is APF-internal to APF's Paper-1/2/4/6 derivation stack. It is not a posteriori counting matched to observed SM + cosmological fractions. Each of the six upstream identifications has its own independent structural justification within APF (gauge-template uniqueness from Paper 2; Higgs admissibility from Paper 4; etc.); the modification catalogue (Appendix A, Table 6) records that no upstream modification yields a PLEC-admissible alternative compatible with observed SM physics.

D.4 Two-tier uniqueness + partition minimality

The bridge theorem and the forcing chain rest on the two-tier $(V_{\text{slot}}, H_{\text{micro}})$ carrier. Is that architecture chosen or mandatory?

Theorem D.4 (Two-tier architecture is structurally mandatory, compressed)**.** *No single-scale carrier can simultaneously support non-trivial π_F (integer structural count), non-trivial π_Q (exponential microstate count with $N \neq d$), and consistent π_C (residual partition with I_2-compatible common denominator). The two-tier architecture $(V_{\text{slot}}, H_{\text{micro}})$ with $N = d_{\text{eff}}^K$ is the unique viable ACC architecture for non-degenerate interfaces. (Full statement + proof: Theorem 4.12.)*

Proposition D.5 (Residual partition uniqueness, compressed)**.** *At the SM interface with the upstream identifications (A)–(F), the residual partition $(K_b, K_c, K_\Lambda) = (3, 16, 42)$ is the unique partition compatible with the T12 interface partition + first-generation Yukawa-row baryonic identification. Nearby partitions (e.g. $3 + 15 + 43$ or $4 + 16 + 41$) fail structural consistency checks detailed in §10.2. (Full statement: Theorem 10.1 + the lookalike-fraction exclusion analysis of §10.2.)*

Centerpiece content. The two-tier architecture is forced (Theorem D.4), and within it the SM partition is uniquely determined (Proposition D.5). Together these place the bridge identification of V_Λ on a structural footing that leaves no architectural freedom unused.

D.5 Brittle cosmological consequences and multivariate confrontation

The composed non-trivial content above generates specific, quantitative cosmological predictions. These are the content of the brittleness / sensitivity analysis.

Theorem D.6 (Upstream-count sensitivity, compressed)**.** *Unit shifts in the upstream integer counts $(K_{\text{SM}}, K_\Lambda, d_{\text{eff}}^{\text{SM}})$ propagate to downstream cosmological predictions at calculable rates:*

- $K_{\text{SM}} \to 62$*:* $\Omega_\Lambda \to 42/62 \approx 0.677$ *($\Delta\Omega_\Lambda \approx -1.1\%$).*
- $K_\Lambda \to 41$*:* $\Omega_\Lambda \to 41/61 \approx 0.672$ *($\Delta\Omega_\Lambda \approx -1.6\%$).*
- $d_{\text{eff}}^{\text{SM}} \to 103$*:* $\rho_\Lambda / M_{\text{Pl}}^4 \to 42/103^{62}$*, a factor-$\sim 1.8$ shift.*
- *Joint* $(60, 101)$*: upstream-chain inconsistency; no admissible ACC record exists.*

(Full statement + proofs: Theorem 11.1.)

Empirical confrontation (Planck 2018, multivariate). The APF prediction $\vec{\Omega}^{\mathrm{APF}} = (3, 16, 42)/61 = (0.04918, 0.26230, 0.68852)$ compared to Planck 2018 $\vec{\Omega}^{\mathrm{obs}} = (0.04897, 0.26449, 0.6847)$ yields (full covariance analysis: §11.4):

Quantity	Value	Interpretation
Marginal residuals / σ_i	$(+0.68, -0.88, +0.52)$	each within 1σ
Multivariate χ^2 (3 d.o.f.)	1.18	$p = 0.76$
Mahalanobis distance	1.09σ	full 3-param space
Bayes factor vs ΛCDM-flat (natural priors)	0.7–1.3	order unity

The APF prediction is statistically compatible with Planck 2018 at the three-observable level. Under natural physical priors ($\sigma_{\mathrm{prior}} \sim 0.1$ per component), APF's zero-free-parameter prediction is competitive with or favoured over flat-prior ΛCDM fits.

Future decisive tests. A factor-3 tightening of the Planck posteriors (achievable with CMB-S4 + LiteBIRD + forthcoming BAO surveys) would push χ^2_{APF} to ~ 10, decisively excluding APF if the central values remain at Planck 2018 values. Conversely, if the central values drift toward $(3/61, 16/61, 42/61)$, APF's zero-parameter prediction would be positively favoured over ΛCDM fits. This is the brittleness of the framework: specific, quantitative, falsifiable on the $\sim$ 5-year horizon.

D.6 Status: what the centerpiece establishes

Proved in this paper.

1. The bridge theorem $V_\Lambda^{\mathrm{cosmo}} = V_\Lambda^{\mathrm{global}} = V_\Lambda^{\mathrm{absorber}}$ (Theorem 6.4, [$\mathrm{P}_{\mathrm{bridge}}$]);
2. V_Λ gauge-uniqueness modulo G_{SM} (Theorem 6.6, [P], new in v2.3);
3. Framework-internal forcing chain for $(61, 102, 3, 16, 42)$ (Theorem A.4, [$\mathrm{P}_{\mathrm{comp}}$]);
4. Single-scale impossibility (Theorem 4.12, [P], new in v2.3);
5. Upstream-count sensitivity (Theorem 11.1, [$\mathrm{P}_{\mathrm{comp}}$]);
6. Multivariate Planck 2018 Bayes-factor compatibility (§11.4, [$\mathrm{P}_{\mathrm{demonstrative}}$], strengthened in v2.3).

Imported from prior APF papers.

- $L_{\mathrm{gauge_template_uniqueness}}$, L_{count}, baryonic Yukawa-row identification: Paper 2 ();
- Higgs-admissibility, 1-of-4680 filter, three-generation forcing: Paper 4 ();
- T12 interface partition, $L_{\mathrm{self_exclusion}}$, horizon-reciprocity decomposition: Paper 6 ().

Open / conjectural.

- Observer-independence of the bridge itself (Conjecture 13.9; formalised but not proved).
- Full posterior-samples ($\Omega_b, \Omega_c, \Omega_\Lambda$ + nuisance) analysis (deferred to companion phenomenology paper).
- Subleading log-A corrections to S_{BH} (APF does not predict them; out of scope by design).

Bottom line. The I_2 bridge, with its gauge-uniqueness strengthening, its derivation chain internal to APF, its structurally-mandatory two-tier architecture, and its brittle downstream consequences that are multivariate-compatible with Planck 2018 at $\chi^2 \approx 1.18$, is the single non-trivial structural mechanism of Paper 8. The rest of the supplement supplies the detailed technical machinery; this centerpiece is the integrated reading.

Acknowledgements

This supplement is released in lockstep with the current Paper 8 main paper. See the main paper for framework-level acknowledgements.

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Prediction	APF value	Current measurement	Falsifier
Ω_Λ	$42/61 = 0.6885$	Planck 2018: 0.6847 ± 0.0073 [1]	Shift of Ω_Λ outside $\sim [0.67, 0.7$ at 3σ of Planck falsifies $K_\Lambda = 42$ at the 1.6% sensitivity of (b).
$\rho_\Lambda / M_{\mathrm{Pl}}^4$	$42/102^{62}$	Planck 2018 equivalent: $10^{-122.94}$; APF: $10^{-122.91}$	An $\mathcal{O}(1)$ multiplicative shift in observed ρ_Λ (within a single e-fold factor) falsifies $d_{\mathrm{eff}} = 102$ at the $\sim 1.8\times$ sensitivity of

Main paper §	**Bank check**	**Module**
§2.3 I_1 holographic	`_identity_I1_holographic`	`unification.py`
§2.3 I_2 gauge–cosmological	`_identity_I2_gauge_cosmological`	`unification.py`
§2.3 I_3 thermo–quantum	`_identity_I3_thermo_quantum`	`unification.py`
§2.3 I_4 action–thermo	`_identity_I4_action_thermo`	`unification.py`
§2.4 T_{ACC} unification	`check_T_ACC_unification`	`unification.py`
§4.3 Three-level unification	`check_T_three_level_unification`	`unification_three_levels.py`
§5.2 F_{horizon}	`check_F_horizon_five_conditions`	`subspace_functors.py`
§5.2 F_{bridge} (Paper 6)	`check_T_interface_sector_bridge`	`gravity.py`
§5.2 F_{quantum}	`check_F_quantum_five_conditions`	`subspace_functors.py`
§5.2 F_{operator}	`check_F_operator_five_conditions`	`subspace_functors.py`
§5.3 Subspace functors unified	`check_T_subspace_functors_unified`	`subspace_functors.py`
§6 FRE	`check_T_FRE_SM_to_entropy_dictionary`	`fractional_reading.py`
§6.2 SM entropy dictionary (Ω_b)	`check_L_Omega_b_is_entropy_fraction`	`fractional_reading.py`
§6.2 SM entropy dictionary (Ω_c)	`check_L_Omega_c_is_entropy_fraction`	`fractional_reading.py`
§6.2 SM entropy dictionary (Ω_Λ)	`check_L_Omega_Lambda_is_entropy_fraction`	`fractional_reading.py`
§7 Conversion-matrix closure	`_identity_composition_tests`	`unification.py`
§8.1 Matter-sector formula	`check_L_Lambda_absolute_numerical_formula`	`fractional_reading.py`
§8.3 ρ_Λ Planck 2018 match	`check_L_Lambda_absolute_vs_Planck_2018`	`lambda_absolute.py`
§8.4 H_0 algebraic inversion	`check_T_Lambda_to_H0_inversion`	`lambda_absolute.py`
§8.5 T_{CMB} absolute	`check_T_T_CMB_absolute_formula`	`thermal_absolute.py`
§8.5 Species-exponent interp.	`check_T_thermal_exponent_interpretation`	`thermal_absolute.py`
§8.6 Composed robustness theorem	`check_T_Lambda_absolute_bulletproof`	`lambda_absolute.py`
§9 Operator-level realisation	`check_T_Lambda_d2_operator_derivation`	`lambda_operator_derivation.py`
§9.2 Identity (A) $Z \to N$	`check_T_Lambda_partition_function_at_beta_one`	`lambda_operator_derivation.py`
§10.2 Σm_ν suggestive fit	`check_L_Sigma_m_nu_suggestive`	`thermal_absolute.py`
§10.3 η out-of-scope	`check_L_eta_does_not_fit_cleanly`	`thermal_absolute.py`

Table 4: Main-paper-to-code correspondence for Paper 8. Every theorem-level claim in the main paper has a registered verification check in the v6.9 codebase, listed here with its `apf/` module. See Paper 13 (*The Minimal Admissibility Core*) for the full theorem registry with dependency edges.

Quantity	Value	Source	Status	If wrong:
Gauge-boson count	12	PDG [4]	[P], standard	SM gauge structure disagrees with PDG.
Higgs-sector count	4	PDG [4] + Paper 2	[P], standard	Higgs content disagrees.
Fermion count	45	PDG [4] + Paper 4	[P+lattice]	Three-generation structure or the 1-of-4680 derivation disagrees.
$K_{\mathrm{SM}} = 61$	61	Paper 2 L_{count}	[P]	$45+4+12 \neq 61$ (arithmetic).
Bilateral count	60	Paper 6 $T_{\mathrm{horizon_reciprocity}}$	[P]	Self-exclusion at horizon wrong.
Vacuum residual C_{vacuum}	42	Paper 6 T12 + $L_{\mathrm{global_interface_is_horizon}}$	[P]	Global-interface dimension or horizon identification wrong.
$d_{\mathrm{eff}}^{\mathrm{SM}} = 102$	102	Paper 6 $L_{\mathrm{self_exclusion}}$	[P]	$60 + 42 \neq 102$ (arithmetic).
$K_b = 3$	3	Paper 2 (baryonic Yukawa row)	[P]	Ordinary-matter slot count wrong.
$K_\Lambda = 42$	42	$= C_{\mathrm{vacuum}}$	[P]	Same falsifier as C_{vacuum}.
$K_c = 16$	16	$K_{\mathrm{SM}} - K_b - K_\Lambda$	[$\mathrm{P}_{\mathrm{comp}}$]	One of $K_{\mathrm{SM}}, K_b, K_\Lambda$ wrong.

Table 5: Epistemic-status summary for the Standard-Model ACC record counts. "Status" uses the taxonomy of §5: [P] proved from Axiom A1 through the stated upstream chain; [P+lattice] proved contingent on lattice-QCD input (here, the three-generation count); [$\mathrm{P}_{\mathrm{comp}}$] proved by composition of upstream results.

Upstream identification	**Modification**	**Downstream effect**
$L_{\text{gauge_template}}$ (gauge count 12)	$SU(3) \to SU(4)_c$	gauge count $12 \to 19$, K_{SM} shifts
$L_{\text{gauge_template}}$	Drop $U(1)_Y$	gauge count $12 \to 11$, K_{SM} shifts
Higgs-admissibility (count 4)	Higgs singlet (count 1)	K_{SM} shifts to 58
Higgs-admissibility	Higgs triplet (count 3 or 5)	K_{SM} shifts to 60 or 62
1-of-4680 filter (fermion count 45)	4 generations (count 60)	K_{SM} shifts to 76

Framework	What is counted	What is derived	What is left open
LQG horizon puncture ensemble [15, 17, 16]	Spin-network punctures on isolated horizon	$S_{\mathrm{BH}} = A/(4\ell_P^2)$ with Immirzi-parameter calibration + log A corrections	Universality of Immirzi parameter across horizon types; non-equilibrium dynamics
String microstate counting [18]	D-brane bound states in extremal / near-extremal BH backgrounds	S_{BH} at leading order + subleading corrections from higher-derivative terms	Generalisation to non-BPS / realistic astrophysical BHs
Supersymmetric localisation [19, 20]	Exact index evaluations on supersymmetric saddles	Exact degeneracies for BPS states + subleading corrections	Non-supersymmetric extensions; dynamical content beyond index theorems
APF ledger (this paper)	Admissibility slots on the ACC record $(K_{\mathrm{horizon}}, d_{\mathrm{eff}})$	I_1: three-way agreement $\pi_G = \pi_T = \ln \pi_Q = S_{\mathrm{BH}}$ at leading order	Subleading (log A) corrections; specific microstate identification; dynamical evolution

Table 7: Comparison of APF's I_1 ledger reading with microscopic horizon-counting frameworks. APF is complementary (provides cross-regime consistency) rather than competing (does not provide microstate identification or subleading corrections).